

Game Theory

From rational choice to mechanism design — Lecture Notes

Alvaro Mendez

2026-10-06

Table of contents

About these notes	1
Scope	1
Reading the notes	1
Notation	1
Sources	2
 I. Part I: Rational decisions	 3
1. Rational decision making	5
1.1. What a game is (and is not)	5
1.2. Some economic background	5
1.3. Preferences and utilities	6
1.4. Decision trees	6
1.5. Lotteries and Nature	7
1.6. Expected utility	7
1.7. Risk attitudes	8
1.7.1. Measuring risk aversion	8
1.8. Decisions over time: backward induction	9
1.9. The value of information	10
1.9.1. Imperfect information and Bayes' rule	10
1.10. Theory versus practice	10
1.11. Exercises	11
1.12. Sources	13
 II. Part II: Static games of complete information	 15
2. Static games and dominance	17
2.1. Static games of complete information	17
2.2. Normal-form representation	17
2.3. A gallery of examples	18
2.4. Pareto efficiency	18
2.5. Strict dominance and IESDS	19
2.6. Best responses and beliefs	19
2.7. Rationalizability and common knowledge	20
2.7.1. Beliefs as probabilities	21
2.7.2. IESDS on a continuum: two classic examples	21
2.8. Exercises	21
2.9. Sources	25
3. Nash equilibrium	27
3.1. Definition	27
3.2. Finding equilibria in a bi-matrix	27
3.3. Relation with dominance	27
3.4. Weak dominance	28
3.5. Equilibrium vs. efficiency	29
3.6. A field guide to small games	29
3.7. Selecting among equilibria	30
3.8. Exercises	31
3.9. Sources	33

4. Applications of Nash equilibrium	35
4.1. Cournot duopoly	35
4.2. Bertrand duopoly	36
4.3. Hotelling's location game	36
4.4. The tragedy of the commons	36
4.5. Selfish routing	37
4.6. Strategic substitutes and complements	37
4.7. Voting: the median voter theorem	38
4.8. Exercises	38
4.9. Sources	41
5. Mixed strategies and existence	43
5.1. Mixed strategies	43
5.2. Equilibrium and dominance with mixtures	43
5.3. Interpreting mixed strategies	44
5.4. Best-response diagrams	45
5.5. Nash's theorem	45
5.6. Fictitious play	46
5.7. Potential games	46
5.8. Computational complexity	47
5.9. Finding mixed equilibria systematically	47
5.10. Evolutionary stability	47
5.11. Exercises	48
5.12. Sources	51
6. Security strategies and zero-sum games	53
6.1. Maximin and minimax	53
6.2. Zero-sum games	54
6.3. Mixed security strategies	54
6.4. The minimax theorem	55
6.5. Linear programming	56
6.6. Constant-sum, symmetric and competitive games	56
6.7. Exercises	57
6.8. Sources	59
7. Social choice and voting	61
7.1. Constitutions	61
7.2. Majority rule and Condorcet cycles	61
7.3. Electoral systems and their paradoxes	62
7.4. Agenda setting and manipulation	63
7.5. Ways out	64
7.6. Voting as a game	64
7.7. Exercises	65
7.8. Sources	66
III. Part III: Dynamic games of complete information	67
8. Extensive-form games	69
8.1. Dynamic games	69
8.2. The game tree	69
8.3. Information sets	70
8.4. Strategies in dynamic games	70
8.5. Mixed and behavioural strategies	71
8.6. From extensive to normal form	72
8.7. How large is the normal form?	72
8.8. Exercises	72
8.9. Sources	75

9. Credibility and subgame perfection	77
9.1. Nash equilibria of dynamic games	77
9.2. Backward induction	77
9.3. Subgame-perfect equilibrium	78
9.4. Stackelberg games	79
9.5. Centipede games and the limits of backward induction	80
9.6. Commitment and the order of moves	80
9.7. Exercises	80
9.8. Sources	84
10. Multistage and repeated games	85
10.1. Why exponential discounting	85
10.2. Multistage games	85
10.3. Finitely repeated games	87
10.4. Infinitely repeated games	87
10.5. Tit for tat and reputation	88
10.6. Beyond Nash reversion	89
10.7. Exercises	89
10.8. Sources	93
11. Bargaining and dynamic duopolies	95
11.1. Alternating-offer bargaining	95
11.2. Bargaining power: patience and proposal rights	96
11.3. Stackelberg duopoly	96
11.4. Collusion in a repeated Cournot duopoly	97
11.5. Exercises	98
11.6. Sources	99
IV. Part IV: Incomplete information	101
12. Bayesian games	103
12.1. Types and the common prior	103
12.2. Type-contingent strategies	104
12.3. Static Bayesian games	104
12.4. Harsanyi's purification of mixed strategies	105
12.5. Examples	105
12.6. Two classic applications	107
12.7. Exercises	107
12.8. Sources	110
13. Perfect Bayesian equilibrium and signaling	111
13.1. Why SPE is not enough	111
13.2. Beliefs and perfect Bayesian equilibrium	111
13.3. Refinements: sequential equilibrium	112
13.4. Signaling games	113
13.4.1. A recipe for pure PBE in signaling games	113
13.5. Worked example: a coffee for Brian	114
13.6. Education as a signal and the intuitive criterion	115
13.7. Reputation building	116
13.8. Cheap talk	116
13.9. Exercises	117
13.10 Sources	120

V. Part V: Mechanisms and cooperation	121
14. Auctions and mechanism design	123
14.1. Sealed-bid auctions	123
14.1.1. First-price auction	123
14.1.2. Second-price (Vickrey) auction	123
14.2. Revenue equivalence and reserve prices	124
14.3. Common values and the winner's curse	125
14.4. Vickrey–Clarke–Groves auctions	125
14.5. Mechanism design	125
14.6. The revelation principle	126
14.7. Exercises	126
14.8. Sources	129
15. Coalitions and axiomatic bargaining	131
15.1. Coalitional games	131
15.2. The core	131
15.3. The Shapley value	132
15.3.1. Two standard families	133
15.4. Nash bargaining	133
15.5. Kalai–Smorodinsky solution	134
15.6. Exercises	134
15.7. Sources	136

About these notes

Scope

These are lecture notes for *Game Theory* (third-semester elective, Master in Physics of Data, University of Padova; course by Prof. Leonardo Badia). Game theory is the mathematics of **multi-agent, multi-objective decision problems**: several rational players, each maximising their own payoff, whose outcomes depend on everyone's joint choices. Born as a branch of micro-economics, it is now a standard tool in engineering — for distributed resource allocation, network protocols, security, the Internet of Things — where the “players” are algorithms and devices whose rationality is far less questionable than that of human beings.

Reading the notes

The notes follow the course's progression from single-agent decisions to games with time, uncertainty, and cooperation:

- **Part I (Chapter 1)**: rational choice — preferences, utilities, lotteries, expected utility, backward induction for one decision maker.
- **Part II (Chapters 2–7)**: static games of complete information — dominance, Nash equilibrium and its applications (duopolies, commons, selfish routing), mixed strategies and Nash's theorem, zero-sum games and the minimax theorem, and the impossibility results of social choice.
- **Part III (Chapters 8–11)**: dynamic games — the extensive form, subgame-perfect equilibrium, multistage and repeated games (the folk theorem), bargaining and dynamic duopolies.
- **Part IV (Chapters 12–13)**: incomplete information — Bayesian games, perfect Bayesian equilibrium, signaling, reputation and cheap talk.
- **Part V (Chapters 14–15)**: auctions and mechanism design, coalitional games and axiomatic bargaining.

Every chapter uses the styled callout environments to separate definitions, theorems, worked examples and remarks from the surrounding prose, and ends with exercises (solutions collapsible in the HTML version) and a *Sources* section. Diagrams are kept deliberately minimal: game trees, payoff regions and best-response curves, drawn only where they carry the argument.

Notation

- $\mathcal{N} = \{1, \dots, n\}$: set of players; $-i$ denotes “all players except i ”.
- S_i : pure-strategy set of player i ; $S = S_1 \times \dots \times S_n$; $s = (s_i, s_{-i}) \in S$.
- $u_i : S \rightarrow \mathbb{R}$: payoff (utility) of player i ; $G = \{S_1, \dots, S_n; u_1, \dots, u_n\}$ a normal-form game.
- ΔS_i : simplex of probability distributions over S_i ; $m_i \in \Delta S_i$ a mixed strategy.
- $BR_i(s_{-i})$: best-response correspondence of player i .
- $\delta \in (0, 1)$: discount factor.
- h_i, H_i : information set of player i and the collection of all of them.
- $t_i \in T_i$: type of player i ; ϕ the common prior; μ a system of beliefs.
- $v(S)$: value of coalition $S \subseteq \mathcal{N}$ in a coalitional game.

Sources

These notes reorganise the *Game Theory* slides by Prof. Leonardo Badia (University of Padova, decks Gth2001–Gth2018 and the exercise sheets gt20s, gt20d, gt20b) and draw explanations and examples from the course textbook:

- Steven Tadelis, *Game Theory: An Introduction*, Princeton University Press, 2013.

Every chapter closes with its own *Sources* section naming the specific slide decks and book chapters from which its content was drawn.

Part I.

Part I: Rational decisions

1. Rational decision making

Before several players interact, we need a precise language for *one* player choosing among alternatives. This chapter builds it: preferences and the utilities that represent them, lotteries and the expected-utility theorem of von Neumann and Morgenstern, risk attitudes, and backward induction for decisions that unfold over time. Everything in later chapters — best responses, equilibria, beliefs — is this single-agent machinery applied to an environment that contains other optimisers.

1.1. What a game is (and is not)

A *game*, in the technical sense, is a **multi-person, multi-objective problem** formulated as a mathematical model. It has nothing to do with videogames, sports or gambling, where outcomes are dominated by chance or skill. The defining features are:

- several **players**, each with an individual objective;
- the **outcome depends on the joint choices** of all players;
- the **rules are known in advance** and players are **rational**.

Engineers usually model single-person problems: maximise a function, meet a goal, achieve efficiency above a threshold. Uncertainty there comes from noise, missing data, approximations. Game theory adds a new kind of uncertainty — each player knows their own move but can only *conjecture* about the moves of the others — and still manages to produce predictions. Those predictions turn out to be more reliable for algorithms and devices than for humans: an autonomous agent does exactly what its utility prescribes.

Remark (Where game theory is used). Classically in economics (micro: trading, auctions, bargaining; intermediate: markets and firms; macro: countries and monetary authorities). In engineering it is a prediction tool for distributed systems: medium access under congestion, cooperation in networks, the Internet of Things (myriads of devices driven only by their own interest), information security (an adversary that is not fully predictable), reputation and trust in online platforms.

1.2. Some economic background

A few notions recur throughout the course.

- **Monopoly vs. market.** A rational monopolist maximises its own profit, not social welfare. Competition drives the *market-clearing price* below the monopoly price, which is why anti-trust authorities punish trusts. Information engineering has notable near-monopolies: utilities, operating systems, wireless telephony, last-mile broadband.
- **Substitute goods.** Demand usually decreases with price (exceptions: Giffen and Veblen goods). Two goods x, y are *substitutes* if the demand for y increases with the price of x (butter/margarine, coffee/tea).
- **Discount factor.** A unit of value today is worth more than the same unit tomorrow — impatience, uncertainty, deterioration, inflation. A payoff of 1 received at time t is valued δ^t today, with $0 < \delta < 1$. An equivalent reading, very natural for networks: δ is the probability that the player is still around for the next round (users join a service only for a while).

1.3. Preferences and utilities

Definition 1.1 (Decision problem). A decision problem consists of a set A of **actions**, a set of **outcomes** produced by those actions, and **preferences** describing which outcomes the decision maker likes better.

Definition 1.2 (Rational preference). A **preference** on a set of alternatives A ($|A| \geq 2$) is a binary relation $\succeq$; $a \succeq b$ reads “ a is at least as good as b ”. It is

- **complete** if for all $a, b \in A$, $a \succeq b$ or $b \succeq a$ (or both);
- **transitive** if $a \succeq b$ and $b \succeq c$ imply $a \succeq c$.

A complete and transitive preference is called **rational**.

Completeness says the decider can always compare; transitivity forbids cycles, which would allow a “money pump” (someone who prefers a to b , b to c and c to a will pay to trade around the cycle forever).

Definition 1.3 (Utility (payoff) function). A function $u : A \rightarrow \mathbb{R}$ **represents** $\succeq$ if for all $a, b \in A$

$$a \succeq b \iff u(a) \geq u(b).$$

Theorem 1.1 (Representation on a finite set). *On a finite set A , a preference $\succeq$ can be represented by a utility function if and only if it is rational.*

Proof. $(\Rightarrow)$ If u represents $\succeq$, completeness and transitivity are inherited from $\geq$ on $\mathbb{R}$. $(\Leftarrow)$ Define $u(a) = |\{b \in A : a \succeq b\}|$, the number of alternatives a beats. By transitivity, $a \succeq b$ implies that everything below b is also below a , so $u(a) \geq u(b)$; by completeness the converse holds. $\square$

Utilities are **ordinal** in deterministic settings: any strictly increasing transformation of u represents the same preference. For a countable good q one typically assumes $u'(q) \geq 0$ (more is better) and $u''(q) \leq 0$ (diminishing returns).

Remark (Rationality). Hidden in the word “utility” is the assumption that players are **rational**: they act in their own interest *and* they are aware of all consequences of their actions. The first point does not mean “selfish” — generosity, fairness or spite can be written into u . The criticism that humans are irrational is a criticism of the *model’s accuracy*, not of rationality itself, and it is largely irrelevant for algorithms and autonomous agents. The second requirement (full awareness of consequences) is the demanding one.

1.4. Decision trees

A decision is drawn as a tree: the **root** is the player who moves, **branches** are the alternatives, **leaves** carry the utilities. Choosing ravioli (r , utility 5) over soup (s , utility 1) is a fork in the road. If the player has further choices (fish or vegetables after each first course), the tree grows more levels — but a single smart agent can always collapse a single-player tree into one layer by listing every complete plan (rf , rv , sf , sv) and picking the best. Single-person decision trees are therefore just optimisation problems; things become interesting only with several players.

1.5. Lotteries and Nature

Real outcomes are often random: the canteen's soup is different every day. Game theory models randomness as the choice of an additional player, **Nature**, who moves according to a known probability distribution.

Definition 1.4 (Lottery). A (simple) **lottery** over the outcomes $X = \{x_1, \dots, x_n\}$ is a probability distribution $p = (p(x_1), \dots, p(x_n))$ with $p(x_k) \geq 0$ and $\sum_k p(x_k) = 1$. If outcomes depend on an action a , the lottery is conditional, $p(x_k | a)$. A certain outcome is a **degenerate lottery** with $p(x_k | a) = 1$ for one k .

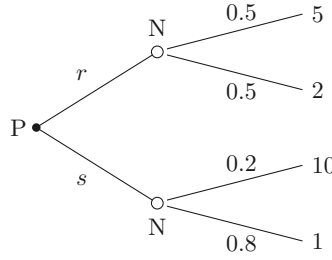


Figure: choosing between two lotteries. Filled dot: the decision maker; open circles: Nature.

Lotteries can also be **compound** (Nature moves several times in a row; probabilities multiply along a path) or **continuous** (a density replaces the distribution, and individual outcomes have probability zero; trees no longer apply but the logic does).

1.6. Expected utility

Which lottery is better: a fair coin between 5 and 2, or a 20% chance of 10 with an 80% risk of 1? The standard answer is to compare **expected payoffs**:

$$\mathbb{E}[u | p] = \sum_{k=1}^n p(x_k) u(x_k).$$

The intuition: if the lottery were repeated $N \rightarrow \infty$ times, the average payoff would approach the expectation. In the example $\mathbb{E}[u | r] = 0.5 \cdot 5 + 0.5 \cdot 2 = 3.5$ and $\mathbb{E}[u | s] = 0.2 \cdot 10 + 0.8 \cdot 1 = 2.8$, so r is preferred.

Is it legitimate to rank lotteries by expected utility? Von Neumann and Morgenstern answered by axiomatising preferences over the set $\mathcal{P}(A)$ of lotteries.

Definition 1.5 (Continuity axiom). For all $p, q, r \in \mathcal{P}(A)$ the sets $\{\alpha \in [0, 1] : \alpha p + (1 - \alpha)q \succeq r\}$ and $\{\alpha \in [0, 1] : r \succeq \alpha p + (1 - \alpha)q\}$ are closed.

Small changes in the probabilities do not flip a strict preference: if I prefer a safe walk in the park to staying home, I still do when there is a tiny probability of being mugged on the way.

Definition 1.6 (Independence axiom). For all $p, q, r \in \mathcal{P}(A)$ and $\alpha \in [0, 1]$: if $p \succeq q$ then $\alpha p + (1 - \alpha)r \succeq \alpha q + (1 - \alpha)r$.

Mixing both options with the same third lottery does not change their ranking: if I prefer betting on football to betting on horses, I also prefer “heads: football, tails: roulette” to “heads: horses, tails: roulette”.

Theorem 1.2 (von Neumann–Morgenstern expected utility). *If a preference $\succeq$ on $\mathcal{P}(A)$ is complete, transitive, continuous and satisfies independence, then there exists $u : A \rightarrow \mathbb{R}$ such that for all $p, q \in \mathcal{P}(A)$*

$$p \succeq q \iff \mathbb{E}[u \mid p] \geq \mathbb{E}[u \mid q].$$

*Such a u is a **vN-M utility**; it is unique up to positive affine transformations $u \mapsto au + b$, $a > 0$.*

The key structural fact is that expected utility is **linear** in the lottery: any p is a convex combination of degenerate lotteries, $p = \sum_k p_k e_k$, and $\mathbb{E}[u \mid p] = \sum_k p_k u(x_k)$.

Warning

Under uncertainty, utilities stop being merely ordinal. Replacing the “tasty soup” payoff 10 with 100 preserves the order of outcomes but can reverse the choice between lotteries. “ $a \succeq b$ ” is no longer enough: we also need to know *by how much*. The same holds later for mixed strategies.

Example 1.1 (Digging a well (continuous action))

Choose a depth $d \geq 0$. The effort is $d^2/2$; the water extracted is $W(d) \sim \mathcal{U}[0, 20d]$. The expected utility is

$$\mathbb{E}[u \mid d] = \mathbb{E}[W(d)] - \frac{d^2}{2} = 10d - \frac{d^2}{2},$$

maximised at $d^* = 10$ with $\mathbb{E}[u] = 50$. Digging 3.2 m, for comparison, yields $32 - 5.12 = 26.88$.

1.7. Risk attitudes

Consider outcomes $x_1 = 0$, $x_2 = 1$, $x_3 = 20$ euro and the lotteries $p_A = (0, 1, 0)$ (one euro for sure) and $p_B = (0.95, 0, 0.05)$. Both have expected *outcome* 1 euro. Their expected *utilities* are $u(x_2)$ and $0.95u(x_1) + 0.05u(x_3)$, which may differ.

Definition 1.7 (Risk attitudes). A player is **risk neutral** if they are indifferent between any two lotteries with the same expected outcome; **risk averse** if they always prefer the degenerate lottery (the sure thing) to a non-degenerate lottery with the same expected outcome; **risk loving** if they prefer the opposite.

In terms of the vN-M utility of money: linear u risk neutral, concave u risk averse (Jensen: $u(\mathbb{E}x) \geq \mathbb{E}u(x)$), convex u risk loving. Monotone transformations such as x , x^2 , $\log x$ all represent the same *ordering* of sure amounts but encode different risk attitudes.

Important

Expected utility theory does **not** claim that “1 euro for sure” equals “a 50/50 gamble on 2 euros”. It claims that a player whose utility of money is $u(x) = x$ is indifferent between them. The risk attitude lives inside u .

1.7.1. Measuring risk aversion

For a lottery L over money, the **certainty equivalent** $CE(L)$ is the sure amount the player considers just as good, $u(CE) = \mathbb{E}[u(x)]$, and the **risk premium** is $\pi = \mathbb{E}[x] - CE$: what the player would give up to remove the risk. A risk-averse player has $\pi > 0$ and is therefore willing to buy fair (and even slightly unfair) insurance.

How *strongly* concave is u ? For a small gamble ε with mean zero and variance σ^2 around wealth w , a second-order expansion gives

$$\pi \approx \frac{1}{2} \sigma^2 A(w), \quad A(w) = -\frac{u''(w)}{u'(w)},$$

the **Arrow–Pratt coefficient of absolute risk aversion**. Unlike u'' alone, A is invariant to affine changes of u , as it must be. Two families recur:

utility	$A(w)$	$R(w) = wA(w)$	behaviour
$u = -e^{-aw}$ (CARA)	a	aw	premium independent of wealth
$u = \frac{w^{1-\rho}}{1-\rho}$, $u = \log w$ for $\rho = 1$ (CRRA)	ρ/w	ρ	premium proportional to wealth

$R(w)$ is the *relative* risk aversion, relevant for gambles proportional to wealth. Most people's absolute risk aversion decreases with wealth: the rich accept bets the poor refuse.

1.8. Decisions over time: backward induction

When the decision maker and Nature alternate, the tree has several decision layers. Suppose ravioli turn out tasty or bland with probability 1/2 each; *after* seeing which, the player may add extra cheese (e) or not (n). Cheese is good with probability 0.6: it raises tasty ravioli from 5 to 10 and bland ravioli from 2 to 3, while bad cheese gives 0. Soup is the same lottery as before.

Definition 1.8 (Backward induction (dynamic programming)). Classify the player's decision nodes: *group 1* contains nodes after which no further decision of the player is possible (only outcomes or Nature's moves follow); *group k* contains nodes followed only by nodes of group $k-1$ or lower. Solve group-1 nodes by maximising expected utility, replace each by its optimal value, and repeat — the former group- k nodes become group $k-1$.

In the example: after tasty ravioli, e gives $0.6 \cdot 10 = 6 > 5$, so add cheese; after bland ravioli, e gives $0.6 \cdot 3 = 1.8 < 2$, so don't. The pruned tree has $\mathbb{E}[u \mid r] = 0.5 \cdot 6 + 0.5 \cdot 2 = 4 > 2.8 = \mathbb{E}[u \mid s]$.

If the later decision happens much later, its payoffs are discounted: replace 10 by 10δ and 3 by 3δ . The optimal plan now depends on δ .

Continuous choices over time. When the decision is how much to consume in each period, $\max \sum_t \delta^{t-1} u(x_t)$ subject to $\sum_t x_t = W$, the Lagrange conditions equate *discounted marginal utilities*: $\delta^{t-1} u'(x_t) = \lambda$. With $u = \log$ this gives $x_t \propto \delta^{t-1}$: impatience tilts consumption toward the present, and any extra wealth is spread over all periods (consumption smoothing). Because δ^t discounting is time consistent, re-optimising later confirms the original plan.

Sequencing risky steps. If a project needs several independent steps, each succeeding with probability p_i at cost c_i , doing them one at a time saves the cost of later steps whenever an earlier one fails. This is an *option value* of waiting. Swapping two adjacent steps shows that step i should come first iff $c_i/(1-p_i) < c_j/(1-p_j)$: cheap tests that are likely to fail go first.

1.9. The value of information

A rational player maximises expected utility, i.e. makes the right choice *on average*. Had they known Nature's choice in advance, they might have chosen differently. How much is that knowledge worth?

Suppose a friend who already ate at the canteen offers to reveal today's quality. Informationally, this **reverses the order of moves**: Nature moves first, then the player chooses knowing the result. With the original lotteries (ravioli 5 or 2 at 1/2; soup 10 or 1 at 0.2/0.8, independent):

ravioli	soup	probability	best choice	payoff
5	10	0.1	s	10
2	10	0.1	s	10
5	1	0.4	r	5
2	1	0.4	r	2

The expected utility with information is $1 + 1 + 2 + 0.8 = 4.8$; without it, the best one can do is 3.5 (choose r). The **value of information** is $4.8 - 3.5 = 1.3$: the most the player should pay the friend.

1.9.1. Imperfect information and Bayes' rule

Information is usually noisy: a test, a forecast, a friend's opinion. Let the state be ω with prior $p(\omega)$ and let a signal s arrive with likelihood $q(s | \omega)$. After observing s , the player should update with **Bayes' rule**

$$p(\omega | s) = \frac{q(s | \omega) p(\omega)}{\sum_{\omega'} q(s | \omega') p(\omega')},$$

and choose the action that is optimal under the posterior. The value of the signal is

$$VI = \sum_s \Pr(s) \max_a \mathbb{E}[u | a, s] - \max_a \mathbb{E}[u | a] \geq 0,$$

non-negative because the player can always ignore it. It is *zero* whenever no signal realisation changes the optimal action: information is valuable only if it can change what you do. Perfect information is the special case $p(\omega | s) \in \{0, 1\}$, and gives an upper bound on the value of any imperfect signal.

1.10. Theory versus practice

Expected utility is a normative benchmark. Experiments document systematic violations, which are worth knowing because they show which axiom fails.

- **Allais paradox.** Most subjects prefer A : 1M for sure to B : (0.89: 1M, 0.10: 5M, 0.01: 0), yet prefer D : (0.10: 5M, 0.90: 0) to C : (0.11: 1M, 0.89: 0). Writing $A \succ B$ as $0.11 u(1) > 0.10 u(5) + 0.01 u(0)$ and $D \succ C$ as the reverse inequality shows no vN-M utility can rationalise both. Both pairs differ by the same common consequence (0.89 of 1M versus 0.89 of 0), so the choices violate **independence**.
- **Framing and reference points.** Prospect theory (Kahneman and Tversky) fits choices better with a value function defined on *gains and losses* relative to a reference point, concave for gains, convex and steeper for losses (loss aversion), and with probability weighting that overweights small probabilities.
- **Present bias.** Exponential discounting δ^t is *time consistent*: a plan made today is still optimal tomorrow. Observed choices often follow $\beta\delta$ (quasi-hyperbolic) discounting, $u_0 + \beta \sum_{t \geq 1} \delta^t u_t$ with $\beta < 1$, which produces procrastination and demand for commitment devices.

Game theory nevertheless keeps the rational benchmark: it is precise, it makes falsifiable predictions, and for engineered agents (algorithms, devices) it is literally true by design.

1.11. Exercises

Problems from Tadelis (T), Chapters 1–2. Badia and Marchioro’s handbook starts directly with games; its lottery problems appear in Chapters 2–5.

Exercise 1.1

(T 1.3, Fruit or Candy) A banana costs 0.50 and a candy 0.25; you have 1.25 and value money one-for-one. The first banana is worth 1.20 to you and each additional one half the previous; the first candy is worth 0.40, each additional one half the previous. (a) How many feasible actions are there? (b) Should you spend all your money? (c) Answer again if candy costs 0.30.

Solution

- (a) Bundles (b, c) with $0.5b + 0.25c \leq 1.25$: $b = 0, c \leq 5$; $b = 1, c \leq 3$; $b = 2, c \leq 1$ — 12 actions.
 (b) Net marginal values: bananas +0.70, +0.10, −0.20; candies +0.15, −0.05. Take every positive item: 2 bananas and 1 candy, costing exactly 1.25, payoff 0.95. Spending everything is a coincidence of prices, not a principle. (c) Now 9 actions, the candy nets +0.10, and 2 bananas plus a candy cost 1.30. The best feasible bundles (2 bananas, or 1 banana and 1 candy) both give 0.80 and leave money unspent.

Exercise 1.2

(T 1.4) Your payoff from drinking a litres is $\theta a - 4a^2$, with $\theta \in [0.2, 6]$ across the population, larger people having larger θ . (a) Is there an amount nobody should drink? (b) Optimal a for $\theta = 1$ and $\theta = 4$? (c) Show smaller people should drink less.

Solution

The first-order condition $\theta - 8a = 0$ gives $a^*(\theta) = \theta/8$, increasing in θ , which is (c). (b) 1/8 and 1/2 litres. (a) Since $\theta \leq 6$, $a^* \leq 0.75$: drinking more than 0.75 litres (in particular a full one-litre bottle) is never optimal.

Exercise 1.3

(T 2.2, Recreation Choices) Payoffs (rain, shine): football (1, 2), boxing (3, 0), hike (0, 1); rain has probability p . (a) Which alternative is never chosen? (b) Find the optimal choice as a function of p .

Solution

- (a) The hike is strictly dominated by football in both states. (b) $\mathbb{E}[F] = 2 - p$, $\mathbb{E}[B] = 3p$: football for $p < 1/2$, boxing for $p > 1/2$, indifferent at $p = 1/2$.

Exercise 1.4

(T 2.3, At the Dog Races) A \$1 bet on Snoopy pays \$2 if he wins (probability 0.7); a \$1 bet on Lassie pays \$11 (probability 0.1); you may also not bet. You are risk neutral. (a) Best action? (b) You are offered “anti-insurance”: \$2 now in exchange for half of any prize. What do you do?

Solution

- (a) Net expected values: Snoopy $0.7 \cdot 2 - 1 = 0.4$, Lassie $0.1 \cdot 11 - 1 = 0.1$, no bet 0. Bet on Snoopy.
 (b) With the contract: Snoopy $2 - 1 + 0.7 \cdot 1 = 1.7$, Lassie $2 - 1 + 0.1 \cdot 5.5 = 1.55$, no bet 2. Accept the payment and do not bet (value 2 > 0.4): the contract halves the upside of every bet, which makes all of them unprofitable.

Exercise 1.5

(T 2.6, Real Estate) An impact score is uniform on $[40, 70]$. Approval has probability 0.35 if the score is below 50, 0.05 if it is in $[50, 55]$, and 0 above. The development is worth \$20M to a risk-neutral developer, and the study is legally required. What is the most the developer should pay for the study?

Solution

$\Pr(\text{approval}) = \frac{10}{30} \cdot 0.35 + \frac{5}{30} \cdot 0.05 = 0.125$, so the study is worth $0.125 \times 20 = \$2.5\text{M}$.

Exercise 1.6

(T 2.8, Juice) A product needs three independent successful steps — ripening (success 0.7, cost 1), market research (0.3, cost 5), cold chain (0.6, cost 0.5), in €M — and then yields $42.5 - 2.5 = 40$. Steps can be run in parallel or sequentially in any order. What should the firm do?

Solution

In parallel: $0.126 \times 40 - 6.5 = -1.46$. Sequentially in order (1, 2, 3) the expected cost is $c_1 + p_1 c_2 + p_1 p_2 c_3$, since later steps are paid only if earlier ones succeed. The success term $0.126 \times 40 = 5.04$ does not depend on the order. Comparing adjacent steps shows that step i should precede j iff $c_i/(1-p_i) < c_j/(1-p_j)$: run cheap, likely-to-fail tests first. The ratios are cold chain 1.25, ripening 3.33, market 7.14. The order cold chain → ripening → market costs $0.5 + 0.6 + 2.1 = 3.2$, giving value $1.84 > 0$, the best of the six orders (the worst, market first, gives -0.37). Sequencing turns a loss-making project into a profitable one: an option value of waiting.

Exercise 1.7

(T 2.10, Surgery) Untreated, a patient lives 6 months. Surgery succeeds with probability 0.7 (12 months) and otherwise kills the patient immediately. Let $v(12) = 1$, $v(0) = 0$. (a) For which $v(6)$ is surgery optimal? From now on $v(6) = 0.8$. (b) A free test is positive with probability 0.9 if the surgery would succeed and 0.1 if it would fail. Find $\Pr(\text{success} \mid +)$ and the optimal decision. (c) Should the patient take the test if it kills with probability 0.005?

Solution

- (a) Surgery gives 0.7, so surgery is optimal iff $v(6) \leq 0.7$. With $v(6) = 0.8$ the patient declines.

(b) $\Pr(+)=0.9 \cdot 0.7+0.1 \cdot 0.3=0.66$ and $\Pr(\text{success} \mid +)=0.63 / 0.66=0.955>0.8$: operate after a positive test. After a negative test $\Pr(\text{success} \mid -)=0.07 / 0.34=0.206$: do not operate. (c) Surviving the test is worth $0.66 \cdot 0.955+0.34 \cdot 0.8=0.902$, so taking it gives $0.995 \cdot 0.902=0.897>0.8$. Take the test: its information is worth more than its risk.

Exercise 1.8

(T 2.11, To Run or Not to Run) Your leg is broken with probability 0.2; you win with probability 0.1, independently. Payoffs: win and healthy 100, win and broken 50, lose and healthy 0, lose and broken -50 ; not running: -10 if broken, 0 otherwise. (a) Optimal choice? (b) Value of perfect information about the leg? (c) About the race?

Solution

(a) Run: $0.8(0.1 \cdot 100)+0.2(0.1 \cdot 50-0.9 \cdot 50)=8-8=0$; don't run: -2 . Run. (b) Healthy (0.8): run, 10; broken (0.2): don't, -10 . Value $8-2=6$, so perfect information about the leg is worth 6. (c) Winning (0.1): run, $0.8 \cdot 100+0.2 \cdot 50=90$; losing (0.9): don't run, -2 (better than running, -10). Value $9-1.8=7.2$, so it is worth 7.2.

Exercise 1.9

(T 2.13, Today, Tomorrow, or the Day After) Allocate 100 over three periods to maximise $\log x_1+\delta \log x_2+\delta^2 \log x_3$, $\delta=0.9$. How does the plan change if a gift of 20 arrives at $t=2$ (no borrowing)?

Solution

The Lagrange conditions give $x_t \propto \delta^{t-1}$: $x=\frac{100}{2.71}(1,0.9,0.81)=(36.90,33.21,29.89)$. With the gift, total wealth 120 gives $(44.28,39.85,35.87)$. This needs no borrowing since $x_1<100$, so the gift is spread over *all* periods, including today: consumption smoothing. Re-optimising at $t=2$ with the remaining 75.72 yields the same $(39.85,35.87)$. Exponential discounting is time consistent.

1.12. Sources

Prof. Badia, *Game Theory* slides, decks Gth2001 (introduction, background, preferences, decision trees) and Gth2002 (lotteries, expected utility, risk, backward induction, value of information). Tadelis, *Game Theory: An Introduction*, Chapters 1–2. Original reference: J. von Neumann and O. Morgenstern, *Theory of Games and Economic Behavior*, Princeton (1944).

Part II.

Part II: Static games of complete information

2. Static games and dominance

The simplest genuinely multi-player interaction is a **static game of complete information**: everyone chooses once, without seeing the others' choices, and everyone knows everyone's payoffs. This chapter sets up the normal-form representation, meets the canonical examples that recur throughout the course, and develops the first two solution ideas — eliminating dominated strategies, and best-responding to beliefs.

2.1. Static games of complete information

Definition 2.1 (Static game of complete information). • **Static**: every player chooses an action without knowledge of the others' actions. Moves need not be literally simultaneous — what matters is the information.
• **Complete information**: every player's payoff function is common knowledge.

Two physically different procedures are both “simultaneous”: (A) two players write their choices on opposite sides of a board at the same time; (B) player 1 writes first while player 2 is blindfolded, then the board is turned and player 2 writes. Only the information available at decision time matters.

Definition 2.2 (Common knowledge). A fact E is **common knowledge** if everybody knows E , everybody knows that everybody knows E , and so on *ad infinitum*.

Complete information means that all possible actions, all resulting outcomes and all players' preferences over outcomes are common knowledge. We also assume that **rationality is common knowledge**: everybody maximises their payoff, everybody knows that everybody does, and so on. This is a powerful and far from innocuous assumption.

2.2. Normal-form representation

The ingredients of a static game are *actions*, *outcomes* and *utilities*. It is convenient to speak of **strategies** rather than actions: a strategy is a complete plan of action (“if these conditions hold, play a ; otherwise a' ”). For now we need only deterministic plans.

Definition 2.3 (Pure strategy and normal form). A **pure strategy** is a deterministic plan of action. A game in **normal form** is

$$G = \{S_1, \dots, S_n; u_1, \dots, u_n\},$$

where S_i is the set of pure strategies of player i and $u_i : S_1 \times \dots \times S_n \rightarrow \mathbb{R}$ is the payoff of player i for each **strategy profile** $s = (s_1, \dots, s_n)$.

With two players and finite strategy sets, G is a **bi-matrix**: rows are player 1's strategies, columns player 2's, and each cell contains the pair (u_1, u_2) .

2.3. A gallery of examples

These games will be revisited many times.

Example 1. Player A has $\{U, M, D\}$, player B has $\{L, R\}$:

	L	R
U	6, 0	0, 5
M	1, 0	4, 3
D	0, 7	2, 0

Example 2. Both players have three strategies:

	L	C	R
U	0, 5	4, 0	7, 3
M	4, 0	0, 5	7, 3
D	3, 7	3, 7	9, 9

Odds & Evens (matching pennies, penalty kicks, bluffing in poker). Two players bet 4 euro and show 0 or 1 fingers; Odd wins if the sum is odd:

Odd \ Even	0	1
0	-4, 4	4, -4
1	4, -4	-4, 4

Rock–paper–scissors has the same structure with three strategies each and a cyclic dominance relation.

Battle of the Sexes. Ann and Brian want to meet at the cinema; Ann prefers the romance (R), Brian the sci-fi movie (S), but both mostly want to be together:

Ann \ Brian	R	S
R	2, 1	0, 0
S	0, 0	1, 2

Prisoner's Dilemma. Two accomplices are questioned separately and may keep **M**um or **F**ink; payoffs are minus the months in jail:

Al \ Bob	M	F
M	-1, -1	-9, 0
F	0, -9	-6, -6

2.4. Pareto efficiency

Definition 2.4 (Pareto dominance and efficiency). A profile s is **Pareto dominated** by s' if $u_i(s') \geq u_i(s)$ for every player i and $u_i(s') > u_i(s)$ for at least one. A profile not Pareto dominated by any other is **Pareto efficient** (Pareto optimal).

There are usually several Pareto-efficient profiles, none dominating the others. In the Prisoner's Dilemma, (M, M) , (M, F) and (F, M) are Pareto efficient while (F, F) is not — it is dominated by (M, M) .

2.5. Strict dominance and IESDS

Definition 2.5 (Strictly dominated strategy). In $G = \{S_1, \dots, S_n; u_1, \dots, u_n\}$, strategy $s_i \in S_i$ is **strictly dominated** by $s'_i \in S_i$ if

$$u_i(s'_i, s_{-i}) > u_i(s_i, s_{-i}) \quad \forall s_{-i} \in S_{-i} = \prod_{j \neq i} S_j.$$

A rational player never plays a strictly dominated strategy: whatever the others do, there is something strictly better. If rationality is common knowledge, everybody can delete everybody's dominated strategies, and the deletion can be iterated on the smaller game.

Definition 2.6 (IESDS). **Iterated elimination of strictly dominated strategies** repeatedly removes, for every player, the strategies that are strictly dominated in the game that survives the previous rounds, until no more can be removed.

Example 2.1 (IESDS in Example 1)

D is strictly dominated by M ($1 > 0$, $4 > 2$). With D gone, L is strictly dominated by R for player B ($5 > 0$, $3 > 0$). With L gone, U is strictly dominated by M ($4 > 0$). The unique survivor is (M, R) with payoffs $(4, 3)$.

Example 2.2 (The Prisoner's Dilemma)

F strictly dominates M for both players ($0 > -1$ and $-6 > -9$). IESDS leaves only (F, F) , which is Pareto dominated by (M, M) . Individually rational behaviour produces a collectively bad outcome — hence the “dilemma”.

Remark (Order irrelevance). For strict dominance in finite games, the final result of IESDS does not depend on the order in which strategies are removed. (This fails for weak dominance, see Chapter 3.)

IESDS often fails to say anything. In Example 2 nothing is dominated, yet (D, R) looks compelling. In Odds & Evens neither strategy dominates the other. In the Battle of the Sexes both (R, R) and (S, S) look sensible but nothing can be removed. We need a sharper concept.

2.6. Best responses and beliefs

For single-agent problems, the solution follows from the setup. In games it depends on what the others do — so we make the dependence explicit.

Definition 2.7 (Best response). Strategy $s_i \in S_i$ is a **best response** to $s_{-i} \in S_{-i}$ if

$$u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i}) \quad \forall s'_i \in S_i.$$

There may be several best responses, all with the same payoff. In the game

	L	C	R
U	3, 3	5, 1	6, 2
M	4, 1	8, 4	3, 6
D	4, 0	9, 6	6, 8

both U and D are best responses of player A to R (payoff 6), and both M and D to L (payoff 4).

Definition 2.8 (Belief and best-response correspondence). A **belief** of player i is a profile of opponents' strategies $s_{-i} \in S_{-i}$ that i conjectures will be played. The **best-response correspondence** $BR_i : S_{-i} \rightrightarrows S_i$ maps each belief to the set of best responses to it. It is single-valued when the best response is unique.

A rational player who holds belief s_{-i} always picks some $s_i \in BR_i(s_{-i})$.

Proposition 2.1 (Dominated strategies are never best responses). *If s_i is strictly dominated, it is not a best response to any $s_{-i} \in S_{-i}$.*

Proof. If s'_i strictly dominates s_i , then $u_i(s'_i, s_{-i}) > u_i(s_i, s_{-i})$ for every s_{-i} , violating the best-response inequality. $\square$

Iterating “never a best response” instead of “strictly dominated” gives **rationalizability**: the strategies that survive are those a rational player could play for *some* belief consistent with common knowledge of rationality. For two players the two procedures coincide once mixed strategies are allowed (Chapter 5).

Note

Dominance arguments only use the first half of what we will need: that each player best-responds to *some* belief. Nash equilibrium adds the second half — that beliefs are *correct*.

2.7. Rationalizability and common knowledge

The solution concepts of this chapter rest on two assumptions with different strength.

Definition 2.9 (Common knowledge). A fact E is **common knowledge** if everybody knows E , everybody knows that everybody knows E , and so on *ad infinitum*.

A game of complete information requires the players, strategy sets and payoffs to be common knowledge. *Rationality* alone justifies deleting strictly dominated strategies **once**; each further round of IESDS requires one more level of “I know that you know that I am rational”. IESDS to the end needs **common knowledge of rationality**.

Definition 2.10 (Dominant-strategy equilibrium). s_i^D is a **strictly dominant** strategy if it strictly dominates every other $s_i \in S_i$. If every player has one, $s^D = (s_1^D, \dots, s_n^D)$ is a **strictly dominant strategy equilibrium**; it is unique, and it is the only profile surviving IESDS.

The Prisoner's Dilemma has one; most games do not. It needs only rationality, not common knowledge — which makes it the most robust prediction, when it exists.

2.7.1. Beliefs as probabilities

A player who is unsure what others will do holds a **probabilistic belief** $\pi_i \in \Delta(S_{-i})$ and maximises expected payoff $\sum_{s_{-i}} \pi_i(s_{-i}) u_i(s_i, s_{-i})$. A strategy is a **never-best response** if it is not optimal for *any* such belief. Strict dominance by a pure strategy is sufficient, but not necessary:

	L	R
U	3	0
M	0	3
D	1	1

D (row player's payoffs shown) is dominated by neither U nor M , yet it is never a best response: against belief $\Pr(L) = q$ it earns $1 < \max(3q, 3 - 3q)$, which is at least 1.5. Equivalently, D is strictly dominated by the *mixed* strategy $\frac{1}{2}U + \frac{1}{2}M$, which earns 1.5 against both columns.

Theorem 2.2 (Pearce (1984)). *In a finite two-player game, a strategy is a never-best response to all beliefs if and only if it is strictly dominated by some (possibly mixed) strategy.*

Rationalizability iterates the deletion of never-best responses. By the theorem, in two-player games it coincides with IESDS when mixed dominators are allowed; with more players (correlated beliefs aside) it can be finer. Either way, it is the set of strategies a player can justify by an infinite chain of beliefs, each consistent with the rationality of the others.

2.7.2. IESDS on a continuum: two classic examples

Cournot duopoly (studied in Chapter 4): firms choose $q_i \geq 0$, price $a - q_1 - q_2$, unit cost c ; the best response is $q_i = \frac{a-c-q_j}{2}$. Quantities above the monopoly output $\frac{a-c}{2}$ are dominated. Given that, quantities below $\frac{a-c}{4}$ are dominated, and so on. The surviving intervals shrink geometrically around $\frac{a-c}{3}$, so IESDS alone pins down the Cournot outcome.

The p -beauty contest: n players pick a number in $[0, 100]$; whoever is closest to $p = 2/3$ of the average wins. No number above $66.\bar{6}$ can be a best response; given that, nothing above $44.\bar{4}$; after k rounds, nothing above $100(2/3)^k$. Only 0 survives. In experiments the average is around 20–35: subjects behave like **level- k thinkers**, performing one to three rounds of reasoning. Common knowledge of rationality is a strong assumption, and the “right” play depends on how rational you believe the *others* are.

2.8. Exercises

Problems from Badia and Marchioro (BM), *Game Theory: A Handbook of Problems and Exercises*, Chapters 1 and 4. The equilibrium parts of these problems are in Chapters 3 and 5.

2. Static games and dominance

Exercise 2.1

(BM 1.2) Mr. Square (S) and Ms. Triangle (T) each pick a point. Both get revenue 10 if the points differ and 0 if they coincide, minus a cost $x + 3y$ for the chosen point (x, y) . S chooses among A (0, 0), B (0, 4), C (4, 4), D (4, 0); T among A (0, 0), E (2, 4), D (4, 0). Write the normal form and remove the dominated strategies.

Solution

	A	D	E
A	0, 0	10, 6	10, -4
B	-2, 10	-2, 6	-2, -4
C	-6, 10	-6, 6	-6, -4
D	6, 10	-4, -4	6, -4

Points with high y are expensive and never worth it. B and C are strictly dominated by A for S, and E by A for T. What remains is the 2×2 anti-coordination game on $\{A, D\}$, where nothing more is dominated.

Exercise 2.2

(BM 1.5) Alice and Bob each create a character: Human (H), Elf (E), Dwarf (D) or Orc (O), costing 0, 5, 4, 3. Humans give nothing to the partner; Orcs give -3; Dwarves give +5; Elves give +15, but only to an Elf partner. Alice gets an extra +5 if she is an Orc and Bob is not. Write the normal form and apply IESDS.

Solution

	H	E	D	O
H	0, 0	0, -5	5, -4	-3, -3
E	-5, 0	10, 10	0, -4	-8, -3
D	-4, 5	-4, 0	1, 1	-7, 2
O	2, -3	2, -8	2, -7	-6, -6

D is strictly dominated by H for both players (a Dwarf pays 4 to help the *other*). Then O is dominated by H for Bob ($0 > -3$, $0 > -8$, $-3 > -6$). Against $\{H, E\}$, Alice's H is dominated by O ($2 > 0$, $2 > 0$). The survivors are $\{E, O\} \times \{H, E\}$, where nothing else is dominated.

Exercise 2.3

(BM 1.6) Two teams choose Aggressive, Spectacular or Relaxed. Aggressive always gives 1; Spectacular gives 3 and adds 1 to the opponent; Relaxed gives 0, unless both relax, in which case both get 5. Write the normal form and simplify it.

Solution

	A	S	R
A	1, 1	2, 3	1, 0
S	3, 2	4, 4	3, 1
R	0, 1	1, 3	5, 5

A is strictly dominated by S ($3 > 1$, $4 > 2$, $3 > 1$). The reduced game on $\{S, R\}$ is a coordination game (a Stag Hunt: (R, R) is Pareto dominant but risky), so no further elimination is possible.

Exercise 2.4

(BM 4.5) Two knights choose Axe, Mace or Sword. Same weapon: White wins with probability 0.6. Axe against Mace: Axe wins with 0.9. Black with the Sword (White without): Black wins with 0.8. White with the Sword (Black without): White wins with 0.7. Winning is worth 10, dying -100 . Write the expected payoffs and solve by IESDS.

Solution

With winning probability π the expected utility is $10\pi - 100(1 - \pi) = 110\pi - 100$ (White's payoff first, White chooses the row):

	A	M	S
A	$-34, -56$	$-1, -89$	$-78, -12$
M	$-89, -1$	$-34, -56$	$-78, -12$
S	$-23, -67$	$-23, -67$	$-34, -56$

Every cell sums to -90 : a constant-sum game. M is strictly dominated by S for both. In the remaining $\{A, S\}$ game A is dominated by S for both. IESDS leaves (S, S) , the unique NE.

Exercise 2.5

(BM 4.4) Two friends choose a course, G, H or I. If they choose different courses both get 0. If they choose the same course j , it turns out good with probability $\rho_G = 0.4$, $\rho_H = 0.8$, $\rho_I = 0.5$. A good course gives Anastasia 1 and Bruno 3, 2, -1 for G, H, I; a bad course gives 0. Write the normal form and apply IESDS, allowing domination by mixed strategies.

Solution

	G	H	I
G	0.4, 1.2	0, 0	0, 0
H	0, 0	0.8, 1.6	0, 0
I	0, 0	0, 0	0.5, -0.5

2. Static games and dominance

No pure strategy strictly dominates Bruno's I : its payoffs $(0, 0, -0.5)$ are only weakly below those of G and H . The mixture $\frac{1}{2}G + \frac{1}{2}H$ gives $(0.6, 0.8, 0) > (0, 0, -0.5)$, so I is strictly dominated. With I gone, Anastasia's I earns $(0, 0)$ against (G, H) , strictly below $\frac{1}{2}G + \frac{1}{2}H = (0.2, 0.4)$. The survivors are $\{G, H\}^2$: a coordination game. This is Pearce's theorem at work: these strategies are never best responses.

Exercise 2.6

(BM 1.7) Two firms assign 0, 1, 2, ... employees (100 k€ each) to a project. Equal staffing pays each 140; otherwise the larger team gets 350 and the smaller 160. (a) Show that nobody assigns more than 2. (b) Write the 3×3 game. (c) Is 2 strictly dominated by a mixture of 0 and 1?

Solution

(a) Assigning 0 guarantees at least 140, while $k \geq 3$ employees yield at most $350 - 300 = 50$. (b) Row firm's payoff (the game is symmetric):

	0	1	2
0	140	160	160
1	250	40	60
2	150	150	-60

(c) No. A mixture $p \cdot 0 + (1-p) \cdot 1$ beats 2 against column 0 iff $140p + 250(1-p) > 150$, i.e. $p < 0.909$, and against column 1 iff $160p + 40(1-p) > 150$, i.e. $p > 0.917$: incompatible. So 2 survives IESDS. (The printed solution claims $0.95 \cdot 0 + 0.05 \cdot 1$ dominates 2; it fails against column 0, $145.5 < 150$.) This matters for Chapter 5: 2 appears in the support of a mixed equilibrium.

Exercise 2.7

(BM 4.2) Two hunters each choose a Bear, Deer or Wolf. Choosing the same prey scares it away (both get 0). A lone hunter succeeds with probability $\rho_B = 0.4$, $\rho_D = 0.9$, $\rho_W = 0.6$, earning +1, and is otherwise killed (-1). Write the normal form and eliminate dominated strategies.

Solution

A lone hunt of j is worth $2\rho_j - 1$: -0.2, 0.8, 0.2.

	B	D	W
B	0, 0	-0.2, 0.8	-0.2, 0.2
D	0.8, -0.2	0, 0	0.8, 0.2
W	0.2, -0.2	0.2, 0.8	0, 0

B is strictly dominated by D (and by W) for both hunters. The reduced game on $\{D, W\}$ is an anti-coordination game, with nothing further dominated.

2.9. Sources

Prof. Badia, *Game Theory* slides, deck Gth2003 (static games of complete information, normal form, examples, Pareto efficiency, strict dominance, IESDS) and the first part of Gth2004 (best responses and beliefs). Tadelis, *Game Theory: An Introduction*, Chapters 3–4. Exercises: L. Badia and T. Marchioro, *Game Theory: A Handbook of Problems and Exercises*, Esculapio (2022), Chapters 1 and 4.

3. Nash equilibrium

Dominance arguments ask only that players best-respond to *some* belief. They often leave many outcomes standing. Nash equilibrium sharpens the prediction by demanding that beliefs also be *correct*: every player best-responds to what the others actually do. This chapter defines the concept, shows how to find equilibria in bi-matrices, relates it to dominance, and introduces the tools — Pareto efficiency, price of anarchy — used to judge the quality of an equilibrium.

3.1. Definition

The guiding idea: game theory should *predict* the outcome of play among rational players, and a prediction is credible only if, once announced, nobody wants to deviate from it. Such a prediction is **self-enforcing** (or *strategically stable*).

Definition 3.1 (Nash equilibrium (pure strategies)). In $G = \{S_1, \dots, S_n; u_1, \dots, u_n\}$ the profile $s^* = (s_1^*, \dots, s_n^*)$ is a **Nash equilibrium** (NE) if, for every player i , s_i^* is a best response to s_{-i}^* :

$$u_i(s_i^*, s_{-i}^*) \geq u_i(s_i, s_{-i}^*) \quad \forall s_i \in S_i,$$

equivalently $s_i^* \in \arg \max_{s_i \in S_i} u_i(s_i, s_{-i}^*)$ for all i .

Motivation by contradiction. Take a profile s' that is *not* a NE. Then some player i has $s_i'' \in S_i$ with $u_i(s_i'', s_{-i}') > u_i(s_i', s_{-i}')$: an incentive to deviate unilaterally. Any outcome other than a NE is therefore unstable.

Remark (A forecast, not a process). A NE is a property of a one-shot game: an outcome after which nobody has regrets about their own choice, given the others'. It is *not* the end point of a sequence of adjustments — a widespread but mistaken reading that dynamic and repeated games will dispel.

3.2. Finding equilibria in a bi-matrix

Brute force checks every cell. More efficiently: for each column, mark player 1's best payoff(s); for each row, mark player 2's best payoff(s). A cell with both entries marked is a NE.

- **Example 1:** (M, R) is the only NE — the same outcome IESDS produced.
- **Example 2:** (D, R) with payoffs $(9, 9)$ is the only NE, although IESDS removed nothing.
- **Odds & Evens:** no cell has both entries marked; there is no pure-strategy NE (Chapter 5 restores one by allowing randomisation).
- **Battle of the Sexes:** both (R, R) and (S, S) are NE. The concept confirms the intuition but cannot predict *which* will occur.
- **Prisoner's Dilemma:** (F, F) is the unique NE.

3.3. Relation with dominance

Theorem 3.1 (IESDS and Nash equilibrium). *In a finite game:*

1. every NE survives IESDS;
2. if IESDS leaves a single profile s^* , then s^* is the unique NE.

The same holds with “rationalizable” in place of “survives IESDS”.

Proof sketch. (1) A strategy played in a NE is a best response to the others’ equilibrium strategies, and a strictly dominated strategy is never a best response; so no equilibrium strategy is removed at the first round, and by induction at any round. (2) By (1) any NE must be the survivor; the survivor is a NE because each component is the only remaining — hence best — response once all others are removed. $\square$

Nash equilibrium is thus a *refinement* of IESDS: whenever IESDS makes a prediction, NE makes the same one; NE often predicts where IESDS cannot.

! Important

Two requirements define a NE: (1) everyone best-responds to their beliefs; (2) everyone’s beliefs are correct. The first follows from rationality. The second is demanding — it requires some external reason for beliefs to coincide with actual play: convention, communication, learning, or an influential player that coordinates expectations.

3.4. Weak dominance

Write $s_{-i} = (s_j)_{j \neq i}$.

Definition 3.2 (Weak dominance). s'_i **weakly dominates** s_i if $u_i(s'_i, s_{-i}) \geq u_i(s_i, s_{-i})$ for every s_{-i} , with strict inequality for at least one s_{-i} . Without the strictness requirement we just say s'_i **dominates** s_i . A strategy that (strictly or weakly) dominates every other strategy of a player is (strictly or weakly) **dominant**.

Proposition 3.2 (Dominant-strategy equilibrium). *If every player i has a dominant strategy s_i^* , then $(s_1^*, \dots, s_n^*)$ is a Nash equilibrium. The converse is false.*

! Warning

Do **not** iteratively eliminate weakly dominated strategies. Add to Odds & Evens (with stakes 5) a third strategy “Punch the opponent” (P), after which both get -5 :

	0	1	P
0	$-5, 5$	$5, -5$	$-5, -5$
1	$5, -5$	$-5, 5$	$-5, -5$
P	$-5, -5$	$-5, -5$	$-5, -5$

P is weakly dominated for both, yet (P, P) is the only pure-strategy NE: no unilateral deviation helps. Eliminating P loses it. Weak elimination is also order-dependent.

3.5. Equilibrium vs. efficiency

Pareto efficiency and Nash equilibrium answer different questions:

- **Pareto efficiency:** no way, in the whole game, to make someone better off without making someone worse off.
- **Nash equilibrium:** no way for a single player to improve by a *unilateral* change.

The Prisoner's Dilemma shows they can be disjoint: the unique NE (F, F) is the only Pareto-inefficient profile. Inefficient equilibria arise because players are driven by self-interest alone. To quantify the cost of decentralised, selfish behaviour, assign each profile a **social cost** $K(s)$, for instance the negative total welfare $K(s) = -\sum_j u_j(s)$ or the min-max fairness cost $K(s) = \max_j(-u_j(s))$.

Definition 3.3 (Price of anarchy and price of stability). The **price of anarchy** is the ratio between the social cost of the *worst* Nash equilibrium and the social optimum:

$$\text{PoA} = \frac{\max_{s^* \in \text{NE}} K(s^*)}{\min_{s \in S} K(s)}.$$

Using the *best* Nash equilibrium in the numerator gives the **price of stability**.

PoA = 1 means selfishness costs nothing; large values mean that a central planner would do much better. For important classes of games (selfish routing with linear latencies, Chapter 4) the PoA admits tight theoretical bounds.

Example 3.1 (A grading game)

An eccentric professor lets a game decide exam grades. You and an unknown classmate secretly pick an integer between 18 and 30. If the numbers match, both get that grade. Otherwise let L be the lower number: whoever proposed L gets $L + R$, the other $L - R$ (below 18 fails, above 30 is honours). For any $R > 1$ the unique NE is $(18, 18)$. At any common $x > 18$, undercutting to $x - 1$ yields $x - 1 + R > x$; at $(18, 18)$, raising yields $18 - R$. Experimentally, with $R = 2$ many students cooperate at high numbers — a well-known criticism of the rationality assumption — while a large R such as 10 dampens cooperation.

3.6. A field guide to small games

Most exercises reduce, after IESDS, to a 2×2 game of one of a few types. Recognising the type gives the equilibria at once.

type	structure	pure NE	mixed NE
Prisoner's Dilemma	a dominant strategy for each player, outcome Pareto dominated	1	none
Coordination (Battle of the Sexes, Stag Hunt)	both want to match	2 (diagonal)	1

3. Nash equilibrium

type	structure	pure NE	mixed NE
Anti-coordination (Chicken, Hawk–Dove, volunteer)	both want to mismatch	2 (off-diagonal)	1
Discoordination (Matching Pennies)	one wants to match, the other to mismatch	0	1

Larger games often hide one of these after dominated rows and columns are removed. With **three or more players**, the normal form is a table of tables: one matrix per action of the third player, and each cell lists three payoffs. A cell is a NE if each player’s payoff is maximal along *their own* direction: the column within the row (player 2), the row within the column (player 1), and the same cell across the matrices (player 3). Games with n symmetric players, such as public goods and volunteer’s dilemmas, are best described by a formula, $u_i(s_i, k)$ depending on how many *others* choose each action.

3.7. Selecting among equilibria

When a game has several equilibria, Nash’s requirement that “beliefs are correct” does not say *which* beliefs. Several criteria are used in practice; none is part of the definition.

- **Focal points** (Schelling, 1960). Asked to meet in New York without communicating, most people pick Grand Central at noon. Culture, labels and history make one equilibrium salient, and the salience itself makes it self-fulfilling.
- **Pareto dominance**. If one equilibrium is better for *everyone*, players who can talk beforehand will agree on it, and the agreement is self-enforcing because it is a NE.
- **Risk dominance** (Harsanyi and Selten, 1988). Pre-play talk may not convince when the efficient equilibrium is risky. Consider the *Stag Hunt*:

	S	H
S	4, 4	0, 3
H	3, 0	3, 3

(S, S) is Pareto dominant, but S is a best response only if $\Pr(\text{other plays } S) \geq 3/4$, while H guarantees 3. In a 2×2 symmetric game, an equilibrium **risk-dominates** the other if the product of the players’ losses from unilateral deviation is larger: here $(4 - 3)^2 = 1 < 9 = (3 - 0)^2$, so (H, H) is risk dominant. Experiments and learning dynamics often converge to the risk-dominant equilibrium. - **Stability and learning**. An equilibrium reached by best-response or fictitious-play dynamics from many starting points (Chapter 5) is more plausible as a long-run convention. Evolutionary game theory makes this precise.

i Note

Multiplicity is not a defect of the theory: many real situations (technology standards, driving on the left or right, bank runs) *do* have several self-fulfilling outcomes, and what selects among them is outside the game.

3.8. Exercises

Problems from Badia and Marchioro (BM), Chapters 1 and 4. The mixed equilibria of these games are found in Chapter 5.

Exercise 3.1

(BM 1.1) Jorah (J) and Khaleesi (K) each attack Meeren (M, value 5) or Yunkai (Y, value 3). An attacked city is always conquered. K's utility is the total value conquered; J's is the same plus 10 if they attack the same city. Write the normal form and find the pure NE.

Solution

	M	Y
M	15, 5	8, 8
Y	8, 8	13, 3

J wants to match K ($15 > 8$, $13 > 8$); K wants to mismatch ($8 > 5$, $8 > 3$). Every cell has a player who regrets, so there is **no pure-strategy NE**: a discoordination game like Matching Pennies.

Exercise 3.2

(BM 1.4) Two students choose a supervisor among Xavier, Yuan and Zingberry, worth 40, 60, 50. Choosing the same supervisor gives each only 70% of the value. Write the normal form and find the pure NE.

Solution

	X	Y	Z
X	28, 28	40, 60	40, 50
Y	60, 40	42, 42	60, 50
Z	50, 40	50, 60	35, 35

X is strictly dominated by Y . In the reduced game the best response to Y is Z ($50 > 42$) and to Z is Y ($60 > 35$). The pure NE are (Y, Z) and (Z, Y) : the students spread out.

Exercise 3.3

(BM 1.8) Andrea books the first week and Brianna the second, each choosing Florence, Rome or Venice. One week in Florence and one in Rome (in either order) lets them tour Tuscany: utility 3 each. Both in Florence: Andrea 1, Brianna 0; both in Rome: Andrea 0, Brianna 1. Each week in Venice adds 1 to both. Find the pure NE.

3. Nash equilibrium

Solution

	F	R	V
F	1, 0	3, 3	2, 1
R	3, 3	0, 1	1, 2
V	2, 1	1, 2	2, 2

Marking best responses gives three pure NE: (F, R) and (R, F) with payoffs $(3, 3)$, and (V, V) with $(2, 2)$. The last is Pareto dominated but still self-enforcing: if the other goes to Venice, Tuscany is out of reach.

Exercise 3.4

(BM 1.11) Siblings Agatha and Bruno each play with the old Train (T) or the New toy (N), while Mom simultaneously buys a Dollhouse (D) or Soldiers (S). If the kids do not play together, everyone gets 0. Together with T, each kid gets 3. Together with N, Agatha gets 6 (D) or 1 (S) and Bruno 1 (D) or 4 (S). Mom's utility is the sum of the kids'. Find the pure NE.

Solution

Payoffs (Agatha, Bruno, Mom):

Mom D	N	T
N	6, 1, 7	0, 0, 0
T	0, 0, 0	3, 3, 6

Mom S	N	T
N	1, 4, 5	0, 0, 0
T	0, 0, 0	3, 3, 6

Mismatched profiles are never NE (each kid would rather join the other). (T, T, D) and (T, T, S) are NE: Mom's choice is then irrelevant. (N, N, D) is a NE, but (N, N, S) is not, since Mom would switch to D ($7 > 5$). Three pure NE.

Exercise 3.5

(BM 4.3) Jean (value 600) and Kate (value 450) each go to the boutique R (20% off the 250 price) or the mall store S. Each shop has one bag; if both go to the same shop, a fair coin decides who gets it. Write the expected-utility normal form and find the pure NE.

Solution

Alone at R: Jean 400, Kate 250; alone at S: 350, 200; together, half of that.

	R	S
R	200, 125	400, 200

S 350, 250 175, 100

The pure NE are (R, S) and (S, R) , an anti-coordination game. Note that the coin toss enters only through expected utility: a 50% chance at $V - P$ is equivalent, for a vN-M player, to a sure $(V - P)/2$.

Exercise 3.6

(BM 4.10, part 1) Find all NE of

	X	Y	Z
A	2, 2	3, 1	0, 0
B	1, 6	7, 4	9, 4
C	0, 1	5, 3	9, 2

Solution

M's best responses: A to X , B to Y , B and C to Z . N's: X to A , X to B , Y to C . Only (A, X) is marked twice, with payoffs $(2, 2)$. A support check confirms it is the *only* NE, mixed ones included, even though (B, Y) and (B, Z) are better for both. The sequential (Stackelberg) versions of this game are solved in Chapter 11.

3.9. Sources

Prof. Badia, *Game Theory* slides, deck Gth2004 (Nash equilibrium, relation with IESDS, weak dominance, Pareto efficiency, price of anarchy, the grading game); exercise sheet gt20s. Tadelis, *Game Theory: An Introduction*, Chapter 5. Original reference: J. Nash, *Equilibrium points in n -person games*, PNAS **36**, 48 (1950). Exercises: L. Badia and T. Marchioro, *Game Theory: A Handbook of Problems and Exercises*, Esculapio (2022), Chapters 1 and 4.

4. Applications of Nash equilibrium

Nash equilibrium is useful in two ways. As a **prediction instrument**, it tells us where a system of self-interested agents will settle (with care when equilibria are multiple). As a **critique of competition**, it shows that self-interest need not produce efficient outcomes. This chapter works out the classic models — Cournot and Bertrand duopolies, Hotelling’s location game, the tragedy of the commons — and closes with selfish routing, where the price of anarchy can be computed exactly.

4.1. Cournot duopoly

Cournot (1838) anticipated Nash’s idea a century early. Two firms produce quantities $q_1, q_2 \geq 0$ of a homogeneous good, $Q = q_1 + q_2$. Production costs $C(q) = cq$ for both; the market-clearing price is

$$P(Q) = \max(a - Q, 0), \quad a > c.$$

Each firm’s strategy set is $S_i = [0, \infty)$ (quantities above a are pointless) and its payoff is its profit

$$u_i(q_i, q_j) = q_i(P(q_i + q_j) - c) = q_i(a - q_i - q_j - c).$$

Proposition 4.1 (Cournot–Nash equilibrium). *The unique Nash equilibrium is*

$$q_1^* = q_2^* = \frac{a - c}{3}, \quad u_1^* = u_2^* = \frac{(a - c)^2}{9}.$$

Proof. Firm i ’s best response maximises a concave parabola in q_i : the first-order condition $a - 2q_i - q_j - c = 0$ gives

$$BR_i(q_j) = \frac{a - q_j - c}{2}.$$

A NE is an intersection of the two best-response lines; solving the symmetric system $q = (a - q - c)/2$ gives $q = (a - c)/3$. $\square$

Monopoly benchmark. A single firm solves $\max_q q(a - q - c)$, giving $q^m = (a - c)/2$ and profit $u^m = (a - c)^2/4$. The monopolist produces *less* than the two duopolists together ($q^m < 2(a - c)/3$); lower production means a higher price and higher total profit.

Why the firms do not collude. If each produced $q^m/2 = (a - c)/4$ they would share u^m , i.e. $(a - c)^2/8 > (a - c)^2/9$ each. But $q^m/2$ is not a best response to $q^m/2$: $BR_i((a - c)/4) = 3(a - c)/8$. With a high price, unilaterally raising output increases one’s own revenue at the other’s expense, and the process continues until no gain is left — at the Cournot point. (Chapter 11 shows how repetition can sustain collusion.)

Asymmetric costs. If firm i has unit cost c_i , the best responses become $q_i = (a - c_i - q_j)/2$, and solving the pair gives

$$q_i^* = \frac{a - 2c_i + c_j}{3}, \quad P^* = \frac{a + c_1 + c_2}{3},$$

valid while both quantities are positive. If $a - 2c_i + c_j \leq 0$, the inefficient firm produces nothing and the other is a monopolist. The cheaper firm produces more and earns more, and a cost advantage of one firm *lowers* the rival’s output — strategic substitutes at work.

4.2. Bertrand duopoly

Bertrand (1883) objected that firms set *prices*, not quantities. Strategies are now $p_i \in [0, \infty)$; consumers buy $q = a - p$ from the cheaper firm (splitting equally at a tie); cost is $c q$.

Proposition 4.2 (Bertrand paradox). *The unique Nash equilibrium is $p_1^* = p_2^* = c$: both firms price at cost and earn zero profit.*

Undercutting is always profitable while $p > c$: a firm stealing the whole market by charging $p - \varepsilon$ nearly doubles its profit. At $p = c$ nobody can gain. Note that $(c + \varepsilon, c)$ is *not* a NE even though firm 1's profit is 0 either way: firm 2, earning 0, could raise its price slightly and earn a positive profit.

The Bertrand outcome is great for consumers but unrealistic. Two classical fixes:

- **Imperfect substitutes.** If $q_i = a - p_i + b p_j$ with $0 < b < 2$, profits are smooth in prices and the first-order conditions give the NE $p_1^* = p_2^* = (a + c)/(2 - b) > c$.
- **Cost asymmetry.** With $c_1 = 1$, $c_2 = 2$ and prices on a grid of step $\varepsilon = 0.01$, firm 1 prices at 1.99 and takes the market: (1.99, 2.00) is a NE. As $\varepsilon \rightarrow 0$ the equilibrium disappears because payoffs are discontinuous — discretising the strategy space is a standard trick to avoid such pathologies.

4.3. Hotelling's location game

Two ice-cream vendors choose a location on a 1 km boulevard, discretised into 101 points $0, 1, \dots, 100$. Customers are uniformly spread and buy from the nearer vendor; prices are equal.

If A stands at 22 and B at 35, A serves everyone up to 28. A has an incentive to move right — best of all to 36, just beyond B, which captures every customer from 36 to 100. Iterating this reasoning, the unique NE is **both vendors at 50**. The result is the classical *median voter theorem*: in a two-party system, platforms converge to the political centre.

4.4. The tragedy of the commons

Since at least Hume (1739) it has been understood that citizens moved by private incentives misuse public resources (Hardin, 1968: “the tragedy of the commons”). An ICT version: n users share a Wi-Fi spectrum; user i activates g_i processes, $G = \sum_i g_i$. Each process costs c and earns a throughput value $v(G)$ that decreases with congestion: $v'(G) < 0$, $v''(G) < 0$, $v(G_{\max}) = 0$. User i 's payoff is $g_i (v(G) - c)$.

Proposition 4.3 (Overuse in equilibrium). *At the symmetric NE $g_i^* = G^*/n$,*

$$v(G^*) + \frac{G^*}{n} v'(G^*) - c = 0,$$

whereas the social optimum G^m , maximising $G(v(G) - c)$, satisfies

$$v(G^m) + G^m v'(G^m) - c = 0.$$

Since v is decreasing and concave, $G^ > G^m$: the resource is overused.*

Interpretation. A user considering one more process compares its cost c with the marginal loss of value of *its* own g_i processes, $v'(G) g_i = v'(G) G/n$. The planner counts the loss suffered by *all* processes, $v'(G) G$. The factor $1/n$ is the externality each user ignores.

4.5. Selfish routing

Pigou’s example (1920). One unit of traffic travels from s to d over two parallel links. The upper link’s cost per unit equals its load x ; the lower link has constant cost 1.

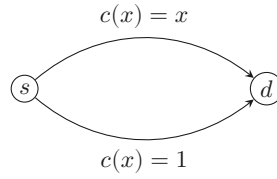


Figure: Pigou’s network.

The upper link is a dominant choice (its cost never exceeds 1), so at equilibrium all traffic uses it and the total cost is 1. Splitting the traffic $\frac{1}{2}-\frac{1}{2}$ gives total cost $\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot 1 = \frac{3}{4}$, which is optimal — but every user on the lower link would deviate. Hence

$$\text{PoA} = \frac{1}{3/4} = \frac{4}{3}.$$

Nonlinear latencies. Replace the upper cost by x^a with a large. The equilibrium is unchanged (total cost 1), while routing a small fraction ε on the lower link costs $(1 - \varepsilon)^{a+1} + \varepsilon \rightarrow 0$ as $a \rightarrow \infty$ and $\varepsilon \rightarrow 0$ suitably. The price of anarchy is **unbounded**.

Braess’s paradox. Adding capacity can make the equilibrium *worse*.

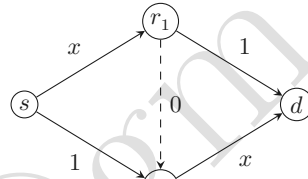


Figure: Braess’s network; the dashed link is the costless shortcut.

Without the dashed link, traffic splits evenly over the two symmetric routes and each unit pays $\frac{1}{2} + 1 = \frac{3}{2}$. With a costless shortcut $r_1 \rightarrow r_2$, the route $s \rightarrow r_1 \rightarrow r_2 \rightarrow d$ becomes dominant: everyone takes it and each unit pays $1 + 0 + 1 = 2$. A “better” network has a worse Nash equilibrium.

Theorem 4.4 (Roughgarden–Tardos). *For selfish routing with **affine latencies** on any network, the cost of a Nash flow is at most $\frac{4}{3}$ times the cost of the optimal flow. Pigou’s two-link network attains the bound.*

The intuition: confronted with two choices, all selfish users take the momentarily better one and overload it. Remarkably, this inefficiency is intrinsic — the worst case does not depend on the topology.

Tip

Selfish routing is a template for many ICT problems: users choosing access points, channels or servers by their own latency. The Pigou bound tells a network designer how much is lost by letting users choose — and Braess’s paradox warns that adding a resource can backfire.

4.6. Strategic substitutes and complements

In games with continuous strategies the slope of the best-response functions summarises the strategic interaction.

4. Applications of Nash equilibrium

- **Strategic substitutes:** BR_i is decreasing. In Cournot, $BR_i(q_j) = (a - c - q_j)/2$: the more the rival produces, the less I want to produce. Aggressive play by one player induces restraint by the other.
- **Strategic complements:** BR_i is increasing. In Bertrand with differentiated products, $BR_i(p_j) = (a + c + bp_j)/2$: a rival's price rise lets me raise mine. Effort in joint projects and arms races behave similarly.

Formally, with smooth payoffs, the sign of $\partial^2 u_i / \partial s_i \partial s_j$ decides which case applies. The distinction matters for (i) **multiplicity**: with complements, best responses can cross several times (coordination failures, as in the investment game of the exercises); (ii) **comparative statics**: a shock that raises one player's action raises (complements) or lowers (substitutes) the other's; (iii) **commitment** in sequential versions (Chapter 9): a leader commits to being aggressive under substitutes but to being soft under complements.

Proposition 4.5 (Uniqueness by contraction). *If each BR_i is a function with $|BR'_i| \leq k < 1$, the composite map $BR = (BR_1, BR_2)$ is a contraction, so the Nash equilibrium exists, is unique, and best-response iteration converges to it from any start.*

Cournot ($k = 1/2$) and Bertrand with $b < 2$ ($k = b/2$) both satisfy it.

4.7. Voting: the median voter theorem

Hotelling's model also describes elections. Voters are spread on a left-right axis $[-a, a]$ with distribution F , and each votes for the candidate whose platform is nearest. Each of two office-seeking candidates chooses a platform to maximise the probability of winning.

Theorem 4.6 (Median voter theorem). *If voter preferences are single-peaked on a line, the unique Nash equilibrium is that both candidates choose the platform of the **median voter** m , $F(m) = 1/2$.*

Proof. If $a_j \neq m$, candidate i wins for sure by locating strictly between a_j and m , close enough to a_j . The voters on the far side of the midpoint then include the median, i.e. more than half. So no profile with $a_j \neq m$ is a NE. At (m, m) each wins with probability $1/2$, and any move away from m loses. $\square$

Convergence to the centre is a strong prediction. It fails when candidates care about policy, when abstention punishes moderation, when there are more than two candidates (with three, no pure equilibrium exists in the uniform case), or when the policy space has more than one dimension (majority cycles, Chapter 7).

4.8. Exercises

Problems from Badia and Marchioro (BM), Chapter 1, and, for oligopoly and contests (not covered by the handbook), Tadelis (T), Chapter 5.

Exercise 4.1

(BM 1.3) A transmitter either concentrates on one frequency f_0 (C, power cost 2) or spreads over the whole spectrum (S, cost 9). The receiver listens on one frequency (A) or two (B), not knowing which one is f_0 . If the transmitter spreads, the receiver's utility is 2 plus the number of frequencies used; if it concentrates, 2 minus that number. The transmitter gets twice the receiver's utility, plus the number of frequencies used by the receiver, minus its power cost. Write the normal form and find the pure NE.

Solution

	A	B
C	1, 1	0, 0
S	-2, 3	1, 4

E.g. (C, A) : receiver $2 - 1 = 1$, transmitter $2 + 1 - 2 = 1$; (S, B) : receiver 4, transmitter $8 + 2 - 9 = 1$. The pure NE are (C, A) and (S, B) : a coordination problem between a cheap narrow-band and an expensive wide-band configuration. Without a protocol fixing the convention, the devices can miscoordinate.

Exercise 4.2

(BM 1.7, continued) In the staffing game of Chapter 2 (row payoffs $(140, 160, 160)$, $(250, 40, 60)$, $(150, 150, -60)$ for 0, 1, 2 employees, symmetric), find the pure NE.

Solution

The best response to 0 is 1 (250); to 1 it is 0 (160); to 2 it is 0 (160). The pure NE are $(1, 0)$ and $(0, 1)$, with payoffs 250 and 160. Exactly one firm commits a worker, a “chicken” structure. The symmetric outcome $(0, 0)$, which would be fair, is not stable.

Exercise 4.3

(BM 1.9) Two TV channels each choose to invest in the Morning, Daytime or Evening news. Same slot: 0 each. Otherwise the earlier slot wins (Evening counts as earlier than the next Morning), the loser gets -1 , and the winner gets 1, 4, 2 for M, D, E. Write the normal form. Is there a pure NE?

Solution

	M	D	E
M	0, 0	1, -1	-1, 2
D	-1, 1	0, 0	4, -1
E	2, -1	-1, 4	0, 0

A weighted rock–paper–scissors: M beats D, D beats E, E beats M. Every pure profile has a player who would switch to the slot that beats the rival’s, so there is no pure NE. Nash’s theorem guarantees a mixed one (Chapter 5).

Exercise 4.4

(BM 1.10) Grandma needs eggs. Each of three grandchildren decides whether to go to the store (G, cost 1) or not (N). If at least one goes, everybody gets 4 from the cake; otherwise everybody gets 0. Write the normal form and find the pure NE.

4. Applications of Nash equilibrium

Solution

$u_i = 4 \cdot \mathbb{1}\{\text{someone goes}\} - \mathbb{1}\{i \text{ goes}\}$. Going gives 3 for sure; staying gives 4 if someone else goes and 0 otherwise. The pure NE are the three profiles where **exactly one** grandchild goes: anyone else going would waste 1, and the one who goes would get $0 < 3$ by staying. This is the *volunteer's dilemma*. The equilibria are asymmetric, and who volunteers is not determined by the game.

Exercise 4.5

(T 5.8, *n*-firm Cournot) n firms with cost cq_i face $P(Q) = a - Q$. Find the Cournot equilibrium and the limit of the price as $n \rightarrow \infty$.

Solution

Firm i maximises $q_i(a - Q_{-i} - q_i - c)$: $q_i = (a - c - Q_{-i})/2$. In the symmetric equilibrium $q = (a - c - (n-1)q)/2$, so $q^* = \frac{a-c}{n+1}$, $Q^* = \frac{n(a-c)}{n+1}$, $P^* = \frac{a+nc}{n+1}$ and each profit is $\frac{(a-c)^2}{(n+1)^2}$. As $n \rightarrow \infty$, $P^* \rightarrow c$: the competitive outcome. Cournot spans the range from monopoly ($n = 1$) to perfect competition.

Exercise 4.6

(T 5.11, *Wasteful Shipping Costs*) Two countries each have demand $p = 90 - q$ and one coal firm with unit cost 10. (a) Without trade, find the outputs. (b) With free trade and shipping cost 10 per unit exported, find the NE. (c) Firm 1 can trigger a regulation raising shipping costs to 40. Will it?

Solution

(a) Monopoly in each country: $q = 40$, $p = 50$, profit 1600. (b) The markets are separate Cournot games with costs 10 (home) and 20 (foreign). With asymmetric costs $q_i = (a - 2c_i + c_j)/3$: home sells 30, foreign 20, $p = 40$. Profits are $900 + 400 = 1300$ per firm, and the equilibrium is unique. Consumers gain, firms lose, and 20×10 per country is wasted on shipping goods that cross in opposite directions. (c) With foreign cost 50, the foreign firm's best response to the monopoly output 40 is $(90 - 50 - 40)/2 = 0$. Trade dies and each firm is back to 1600 $>$ 1300. Firm 1 releases the report.

Exercise 4.7

(T 5.18, *Political Campaigning*) Candidate i spends $s_i \geq 0$ and wins with probability $s_i/(s_1 + s_2)$; winning is worth v . Find the best responses and the NE. What if candidate 2 values winning at kv , $k > 1$?

Solution

$u_i = v \frac{s_i}{s_1 + s_2} - s_i$; the FOC $v \frac{s_j}{(s_1 + s_2)^2} = 1$ gives $BR_i(s_j) = \sqrt{vs_j} - s_j$. Symmetric NE: $s^* = v/4$, increasing linearly in v . Total spending dissipates half the prize. With values v and kv the FOCs give $vs_2 = kvs_1$, so $s_2 = ks_1$, and $s_1^* = \frac{kv}{(1+k)^2}$, $s_2^* = \frac{k^2v}{(1+k)^2}$. The candidate who cares more spends k times more and wins with probability $k/(1+k)$. This is the Tullock contest, a model of lobbying and rent-seeking.

4.9. Sources

Prof. Badia, *Game Theory* slides, deck Gth2006 (Cournot, Bertrand, Hotelling, tragedy of the commons, selfish routing, Braess paradox, price of anarchy). Tadelis, *Game Theory: An Introduction*, Chapter 5.2. Original references: A. Cournot (1838); J. Bertrand (1883); H. Hotelling, *Stability in competition*, Econ. J. **39**, 41 (1929); G. Hardin, *The tragedy of the commons*, Science **162**, 1243 (1968); T. Roughgarden and É. Tardos, *How bad is selfish routing?*, J. ACM **49**, 236 (2002). Exercises: L. Badia and T. Marchioro, *Game Theory: A Handbook of Problems and Exercises*, Esculapio (2022), Chapter 1.

5. Mixed strategies and existence

Odds & Evens has no Nash equilibrium in pure strategies, which is unsatisfying if the equilibrium is meant to be *the* prediction of a game. The fix is to let players randomise. With mixed strategies every finite game has an equilibrium — Nash’s theorem — and new tools appear: the indifference principle for computing equilibria, best-response diagrams, and dominance by mixtures. The chapter ends with dynamics (fictitious play), a class of games where equilibria are easy (potential games), and the bad news that in general they are hard to compute.

5.1. Mixed strategies

Enlarging Odds & Evens with a “flip a fair coin” strategy $\frac{1}{2}$ for both players shows a cell, $(\frac{1}{2}, \frac{1}{2})$, from which nobody wants to move. Let us formalise this.

Definition 5.1 (Mixed strategy). For a non-empty finite set A , a probability distribution on A is $p : A \rightarrow [0, 1]$ with $\sum_{x \in A} p(x) = 1$; the set of all of them is the **simplex** ΔA . A **mixed strategy** of player i in $G = \{S_1, \dots, S_n; u_1, \dots, u_n\}$ is $m_i \in \Delta S_i$. Its **support** is $\{s_i \in S_i : m_i(s_i) > 0\}$.

A pure strategy s_i is identified with the degenerate distribution putting probability 1 on it. Players randomise independently, and payoffs are extended to mixed profiles by expectation (they are vN-M utilities):

$$u_i(m_1, \dots, m_n) = \sum_{s \in S} m_1(s_1) m_2(s_2) \cdots m_n(s_n) u_i(s).$$

Example 5.1 (*Odds and Evens*)

Let Odd play 0 with probability q and Even play 0 with probability r . Odd’s expected payoff is

$$u_{\text{Odd}}(q, r) = -4qr + 4q(1-r) + 4(1-q)r - 4(1-q)(1-r) = -4(2q-1)(2r-1),$$

and Even’s is its opposite. If $q = \frac{1}{2}$, Even’s payoff is 0 whatever r ; if $r = \frac{1}{2}$, the same holds for Odd. So $(q, r) = (\frac{1}{2}, \frac{1}{2})$ is an equilibrium, and one checks it is the only one: if Odd’s payoff were negative (or positive) at another profile, Odd (or Even) could profitably move.

5.2. Equilibrium and dominance with mixtures

Definition 5.2 (Mixed Nash equilibrium). A profile $m^* \in \Delta S_1 \times \cdots \times \Delta S_n$ is a **Nash equilibrium** if for every i

$$u_i(m_i^*, m_{-i}^*) \geq u_i(m_i, m_{-i}^*) \quad \forall m_i \in \Delta S_i.$$

Dominance is defined as before with mixed strategies on both sides. Although there are a continuum of opponents’ mixtures, it suffices to check pure ones: m'_i strictly dominates m_i iff $u_i(m'_i, s_{-i}) > u_i(m_i, s_{-i})$ for every *pure* $s_{-i} \in S_{-i}$ (expected payoffs are linear in each opponent’s mixture).

Example 5.2 (*A pure strategy dominated only by a mixture*)

	L	C	R
T	7, 4	5, 0	8, 1
D	6, 0	3, 4	9, 1

R is not dominated by L or by C , but $\frac{1}{2}L + \frac{1}{2}C$ gives player B a payoff of 2 against both T and D , more than R 's 1. Eliminating R , T dominates D for A; then L dominates C for B. The unique NE is (T, L) .

IESDS in mixed strategies (IESDS_m) keeps the good properties of the pure case: Nash equilibria survive it and the order of elimination is irrelevant. As before, only *strict* dominance may be used — a weakly dominated strategy can be an equilibrium or belong to the support of one.

Theorem 5.1 (Indifference principle). *A mixed profile m is a Nash equilibrium if and only if, for every player i ,*

$$u_i(m) = u_i(s_i, m_{-i}) \quad \forall s_i \in \text{supp}(m_i), \quad u_i(m) \geq u_i(s_i, m_{-i}) \quad \forall s_i \notin \text{supp}(m_i).$$

Every pure strategy in the support of an equilibrium mixture earns the *same* payoff, and no strategy outside earns more. (If one strategy in the support earned more, shifting probability onto it would raise the payoff.) In particular a pure NE remains a NE when mixed strategies are allowed.

Example 5.3 (*Battle of the Sexes: the third equilibrium*)

Let Ann play R with probability q and Brian with probability r :

$$u_A(q, r) = 2qr + (1 - q)(1 - r), \quad u_B(q, r) = qr + 2(1 - q)(1 - r).$$

A fully mixed NE requires Ann to be indifferent between R and S : $u_A(1, r) = u_A(0, r)$, i.e. $2r = 1 - r$, so $r = \frac{1}{3}$. Brian's indifference, $q = 2(1 - q)$, gives $q = \frac{2}{3}$. The third equilibrium is $(q, r) = (\frac{2}{3}, \frac{1}{3})$, with expected payoffs $(\frac{2}{3}, \frac{2}{3})$ — worse for both than either pure equilibrium, because of miscoordination.

i Note

In a mixed equilibrium, each player's mixture is pinned down by the *opponent's* indifference, not their own payoffs. Raising Ann's payoff at (R, R) changes Brian's equilibrium mixture, not Ann's.

5.3. Interpreting mixed strategies

In the end players take pure actions. Three readings of “randomisation” are common:

- **Large numbers:** in M independent one-shot plays, q is the fraction of times a pure strategy is chosen.
- **Fuzzy action:** the player genuinely does not settle on an action.
- **Beliefs:** q is the uncertainty *the opponent* has about my (pure) choice. Harsanyi's purification theorem (Chapter 12) makes this precise.

5.4. Best-response diagrams

Beliefs are now elements of ΔS_{-i} and $BR_i : \Delta S_{-i} \rightrightarrows \Delta S_i$. Consider

Art \ Bea	F	G
U	6, 1	0, 4
D	2, 5	4, 0

Let Art play U with probability q , Bea play F with probability r . Art gets $6r$ from U and $2r + 4(1 - r)$ from D : his best response is $q = 1$ if $r > \frac{1}{2}$, $q = 0$ if $r < \frac{1}{2}$, anything at $r = \frac{1}{2}$. Bea gets $q + 5(1 - q)$ from F and $4q$ from G : her best response is $r = 1$ if $q < \frac{5}{8}$, $r = 0$ if $q > \frac{5}{8}$.

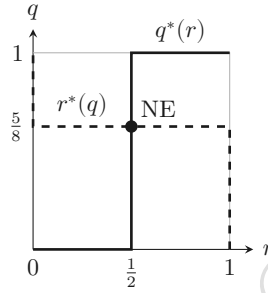


Figure: best-response correspondences of Art (solid) and Bea (dashed); they cross only at the mixed equilibrium.

The unique NE is $(q, r) = (\frac{5}{8}, \frac{1}{2})$. For the Battle of the Sexes the same construction gives two step functions crossing **three** times: the two pure equilibria at the corners and the mixed one at $(\frac{2}{3}, \frac{1}{3})$. In every 2×2 game the two staircases, running across the unit square, must cross at least once.

5.5. Nash's theorem

Theorem 5.2 (Nash, 1950). *Every game with finitely many players and finite strategy sets has at least one Nash equilibrium, possibly in mixed strategies.*

Define $BR_i(m_{-i}) = \arg \max_{m_i \in \Delta S_i} u_i(m_i, m_{-i})$ and the aggregate correspondence on $\Delta S = \prod_i \Delta S_i$,

$$BR(m) = BR_1(m_{-1}) \times \cdots \times BR_n(m_{-n}).$$

A profile is a NE iff it is a **fixed point**, $m \in BR(m)$. $BR_i(m_{-i})$ is never empty and always contains a pure strategy; by linearity of u_i in m_i it is convex (the set of mixtures over the maximising pure strategies).

Theorem 5.3 (Brouwer and Kakutani fixed-point theorems). **Brouwer:** *a continuous function from a closed interval (more generally a compact convex set) into itself has a fixed point. In one dimension this is Bolzano's theorem applied to $f(x) - x$ on $[0, 1]$.*

Kakutani: *let $A \subset \mathbb{R}^n$ be non-empty, compact and convex and $F : A \rightrightarrows A$ a correspondence with non-empty convex values and closed graph ($x_k \rightarrow x, y_k \rightarrow y, y_k \in F(x_k)$ imply $y \in F(x)$). Then there exists $x^* \in F(x^*)$.*

Proof of Nash's theorem. ΔS is non-empty, compact and convex; BR has non-empty convex values; its graph is closed because u_i is continuous. Kakutani's theorem gives a fixed point. $\square$

5.6. Fictitious play

In **fictitious play** (Brown, 1951) the “regrets” of a NE become actual moves: each player assumes the opponents keep playing the empirical frequencies of their past actions and best-responds to them. Full rationality is relaxed — predictions may be wrong and are corrected over time. Nash equilibria are absorbing points of the dynamics, but they need not be reached: in rock–paper–scissors (with Shapley’s variant as the classical counterexample) play cycles forever. Fictitious play does converge for games solvable by IESDS, for potential games, for zero-sum games, and for $2 \times N$ games with generic payoffs.

5.7. Potential games

Definition 5.3 (Potential games). A function $\Omega : S \rightarrow \mathbb{R}$ is an **exact potential** for G if for every i , s_i, s'_i and s_{-i}

$$\Omega(s'_i, s_{-i}) - \Omega(s_i, s_{-i}) = u_i(s'_i, s_{-i}) - u_i(s_i, s_{-i}),$$

a **weighted potential** with weights $w_i > 0$ if the right-hand side is multiplied by w_i , and an **ordinal potential** if

$$\Omega(s'_i, s_{-i}) > \Omega(s_i, s_{-i}) \iff u_i(s'_i, s_{-i}) > u_i(s_i, s_{-i}).$$

A game admitting one is a (exact, weighted, ordinal) **potential game**.

If G has an ordinal potential, its Nash equilibria coincide with those of the game in which *every* player’s payoff is Ω : a multi-player problem collapses into a single-objective optimisation, which is what makes potential games attractive for distributed optimisation.

Example 5.4 (Potentials)

The Prisoner’s Dilemma (with $-1, -9, 0, -6$) has the exact potential $\Omega(M, M) = 0$, $\Omega(M, F) = \Omega(F, M) = 1$, $\Omega(F, F) = 4$; the players maximise Ω , not their joint welfare. The Cournot duopoly has the ordinal potential $\Omega(q_1, q_2) = q_1 q_2 (a - q_1 - q_2 - c)$, since $u_i = \Omega/q_j$ for $q_j > 0$.

Theorem 5.4 (Pure equilibria of potential games). *Every finite ordinal potential game has at least one pure-strategy Nash equilibrium, and it can be found deterministically.*

Proof. Let players improve their payoff one at a time. Each improvement strictly increases Ω , which takes finitely many values, so the process stops — at a profile where nobody can improve: a pure NE (a local maximum of Ω). $\square$

Congestion games (players choose the least congested resources: routes, channels, servers) are potential games (Rosenthal, 1973), and conversely every potential game is isomorphic to a congestion game (Monderer and Shapley, 1996).

Coordination games reward players for making the same choice and have several pure equilibria, which are maxima of a potential. In Rousseau’s *Stag Hunt*

	S	H
S	10, 10	0, 7
H	7, 0	7, 7

an exact potential is $\Omega(S, S) = 3$, $\Omega(S, H) = \Omega(H, S) = 0$, $\Omega(H, H) = 7$. **Anti-coordination** games (Hawk–Dove, Chicken) reward different choices; **discoordination** games (Odds & Evens) pit a player who

wants a match against one who wants a mismatch. Every potential game is the sum of a **pure coordination game** (identical payoffs) and a **dummy game** (each player's payoff depends only on the others' strategies).

5.8. Computational complexity

Nash's theorem guarantees existence, but how hard is it to *find* an equilibrium? Since a solution always exists, the problem cannot be NP-complete in the usual sense. Daskalakis, Goldberg and Papadimitriou (2006, extended to two players by Chen and Deng) proved that **NASH is PPAD-complete**.

PPAD (*Polynomial Parity Arguments on Directed graphs*, Papadimitriou 1994) contains search problems whose solution is guaranteed by a parity argument, the prototype being **END-OF-LINE**: in a directed graph where every node has in- and out-degree at most one, given an unbalanced node, find another one. One must exist, but finding it may require walking the whole path. Computing a fixed point of a continuous map — hence of *BR* — reduces to such a walk (via Sperner's lemma). Roughly, $P \subseteq \text{PPAD} \subseteq \text{NP}$, and PPAD is believed to be hard.

Tip

For engineers this is less gloomy than it sounds: many games of practical interest (potential games, zero-sum games) have constructively computable equilibria, and often an **ε -Nash equilibrium** — where unilateral improvements smaller than ε are ignored — is good enough.

5.9. Finding mixed equilibria systematically

For games larger than 2×2 , the indifference principle becomes an algorithm, **support enumeration**:

1. Pick candidate supports $T_1 \subseteq S_1$, $T_2 \subseteq S_2$ (after removing strictly dominated strategies; for nondegenerate games $|T_1| = |T_2|$).
2. Solve the linear system: player 2's mixture makes every strategy of T_1 give player 1 the same payoff v_1 , and vice versa, with probabilities summing to one.
3. Keep the solution only if all probabilities are non-negative **and** no strategy *outside* the support earns more than v_i .

Step 3 is the one most often forgotten. In the staffing game of the exercises, mixing over $\{0, 1\}$ solves step 2, but the excluded strategy 2 earns more than the support, so that candidate is not an equilibrium.

Theorem 5.5 (Oddness (Wilson, 1971)). *Almost every finite game (all except a set of payoff tables of measure zero) has a **finite and odd** number of Nash equilibria.*

A 2×2 coordination game has three (two pure, one mixed); Matching Pennies has one. If you have found an even number of equilibria of a generic game, one is missing. Degenerate games (ties, as in the Pizza game) can have continua.

5.10. Evolutionary stability

Mixed equilibria have a population reading. Consider a large population of individuals randomly matched to play a symmetric game with payoff $u(s, s')$. A “strategy” is now an inherited behaviour, and a mixed strategy is the population share using each pure action.

Definition 5.4 (Evolutionarily stable strategy (Maynard Smith and Price, 1973)). m^* is an **ESS** if for every mutant $m \neq m^*$: either $u(m^*, m^*) > u(m, m^*)$, or $u(m^*, m^*) = u(m, m^*)$ and $u(m^*, m) > u(m, m)$.

An ESS is a symmetric NE (first condition) that also resists invasion by alternative best responses (second condition). In the Hawk–Dove game with $v = 10$ and injury $k = 6$ (Hawk payoff -1 against Hawk, 10 against Dove; Dove 0 and 1), the mixed NE has a share 0.9 of Hawks and is an ESS: a mutant with more Hawks fares worse against itself.

Replicator dynamics make the population view dynamic: the share x_s of strategy s grows at a rate equal to its payoff advantage over the population average,

$$\dot{x}_s = x_s(u(s, x) - u(x, x)).$$

Every ESS is an asymptotically stable rest point, and every stable rest point is a NE. In coordination games the dynamics select one of the pure equilibria depending on the starting shares; the mixed one is unstable — another argument for treating mixed equilibria of coordination games with care.

5.11. Exercises

Problems from Badia and Marchioro (BM), Chapters 1 and 4: the mixed equilibria of the games whose normal forms were derived in Chapters 2–4.

Exercise 5.1

(BM 1.1) Find the NE of the Jorah–Khaleesi game, $(M, M) = (15, 5)$, $(M, Y) = (8, 8)$, $(Y, M) = (8, 8)$, $(Y, Y) = (13, 3)$.

Solution

No pure NE (Chapter 3), so there is a unique fully mixed one. Let J play M with probability p and K with q . K must make J indifferent: $15q + 8(1 - q) = 8q + 13(1 - q)$, so $q = 5/12$. J must make K indifferent: $5p + 8(1 - p) = 8p + 3(1 - p)$, so $p = 5/8$. NE: $(p, q) = (5/8, 5/12)$, with payoffs $(10.92, 6.13)$.

Exercise 5.2

(BM 1.2) Find all NE of the reduced Square–Triangle game $(A, A) = (0, 0)$, $(A, D) = (10, 6)$, $(D, A) = (6, 10)$, $(D, D) = (-4, -4)$.

Solution

Pure: (A, D) and (D, A) . Mixed: by symmetry both play A with probability α , and indifference gives $10(1 - \alpha) = 6\alpha - 4(1 - \alpha)$, so $\alpha = 0.7$, with payoff 3 each. Three NE, as oddness predicts.

Exercise 5.3

(BM 1.4) Find the mixed NE of the supervisor game.

Solution

X is strictly dominated, so mix over $\{Y, Z\}$. If the other picks Y with probability p : $42p + 60(1 - p) = 50p + 35(1 - p)$, so $p = 25/33$. Each student picks Y with $25/33$ and Z with $8/33$, with expected payoff 46.4. X (payoff 40) is not a profitable deviation, so this is a NE.

Exercise 5.4

(BM 1.5) Find the mixed NE in the reduced War Fantasy World game, Alice $\in \{E, O\}$, Bob $\in \{H, E\}$: $(E, H) = (-5, 0)$, $(E, E) = (10, 10)$, $(O, H) = (2, -3)$, $(O, E) = (2, -8)$.

Solution

Pure NE: (O, H) and (E, E) . Mixed: Alice plays E with p and Bob plays H with q . Alice's indifference: $-5q + 10(1 - q) = 2$, so $q = 8/15$. Bob's: $-3(1 - p) = 10p - 8(1 - p)$, so $p = 1/3$. Payoffs $(2, -2)$. Check the eliminated strategies: against Bob's mixture Alice's H gives $0 < 2$, and against Alice's mixture Bob's O gives $-5 < -2$. So $(\frac{1}{3}E + \frac{2}{3}O, \frac{8}{15}H + \frac{7}{15}E)$ is a NE.

Exercise 5.5

(BM 1.6) Find the mixed NE of the reduced exhibition-match game, $(S, S) = (4, 4)$, $(S, R) = (3, 1)$, $(R, S) = (1, 3)$, $(R, R) = (5, 5)$.

Solution

If the other plays S with probability p : S gives $4p + 3(1 - p) = 3 + p$ and R gives $p + 5(1 - p) = 5 - 4p$. They are equal at $p = 2/5$. Each team plays S with $2/5$, earning 3.4, less than in either pure equilibrium. Under replicator dynamics this mixed point is unstable.

Exercise 5.6

(BM 1.7, continued) Find the mixed NE of the staffing game (rows $(140, 160, 160)$, $(250, 40, 60)$, $(150, 150, -60)$, symmetric).

Solution

Since 2 is not dominated (Chapter 2), supports must include it. The symmetric fully mixed NE equalises the three rows: $140a + 160b + 160c = 250a + 40b + 60c = 150a + 150b - 60c$ with $a + b + c = 1$, giving $(a, b, c) = (254, 232, 1)/487$ and payoff $72840/487 \approx 149.6$. There are also two asymmetric ones: one firm mixes $\frac{10}{21} \cdot 0 + \frac{11}{21} \cdot 2$ while the other mixes $\frac{1}{2} \cdot 0 + \frac{1}{2} \cdot 1$. With the two pure NE this makes five (odd). The printed solution's mixture $\frac{12}{23} \cdot 0 + \frac{11}{23} \cdot 1$ fails step 3 of support enumeration: against it, 2 earns $150 > 149.6$.

Exercise 5.7

(BM 1.8) Find a fully mixed NE of the Italian-trip game (rows/columns F, R, V: $(1, 0)$ $(3, 3)$ $(2, 1)$ / $(3, 3)$ $(0, 1)$ $(1, 2)$ / $(2, 1)$ $(1, 2)$ $(2, 2)$), and the mixed NE that ignores Venice.

Solution

Andrea plays $(f, r, 1 - f - r)$ making Brianna indifferent among F, R, V: $3r + (1 - f - r) = 3f + r + 2(1 - f - r) = f + 2r + 2(1 - f - r)$, so $f = 1/4$, $r = 1/2$. Symmetrically Brianna plays $(1/2, 1/4, 1/4)$. Both earn 1.75. Ignoring Venice, the game on $\{F, R\}$ is anti-coordination: Andrea plays F with 0.4,

5. Mixed strategies and existence

Brianna with 0.6, earning 1.8 each. Venice then gives $2 > 1.8$, so this is **not** a NE of the full game. Further NE exist, e.g. Andrea $\frac{1}{2}R + \frac{1}{2}V$ against Brianna $\frac{1}{2}F + \frac{1}{2}V$.

Exercise 5.8

(BM 1.9) Find the mixed NE of the TV-channel game (M, D, E; row payoffs (0, 1, -1), (-1, 0, 4), (2, -1, 0), the column player's symmetric).

Solution

By symmetry both use (m, d, e) . Equal payoffs: $d - e = -m + 4e = 2m - d$ with $m + d + e = 1$, which gives $(m, d, e) = (9/24, 11/24, 4/24)$ and an expected payoff of $7/24 \approx 0.29$. Despite D 's large prize, the slot that beats D (Morning) is played most: in mixed equilibria, *your* probabilities are set by the *opponent's* payoffs.

Exercise 5.9

(BM 1.10) Find the symmetric fully mixed NE of the grandma's-eggs game.

Solution

Each grandchild goes with probability p . Going yields 3; staying yields $4(1 - (1 - p)^2)$. Indifference gives $(1 - p)^2 = 1/4$, so $p = 1/2$. The cake fails with probability $(1/2)^3 = 1/8$. With n grandchildren, $(1 - p)^{n-1} = 1/4$, so each volunteers less and the failure probability $(1/4)^{n/(n-1)}$ *increases* toward $1/4$: the bystander effect.

Exercise 5.10

(BM 1.11) Find the additional (mixed) NE of the siblings-and-toys game.

Solution

If both kids play T , Mom is indifferent: $(T, T, \mu D + (1 - \mu)S)$ is a NE for every μ . Whenever both kids play N with positive probability, Mom strictly prefers D . With D fixed, the kids play a Battle of the Sexes: if Agatha plays N with p and Bruno with q , then $6q = 3(1 - q)$ gives $q = 1/3$ and $p = 3(1 - p)$ gives $p = 3/4$. NE: $(\frac{3}{4}N + \frac{1}{4}T, \frac{1}{3}N + \frac{2}{3}T, D)$.

Exercise 5.11

(BM 4.1) Luca and Marco apply to company A (worth 100) or B (worth 110). A lone applicant is hired with probability 0.5. If both apply to the same company, Luca is hired with 0.5 and Marco with 0.2, independently. Utility doubles if both end up at the same company; rejection gives 0. Write the normal form and find all NE.

Solution

Same company A: Luca $0.5 \cdot 0.8 \cdot 100 + 0.5 \cdot 0.2 \cdot 200 = 60$, Marco $0.2 \cdot 0.5 \cdot 100 + 0.2 \cdot 0.5 \cdot 200 = 30$; at B multiply by 1.1.

	A	B
A	60, 30	50, 55

B 55, 50 66, 33

Luca wants to match, Marco to mismatch: no pure NE. Luca's indifference fixes Marco's mixture: $60\mu + 50(1-\mu) = 55\mu + 66(1-\mu)$, so $\mu = 16/21$. Marco's fixes Luca's: $30\lambda + 50(1-\lambda) = 55\lambda + 33(1-\lambda)$, so $\lambda = 17/42$.

Exercise 5.12

(BM 4.3) Find the mixed NE of the handbag game, $(R, R) = (200, 125)$, $(R, S) = (400, 200)$, $(S, R) = (350, 250)$, $(S, S) = (175, 100)$.

Solution

Jean plays R with a and Kate with b . Kate's indifference: $125a + 250(1-a) = 200a + 100(1-a)$, so $a = 2/3$. Jean's: $200b + 400(1-b) = 350b + 175(1-b)$, so $b = 3/5$. Expected payoffs (280, 166.7).

Exercise 5.13

Find all Nash equilibria (pure and mixed) of

	f	g
F	9, 3	2, 2
G	0, 0	3, 9

Solution

Pure: (F, f) and (G, g) . Mixed: let A play F with q , B play f with r . B indifferent: $3q = 2q + 9(1-q) \Rightarrow q = 9/10$. A indifferent: $9r + 2(1-r) = 3(1-r) \Rightarrow r = 1/10$. The mixed NE is $(q, r) = (9/10, 1/10)$.

Exercise 5.14

Show that rock-paper-scissors (win 1, lose -1 , tie 0) has the unique equilibrium in which both players mix uniformly.

Solution

At the uniform profile every pure strategy earns 0, so the indifference conditions hold. Conversely, if a player's mixture is not uniform, some pure strategy (say R) has probability above $\frac{1}{3}$ and the opponent's best response to it (P) earns a positive payoff; in a zero-sum game this means the first player is not best-responding, since uniform mixing would guarantee 0. A full proof checks all supports with the indifference principle.

5.12. Sources

Prof. Badia, *Game Theory* slides, deck Gth2007 (mixed strategies, indifference principle, best-response diagrams, Nash's theorem, fictitious play, potential and congestion games, coordination games, PPAD); exercise sheet gt20s. Tadelis, *Game Theory: An Introduction*, Chapter 6. Original references: J. Nash, PNAS **36**, 48 (1950); G. W. Brown, *Iterative solution of games by fictitious play* (1951); D. Monderer and L. Shapley, *Potential games*, Games Econ. Behav. **14**, 124 (1996); C. Daskalakis, P. Goldberg and

5. Mixed strategies and existence

C. Papadimitriou, *The complexity of computing a Nash equilibrium*, SIAM J. Comput. **39**, 195 (2009).
Exercises: L. Badia and T. Marchioro, *Game Theory: A Handbook of Problems and Exercises*, Esculapio (2022), Chapters 1 and 4.

6. Security strategies and zero-sum games

Nash equilibrium asks players to hold correct beliefs about each other. A more pessimistic, optimisation-flavoured approach asks instead: what can a player *guarantee* regardless of what the others do? This chapter defines security (maximin) strategies and the minimax payoff, relates them to Nash equilibrium, and shows that for **zero-sum games** the two viewpoints coincide — von Neumann’s minimax theorem — so that equilibria can be computed by linear programming.

6.1. Maximin and minimax

Consider player i against “the rest”, $-i$.

Definition 6.1 (Security strategy and maximin). Let $f_i(s_i) = \min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i})$ be the worst payoff of s_i . A strategy $s_i^* \in \arg \max_{s_i} f_i(s_i)$ is a **security strategy** (maximinimizer) and

$$w_i = \max_{s_i \in S_i} \min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i})$$

is the **maximin** or **security payoff** of i .

A security strategy is the conservative choice: the best payoff in case the opponents make the worst possible move for i .

Definition 6.2 (Minimax). Let $F_i(s_{-i}) = \max_{s_i} u_i(s_i, s_{-i})$ be i ’s best payoff against s_{-i} . The **minimax** of i is

$$z_i = \min_{s_{-i} \in S_{-i}} \max_{s_i \in S_i} u_i(s_i, s_{-i}).$$

If i could move *after* seeing $-i$ ’s choice, the minimax is the payoff i would be guaranteed: the opponents’ most punishing choice, against which i best-responds.

Example 6.1 (Maximin vs. minimax)

Only player A’s payoffs matter here:

	L	C	R	f_A (row min)
T	5	3	4	3
D	2	6	1	1
F_A (column max)	5	6	4	

A’s maximin is 3 (security strategy T); A’s minimax is 4 (B’s most punishing column is R).

Proposition 6.1 (Maximin, minimax and equilibrium). *For every player i : (1) $\maximin_i \leq \minimax_i$;*

6. Security strategies and zero-sum games

(2) if s is a Nash equilibrium, then $\minimax_i \leq u_i(s)$.

(1) holds because moving second can only help. (2) holds because at a NE, $u_i(s) = \max_{s_i} u_i(s_i, s_{-i}) = F_i(s_{-i}) \geq \min F_i$. Equality of maximin and minimax does not, however, guarantee a pure NE in general games.

6.2. Zero-sum games

Definition 6.3 (Zero-sum game). A two-player game is **zero-sum** if $u_1(s) = -u_2(s)$ for every profile s . More generally a game is **strictly competitive** (adversarial) if u_1 increases exactly when u_2 decreases.

Odds & Evens, rock–paper–scissors and chess are zero-sum. Since utilities matter up to affine transformations, a constant-sum game (chess scored 1 for a win, $\frac{1}{2}$ each for a draw) is equivalent to a zero-sum one; football league points (3 for a win, 1 each for a draw) are not. A zero-sum game is described by a single matrix $u = u_1$.

Theorem 6.2 (Minimax theorem, pure strategies). *Let G be a finite two-player zero-sum game.*

1. G has a pure Nash equilibrium iff $\maximin_i = \minimax_i$ for each player i .
2. All Nash equilibria give the same payoff, equal to $\maximin_i$.
3. Nash equilibria are exactly the pairs of security strategies.

Example 6.2 (A saddle point)

	L	C	R
T	−9	8	−5
M	−2	6	2
D	−1	3	4

Row minima are −9, −2, −1: A's maximin is −1 (play D). Column maxima are −1, 8, 4: A's minimax is −1 (B plays L). They coincide, and (D, L) is a NE with $u_A = -1$.

In a zero-sum game it is enough to check one player, since $\maximin_1 = -\minimax_2$ and $\minimax_1 = -\maximin_2$. The common value is the **value of the game**, and a joint security strategy is a **saddle point** of the payoff matrix — the minimum of its row and the maximum of its column. Some games with infinite strategy sets have no value.

6.3. Mixed security strategies

Definition 6.4 (Mixed maximin and minimax). With $f_i(m_i) = \min_{m_{-i}} u_i(m_i, m_{-i})$, a maximiser of f_i over ΔS_i is a **mixed security strategy** and its value the **mixed maximin**. Symmetrically, $\min_{m_{-i}} \max_{m_i} u_i(m_i, m_{-i})$ is the **mixed minimax**.

By linearity, $f_i(m_i)$ can be computed against the opponents' *pure* strategies only, and the inner maximum in the minimax against i 's pure strategies only. Moreover, the payoff is continuous and bilinear, so mixed

maximin and mixed minimax always exist and are equal; overall

$$\maximin_i \leq \maximin_i^m = \minimax_i^m \leq \minimax_i,$$

and at any mixed NE m , $\minimax_i^m \leq u_i(m)$.

Example 6.3 (*Art's guarantee*)

In the Art–Bea game of Chapter 5 (Art: U gives 6 or 0, D gives 2 or 4), if Art plays U with probability q ,

$$f_A(q) = \min\{6q + 2(1 - q), 4(1 - q)\} = \min\{2 + 4q, 4 - 4q\},$$

maximised where the two lines cross, at $q = \frac{1}{4}$, with mixed maximin 3.

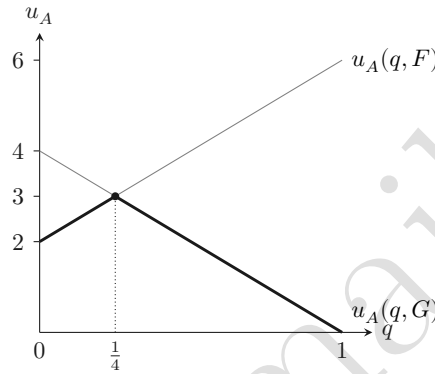


Figure: Art's payoff against Bea's two pure strategies; the bold lower envelope is $f_A(q)$, maximised at $q = \frac{1}{4}$.

One checks that Art's mixed minimax is also 3, which is his payoff at the NE $(\frac{5}{8}, \frac{1}{2})$; for Bea, mixed maximin = mixed minimax = 2.5 = $u_B(\text{NE})$.

Warning

In a non-zero-sum game a security strategy need not be an equilibrium strategy. In the Battle of the Sexes, pure maximin is 0 and pure minimax is 1 for both players; the mixed maximin is $\frac{2}{3}$, achieved by Ann playing R with probability $\frac{1}{3}$ — not the $\frac{2}{3}$ of the mixed NE. The three equilibria give Ann 2, 1 and $\frac{2}{3}$.

Example 6.4 (*Odds and Evens*)

$\maximin = -4 < \maximin^m = \minimax^m = 0 < \minimax = 4$. The value 0 is the payoff of the mixed equilibrium.

6.4. The minimax theorem

Theorem 6.3 (von Neumann, 1928). *Let G be a finite two-player zero-sum game. Then:*

1. *for each player, $\maximin_i^m = \minimax_i^m$, and G has a Nash equilibrium in mixed strategies;*
2. *the mixed Nash equilibria are exactly the pairs of mixed security strategies, and all yield player i the payoff $\maximin_i^m$.*

All equilibria of a zero-sum game are therefore *equivalent* and *interchangeable*: they share the same payoff, and combining one player's equilibrium strategy with the other's from a different equilibrium is again an

equilibrium. The common $\max_1^m = -\min_2^m$ is the **value of the game**. Unlike Nash's theorem, the proof is constructive — it tells us how to *compute* equilibria.

6.5. Linear programming

Let player 1 have pure strategies $A_1, \dots, A_L$ and player 2 have $B_1, \dots, B_M$; write $u(A_j, B_k)$ for player 1's payoff. A mixed security strategy $a = (a_1, \dots, a_L)$ of player 1 solves

$$\begin{aligned} \max_{a, W} \quad & W \\ \text{s.t.} \quad & \sum_j a_j u(A_j, B_k) \geq W \quad k = 1, \dots, M, \\ & a_j \geq 0, \quad \sum_j a_j = 1. \end{aligned}$$

The M constraints check a against player 2's pure strategies only. At the optimum W cannot be raised because some constraints are *active*; these identify the support of player 2's optimal mixture. Symmetrically, player 2's strategy b solves

$$\min_{b, W} \quad W \quad \text{s.t.} \quad \sum_k b_k u(A_j, B_k) \leq W \quad \forall j, \quad b \geq 0, \quad \sum_k b_k = 1.$$

The two programs are **duals** of each other, and LP duality is precisely the minimax theorem. LPs are solved in polynomial time by interior-point methods; the simplex method (CPLEX, `lp_solve`) is exponential in the worst case but fast in practice. Meta-heuristics (genetic algorithms, tabu search) can be faster still but carry no guarantee.

Note

Minimaximisation is the backbone of classical game-playing AI. Chess is zero-sum (though sequential), and the minimax search with alpha-beta pruning evaluates positions exactly as a security-strategy computation on the game tree.

6.6. Constant-sum, symmetric and competitive games

Three facts that shorten many exercises:

- **Constant-sum games.** If $u_1 + u_2 = c$ in every cell, subtracting $c/2$ from each payoff gives a zero-sum game with the same best responses, hence the same equilibria. The duel of two knights (each cell summing to -90) is an example.
- **Symmetric zero-sum games have value 0.** If the payoff matrix is skew-symmetric, $U^\top = -U$ (rock-paper-scissors and its weighted variants), then any strategy that guarantees W to the row player guarantees W to the column player too; with $W + W = 0$ the value is 0. The equilibrium mixtures still depend on the payoffs.
- **Saddle points.** A pure profile is a NE of a zero-sum game iff its entry is the minimum of its row and the maximum of its column. Pure maximin = pure minimax exactly when a saddle point exists. Otherwise the gap $\max_{\min} < \min_{\max}$ measures the value of knowing the opponent's move, and only mixing closes it.

Strictly competitive (ordinally zero-sum) games, where every change that helps one player hurts the other, share the interchangeability of equilibria but not the LP machinery, since only the ordering of payoffs is opposed.

6.7. Exercises

Problems from Badia and Marchioro (BM), Chapter 2 (only the row player's payoffs are shown).

Exercise 6.1

(BM 2.1) Coral holds cards 9, 6, 2 and Dwayne holds 9, 8, 2; each plays one card. Equal cards F : Coral gets F . Different cards: Coral gets $F_C - F_D$. Dwayne gets the opposite. Write the game, and find the NE, maximin and minimax.

Solution

	9	8	2
9	9	1	7
6	-3	-2	4
2	-7	-6	2

9 is dominant for Coral. Against it Dwayne minimises: 8. The entry 1 is the minimum of its row and the maximum of its column, a **saddle point**. NE (9,8); maximin = minimax = 1 for Coral and -1 for Dwayne.

Exercise 6.2

(BM 2.2) Solve the zero-sum game with rows A, B, C and columns X, Y: (9, -5), (-2, 7), (8, -1). What is the support of player 1's equilibrium mixture?

Solution

Pure maximin -1 and minimax 7 differ: there is no saddle point. Let player 2 play X with probability b . Player 1's payoffs are $14b - 5$ (A), $7 - 9b$ (B), $9b - 1$ (C), and player 2 minimises their upper envelope. The minimum is where B and C cross, $7 - 9b = 9b - 1$, so $b = 4/9$ and the value is 3 (A gives $1.2 < 3$ there). So the support is $\{B, C\}$, and player 1's weights make player 2 indifferent: $-2a + 8(1 - a) = 7a - (1 - a)$, so $a = 1/2$. NE: $((0, \frac{1}{2}, \frac{1}{2}), (\frac{4}{9}, \frac{5}{9}))$, value 3.

Exercise 6.3

(BM 2.3) Coaches choose Counter-offensive, Defensive or Extremely aggressive. C beats E (+3), D beats C (+1), E beats D (+8); same choice is a tie; the loser gets the opposite. Find the value and the NE.

Solution

	C	D	E
C	0	-1	3
D	1	0	-8
E	-3	8	0

6. Security strategies and zero-sum games

The matrix is skew-symmetric, so the value is 0. The cycle $C \rightarrow E \rightarrow D \rightarrow C$ rules out pure strategies and two-strategy supports, so both play all three: $d - 3e = 0$, $-c + 8e = 0$, $3c - 8d = 0$, giving $(c, d, e) = (2/3, 1/4, 1/12)$ for both. Pure maximin is -1 and pure minimax 1 .

Exercise 6.4

(BM 2.4) In a magic duel, Fire beats Ice, Ice beats Rock, Rock beats Wind, Wind beats Fire. Absolom wins ties; Balthazar wins when neither element beats the other (Fire–Rock, Ice–Wind). Absolom may not use Fire. Payoffs ± 1 . Write the LP for Absolom and solve it.

Solution

	F	I	R	W
I	−1	1	1	−1
R	−1	−1	1	1
W	1	−1	−1	1

max W subject to $a_1 + a_2 + a_3 = 1$, $a \geq 0$, and one constraint per column: $-a_1 - a_2 + a_3 \geq W$, $a_1 - a_2 - a_3 \geq W$, $a_1 + a_2 - a_3 \geq W$, $-a_1 + a_2 + a_3 \geq W$. Balthazar can guarantee 0 (e.g. by mixing uniformly), so $W \leq 0$; with $W = 0$ the first and third constraints give $a_3 = a_1 + a_2$, and the second and fourth give $a_1 = a_2 + a_3$. Hence $a = (\frac{1}{2}, 0, \frac{1}{2})$: Ice or Wind with equal probability, never Rock. Balthazar's optimal mixtures form a set, $b_F \geq b_R$, $b_I \geq b_W$, $b_F - b_I = b_R - b_W$ (e.g. uniform, or $(\frac{1}{2}, \frac{1}{2}, 0, 0)$), all interchangeable with Absolom's.

Exercise 6.5

(BM 2.5) Red and Blue each pick 1, 2 or 3. Equal numbers: Blue wins. Numbers differing by 1: Red wins. Differing by 2: no winner. The winner gets 1 from the loser. Find all NE.

Solution

	1	2	3
1	−1	1	0
2	1	−1	1
3	0	1	−1

No saddle point (pure maximin -1 , minimax 1). Full-support indifference gives Red $(2/7, 3/7, 2/7)$ and, coincidentally, Blue the same mixture, with value $1/7$ for Red. Smaller supports fail: with both mixing $\frac{1}{2} - \frac{1}{2}$ on $\{1, 2\}$, Red's strategy 3 earns $\frac{1}{2} > 0$. The LP with $W = 1/7$ has all three constraints active, which confirms that the NE is unique.

From the course material:

Exercise 6.6

Player Jim (rows) faces Joe (columns); only Jim's payoffs are shown:

	S	C
T	3	0
M	1	2

Compute Jim's pure maximin and minimax, and his mixed maximin and minimax.

Solution

Pure: row minima $0, 1 \rightarrow$ maximin 1 ; column maxima $3, 2 \rightarrow$ minimax 2 . Mixed: with T played with probability p , Jim gets $3p + (1 - p) = 1 + 2p$ against S and $2(1 - p)$ against C ; equalising gives $p = \frac{1}{4}$ and $\text{maximin}^m = 1.5$. Joe's most punishing mixture plays S with probability r such that $3r = r + 2(1 - r)$, i.e. $r = \frac{1}{2}$, giving $\text{minimax}^m = 1.5$. Indeed $1 \leq 1.5 = 1.5 \leq 2$.

Exercise 6.7

Write the LP that computes the value of rock-paper-scissors for the row player and verify that $a = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, $W = 0$ is optimal.

Solution

Maximise W subject to $a_S - a_P \geq W$ (against R), $a_R - a_S \geq W$ (against P), $a_P - a_R \geq W$ (against S), $a \geq 0$, $\sum a = 1$. Summing the three constraints gives $0 \geq 3W$, so $W \leq 0$; the uniform mixture attains $W = 0$ with all constraints active, hence it is optimal and the value is 0 .

6.8. Sources

Prof. Badia, *Game Theory* slides, deck Gth2010 (maximin, minimax, zero-sum games, minimax theorem in pure and mixed strategies, linear programming). Tadelis, *Game Theory: An Introduction*, Chapter 6.4. Original reference: J. von Neumann, *Zur Theorie der Gesellschaftsspiele*, Math. Ann. **100**, 295 (1928). Exercises: L. Badia and T. Marchioro, *Game Theory: A Handbook of Problems and Exercises*, Esculapio (2022), Chapter 2.

7. Social choice and voting

Chapter 1 showed that a single rational decision maker has a utility function. Can a *group* of rational individuals be summarised by a rational collective preference? The question matters well beyond elections: scheduling packets under several criteria, choosing among solutions of a multi-objective optimisation, or allocating resources by competing priorities are all “votes”. The answer, in the form of Arrow’s impossibility theorem, is essentially negative, and the paradoxes of voting systems show why.

7.1. Constitutions

Let $\mathcal{R}(A)$ be the set of rational preferences on a set of alternatives A , and let n individuals have preferences $(\succeq_1, \dots, \succeq_n)$, a **profile**.

Definition 7.1 (Constitution). A **constitution** (social welfare function) is a map $f : \mathcal{R}(A)^n \rightarrow \mathcal{R}(A)$ turning every profile into a social preference $\succeq = f(\succeq_1, \dots, \succeq_n)$. For $Y \subseteq A$, $\succeq|_Y = \succeq \cap (Y \times Y)$ denotes the restriction to Y .

Definition 7.2 (Properties of constitutions). • **Independence of irrelevant alternatives (IIA):** if two profiles agree on the pair $\{a, b\}$ for every individual, the social preferences agree on $\{a, b\}$. Adding or removing other alternatives does not change how society ranks a against b .
• **Pareto efficiency:** if $a \succeq_i b$ for every i , then $a \succeq b$.
• **Dictatorship:** there is an individual i such that $a \succeq_i b$ implies $a \succeq b$ for every profile.
• **Monotonicity:** if one individual raises an alternative in their ranking, society does not rank it lower.
• **Non-imposition:** every rational preference is the output of some profile (f is surjective).

Theorem 7.1 (Arrow, 1951). *With at least three alternatives, no constitution is simultaneously non-dictatorial, monotonic, IIA and non-imposed. Equivalently (1963 version): every Pareto-efficient constitution satisfying IIA is a dictatorship.*

7.2. Majority rule and Condorcet cycles

With two candidates, majority rule is unproblematic. With three it can fail spectacularly:

voter	1	2	3
best	A	C	B
		B	A
worst	C	B	A

Pairwise, A beats B (voters 1, 2), B beats C (voters 1, 3), and C beats A (voters 2, 3). Majority preference is **not transitive**.

Definition 7.3 (Condorcet winner and cycle). A candidate beating every other in pairwise majority contests is a **Condorcet winner**; one beaten by every other is a **Condorcet loser**. If no Condorcet winner exists, majority preference contains a **Condorcet cycle**.

Three-candidate profiles arise from two-candidate ones by inserting the third candidate somewhere in each voter's ranking; depending on where it goes, the newcomer can become a Condorcet winner, a Condorcet loser, or close a cycle. Cycles cannot occur with two alternatives, but with random preferences their probability grows with the number of candidates:

candidates \ voters	3	5	7	9	∞
3	5.6%	6.9%	7.5%	7.8%	8.8%
5	16.0%	20.0%	21.5%	23.0%	25.1%
7	23.9%	29.9%	30.5%	34.2%	36.9%
∞	100%	100%	100%	100%	100%

Real examples abound: fiscal policies ranked by liberals, anti-deficit voters and conservatives over combinations of taxes and spending; or traffic classes (voice over IP, video streaming, best-effort data) ranked by delay, loss and fairness criteria, each criterion being a “voter”.

7.3. Electoral systems and their paradoxes

Plurality voting: the candidate with the most first places wins. With 9 voters

voters	4	3	2
best	A	B	C
		B	C
worst	C	A	A

A wins with 4 first places, yet a majority prefers B to A and C to A , and B is the Condorcet winner.

Two-round run-off: the top two meet in a ballot. With

voters	4	3	2
best	A	B	C
		C	C
worst	B	A	A

A and B reach the ballot and B wins 5–4, while the Condorcet winner C does not even make it to the second round. Plurality and run-off favour polarised candidates over compromise ones.

Borda count: with M candidates each voter gives $M - 1$ points to the first, $M - 2$ to the second, ..., 0 to the last. With

voters	5	3	1
best	A	B	C

voters	5	3	1
	B	C	B
worst	C	A	A

A gets 10, B 12, C 5: B wins although A is the Condorcet winner. Borda also violates IIA dramatically. With four candidates

voters	5	2	2
best	D	A	A
	C	D	B
	B	B	D
worst	A	C	C

the ranking is D (21), A (12), B (11), C (10). If D withdraws, the scores become C 10, B 9, A 8: the order of the remaining three is *entirely reversed*.

Approval voting: each voter approves N candidates, one point each ($N = 1$ is plurality). With

voters	3	3	2	1
best	A	D	B	A
	C	B	D	B
	D	C	C	C
worst	B	A	A	D

top-2 approval elects B (A 4, B 6, C 3, D 5), top-3 approval elects C (A 4, B 6, C 9, D 8), plurality elects A , and Borda elects D (16 vs. 14, 12, 12). Four systems, four winners.

Instant run-off: voters rank all candidates; the candidate with the fewest first places is repeatedly eliminated and their votes transferred. With

voters	6	5	4	2
best	A	C	B	B
	B	A	C	A
worst	C	B	A	C

C is eliminated first and A wins with 11 votes. If the last two voters *raise* A to first place, B is eliminated first, its four votes go to C , and C beats A 9–8. **A loses because of increased support** — a violation of monotonicity.

7.4. Agenda setting and manipulation

Sequential pairwise votes look fair but are not under a Condorcet cycle. With $A > B > C > A$, “first A vs. B , then the winner vs. C ” elects C ; “first B vs. C , then the winner vs. A ” elects A . Whoever sets the agenda chooses the winner. The power is not unlimited: a Condorcet loser D can never win, and a Condorcet winner always does.

Voters can also misreport. With

voters	4	3	2
best	A	B	C
	D	A	B
	C	D	D
worst	B	C	A

A wins under plurality. Voters 8–9, for whom A is the worst, can report B first and make B win. Voters 1–4 can pre-empt this by reporting C , who then wins with only two sincere supporters; B 's supporters can in turn switch to A . The outcome depends on who cheats first.

Definition 7.4 (Strategy-proofness). A social choice rule is **strategy-proof** (non-manipulable) if for every profile no individual j obtains an outcome they strictly prefer by reporting $\succeq'_j$ instead of their true $\succeq_j$.

Theorem 7.2 (Gibbard–Satterthwaite). *With at least three alternatives, every strategy-proof social choice rule whose range includes every alternative is a dictatorship.*

7.5. Ways out

No universally good system exists, but there are escape routes:

- **Weaken the axioms** (e.g. drop IIA, as Borda does).
- **Restrict the domain** of admissible profiles. Majority rule, $a \succeq b \iff |\{i : a \succeq_i b\}| \geq |\{i : b \succeq_i a\}|$, is Pareto efficient, IIA and non-dictatorial, but in general not transitive because of Condorcet cycles — so it is not a constitution. On domains where cycles cannot occur it becomes one: Black (1948) showed this for **single-peaked** preferences (alternatives on a line, each voter preferring options closer to their ideal point), Sen (1966) gave more general conditions.
- **Change the ballot**: approval voting with free choice of N , or voting “for” and “against”.

💡 Tip

For engineering, the lesson is that aggregating several performance criteria into a single ranking is never neutral. Every scalarisation (weighted sums are Borda-like) embodies choices that can reverse rankings when an irrelevant option is added or removed.

7.6. Voting as a game

An election can itself be modelled as a static game: voters are players, ballots are strategies, and the rule maps profiles to outcomes. Three observations follow.

- **Two candidates**: voting sincerely is weakly dominant, since your ballot matters only when it is pivotal, and then you want your favourite to win. The median voter result of Chapter 4 describes the candidates' side of this game.
- **Three or more candidates**: sincere voting is generally *not* an equilibrium. Under plurality, supporters of a hopeless candidate prefer to vote for the lesser evil among the front-runners, which is why plurality systems tend to two parties (**Duverger's law**). Run-off systems invite subtler manipulation: supporting a weak opponent in the first round to meet them in the second.

- **Equilibrium is too permissive:** with many voters no single ballot is pivotal, so *every* profile in which a candidate wins by two or more votes is a Nash equilibrium, including everyone voting for their least favourite. One must first delete weakly dominated ballots, or study coalitions of like-minded voters, as in the exercises.

The Gibbard–Satterthwaite theorem guarantees that some profile always rewards misreporting. The useful question for a given rule is therefore *who* can manipulate, and whether the others can counter it. A **Condorcet winner**, when it exists, is robust: any coalition preferring another candidate is a minority in the head-to-head comparison.

7.7. Exercises

Problems from Badia and Marchioro (BM), Chapter 4.

Exercise 7.1

(BM 4.8) Thirteen students elect a class representative by plurality among A, B, C. Preferences: students 1–5: $A \succ B \succ C$; 6–9: $B \succ C \succ A$; 10–12: $C \succ B \succ A$; 13: $C \succ A \succ B$. (a) Who wins with sincere voting? (b) How can students 10–12 manipulate the result? (c) Can A’s supporters counter?

Solution

(a) A 5, B 4, C 4: A wins. (b) Students 10–12 prefer B to A: voting B gives B 7 votes and B wins. (c) A’s supporters could only block B by voting C, which would at best tie and which they dislike more than B. They have no counter. B is the **Condorcet winner** (B beats A 7–6 and C 9–4), so the manipulation actually restores the majority’s choice.

Exercise 7.2

(BM 4.9) A two-round run-off among A (42%, $B \succ C \succ D$ after A), B (11%, $A \succ C \succ D$), C (27%, $B \succ D \succ A$), D (20%, $C \succ B \succ A$). (a) Sincere outcome? (b) Best strategic vote for D’s supporters? (c) Who wins then, and can A’s supporters respond?

Solution

(a) A and C reach the ballot; B’s voters prefer A, so A wins 53–47. (b) D’s supporters prefer B to A: voting B in round one sends A and B to the ballot (B 31%, C 27%). (c) B wins the ballot with 58% (everyone but A’s supporters): B is the Condorcet winner despite the fewest first preferences. A’s supporters cannot beat B head-to-head, so they must keep B out of the ballot. If 7 of their 42 points vote C in round one (A 35, C 34, B 31), A meets C again and wins 53–47. Manipulation invites counter-manipulation, and the outcome depends on who coordinates better.

From the course material:

Exercise 7.3

Verify that in the instant run-off example the two voters who switch to $A \succ B \succ C$ cause A to lose, and compute the Condorcet winner (if any) in both profiles.

Solution

Original: first places A 6, C 5, B 6 $\rightarrow C$ out, its 5 votes go to $A \rightarrow A$ 11. Pairwise: A vs B : $6 + 5 = 11$ prefer A ; A vs C : $6 + 2 = 8$ prefer A vs 9 prefer C , so C beats A ; B vs C : $6 + 4 + 2 = 12$ prefer B . $B > C > A$ and $A > B$: a cycle, no Condorcet winner. Modified profile: first places A 8, C 5, B 4 $\rightarrow B$ out, its 4 votes go to $C \rightarrow C$ 9 beats A 8. Pairwise: $A > B$ (13–4), $C > A$ (9–8), $B > C$ (12–5): again a cycle.

Exercise 7.4

Show that majority rule over three alternatives is transitive whenever all voters' preferences are single-peaked on the order $A < B < C$.

Solution

Single-peakedness on $A < B < C$ forbids B being ranked last, leaving the orders ABC , BAC , BCA , CBA . A cycle $A > B > C > A$ would need a strict majority with A over B (only ABC voters) and a strict majority with C over A (only BCA , CBA voters): two disjoint groups, each larger than $n/2$ — impossible. The reverse cycle $A > C > B > A$ needs a majority with C over B (only CBA) and one with A over C (ABC , BAC), again disjoint. Hence no cycle exists and majority rule is transitive; in fact the median voter's ranking decides every pairwise contest.

7.8. Sources

Prof. Badia, *Game Theory* slides, deck Gth2005 (constitutions, Arrow's theorem, Condorcet cycles, electoral systems, agenda setting, manipulation, Gibbard–Satterthwaite, majority rule). Original references: K. Arrow, *Social Choice and Individual Values*, Wiley (1951); A. Gibbard, *Econometrica* **41**, 587 (1973); M. Satterthwaite, *J. Econ. Theory* **10**, 187 (1975); D. Black, *J. Polit. Econ.* **56**, 23 (1948). Exercises: L. Badia and T. Marchioro, *Game Theory: A Handbook of Problems and Exercises*, Esculapio (2022), Chapter 4.

Part III.

Part III: Dynamic games of complete
information

8. Extensive-form games

In many interactions some players move first and others react. The bi-matrix hides this time structure, so we need a richer representation: the **extensive form**, a game tree annotated with who moves where and what they know. This chapter introduces the tree, information sets, the redefinition of *strategy* as a complete contingent plan, behavioural strategies, and the correspondence between extensive and normal forms.

8.1. Dynamic games

A **dynamic game** has players moving at different times. Complete information is kept — payoffs are common knowledge — but a new distinction appears:

- **perfect information:** every player, when moving, knows everything that happened before;
- **imperfect information:** some moves are effectively simultaneous, or Nature moves and its choice is not observed.

The sequential Battle of the Sexes. In the static version Ann and Brian choose without knowing each other's choice — not very realistic. Suppose instead Ann picks the movie first and phones Brian. Being fully informed, Ann knows that whatever she picks, Brian's best response is to follow; since she prefers R, she picks R. There is no coordination problem any more. To represent this we need to connect each choice to the knowledge available when it is made — Brian's call might not go through, for example.

8.2. The game tree

Definition 8.1 (Extensive form). An extensive-form game specifies:

1. the set of players;
2. their payoff functions over terminal outcomes;
3. the order of moves;
4. the actions available to each player when it is their turn;
5. the information each player has when moving;
6. the probability distributions of Nature's moves;
7. the common knowledge of all the above.

Items 1–2 are shared with the normal form, 3–5 add the time dimension, 6–7 express complete information. Graphically: decision nodes are labelled by the player who moves, branches by actions, and terminal nodes by payoff vectors.

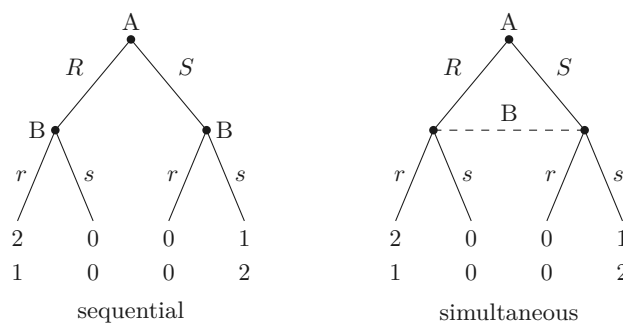


Figure: the Battle of the Sexes in extensive form. Payoffs are (Ann, Brian). Left: Brian observes Ann's choice. Right: the dashed line joins nodes Brian cannot distinguish — one information set.

Dummy players. Players with a single available action (or who never move) can be omitted from the tree; they still appear in the payoff vectors.

8.3. Information sets

Label nodes x_0 (the root), $x_1, x_2, \dots$; each node is either terminal (carrying payoffs) or the parent of other nodes. In the sequential Battle of the Sexes, Brian moves at x_1 (after R) or x_2 (after S) with the same actions but different consequences — and he *knows* which node he is at.

Definition 8.2 (Information set). Each player i has a collection H_i of **information sets**. The information set $h_i(x) \in H_i$ associated with a node x where i moves contains all nodes that i cannot distinguish from x ; by convention $x \in h_i(x)$. If $h_i(x) = \{x\}$ is a singleton, i knows exactly where they are. All nodes in the same information set must have the same set of available actions: $x' \in h_i(x) \Rightarrow A(x') = A(x)$.

In the simultaneous Battle of the Sexes, Brian's information set is $\{x_1, x_2\}$: he does not know Ann's choice. Graphically, nodes in the same set are joined by a dashed line (or enclosed in an oval).

Remark (Equalising action sets). If two indistinguishable nodes seem to offer different actions, add “forbidden” actions with payoff $-M$, where M is huge (formally $-\infty$, or below the minimum payoff). The game changes, but every strategy using the forbidden action becomes strictly dominated, so nothing of substance is altered.

Definition 8.3 (Perfect and imperfect information (formal)). A dynamic game has **perfect information** if all information sets are singletons and Nature never moves. Otherwise it has **imperfect information**: either *exogenous* uncertainty (Nature's moves, as in single-person problems with lotteries) or *endogenous* uncertainty (unobserved, effectively simultaneous moves of other players). In both cases players must form beliefs.

8.4. Strategies in dynamic games

In static games a pure strategy is an action. In dynamic games the history matters through information sets.

Definition 8.4 (Pure strategy (extensive form)). A pure strategy of player i specifies an action at **every** information set of i — including those that will not be reached given the strategy itself. It is a complete

program: a countermove for every possible situation.

In the sequential Battle of the Sexes Brian has two actions but **four** strategies, written $(a_R, a_S) = (\text{reply to } R, \text{reply to } S)$: (s, s) “go to S no matter what”, (r, r) “go to R no matter what”, (r, s) “do what Ann does”, (s, r) “avoid Ann”.

If the static Battle of the Sexes is played on two consecutive nights with the first outcome observed, a strategy is a quintuple $(a_1, a_2^{Rr}, a_2^{Rs}, a_2^{Sr}, a_2^{Ss})$: the first-night move and the second-night reply to each of the four possible first-night outcomes. For instance (r, s, r, r, r) means “R on the first night; S on the second if the first outcome was Rr , otherwise R”. The count explodes quickly: two sequential moves with $|A_1| = |A_2| = 3$ already give player 2 $3^3 = 27$ strategies. Beyond small examples strategies are described implicitly.

8.5. Mixed and behavioural strategies

A **mixed strategy** is still a probability distribution over the (now large) set of pure strategies: the randomisation happens once, at the start, and the player then sticks to the drawn plan. Alternatively, a player can randomise *as the game unfolds*.

Definition 8.5 (Behavioural strategy). A **behavioural strategy** of player i specifies, for each information set $h_i \in H_i$, an independent probability distribution $\sigma_i(\cdot | h_i)$ over the actions $A_i(h_i)$ available there.

Think of a strategy as a handbook of instructions. A mixed strategy picks one of several handbooks at random; a behavioural strategy is a single handbook whose every page gives a random instruction. In the sequential Battle of the Sexes, a mixed strategy of Brian assigns $p_{rr}, p_{rs}, p_{sr}, p_{ss}$; the equivalent behavioural strategy has

$$\sigma_B(r | x_1) = p_{rr} + p_{rs}, \quad \sigma_B(r | x_2) = p_{rr} + p_{sr}.$$

Theorem 8.1 (Kuhn, 1953). *In a finite game with **perfect recall** — no player ever forgets information they previously knew, including their own past actions — every mixed strategy has an outcome-equivalent behavioural strategy and vice versa.*

Perfect recall holds in virtually every game of interest. When it fails, the two notions can differ.

Example 8.1 (The absent-minded driver)

Andrew drives home on a highway at night. Exiting at the first junction leads into a bad neighbourhood (payoff 0); exiting at the second takes him home (payoff 4); missing both leads to a long detour (payoff 1). Andrew is tired and cannot tell the two junctions apart: his single information set is $\{x_0, x_1\}$, so the same action must be chosen at both.

- **Pure and mixed strategies.** “Exit” gives 0, “continue” gives 1; any mixture yields at most 1.
- **Behavioural strategy.** Exit with probability p at each junction, independently:

$$\mathbb{E}[u] = p \cdot 0 + (1 - p)p \cdot 4 + (1 - p)^2 \cdot 1 = -3p^2 + 2p + 1,$$

maximised at $p^* = \frac{1}{3}$ with $\mathbb{E}[u] = \frac{4}{3} > 1$.

Randomising at each node beats every mixture over plans. A further paradox appears at the *action stage*: if Andrew believes he is at the first junction with probability q , the expected payoff becomes $-3qp^2 - qp + 3p + 1$, whose maximiser equals the planning-stage $p^* = \frac{1}{3}$ only when $q = 1$.

8.6. From extensive to normal form

Enumerating strategies turns any extensive-form game into a normal-form one. For the sequential Battle of the Sexes:

Ann \ Brian	rr	rs	sr	ss
R	2, 1	2, 1	0, 0	0, 0
S	0, 0	1, 2	0, 0	1, 2

The simultaneous version gives back the familiar 2×2 matrix. The correspondence is many-to-one: different trees can share a normal form, for instance when some terminal nodes carry identical payoffs, so that the order of moves does not matter.

i Note

Going back to the normal form seems to throw away the dynamics. Yet it is exactly what is needed for an inherently static concept like Nash equilibrium, which is the subject of the next chapter — together with the reason why it is not enough.

8.7. How large is the normal form?

A pure strategy specifies an action at **every** information set of the player, reached or not. If player i has information sets $h_1, \dots, h_K$ with $|A(h_k)|$ actions each, then

$$|S_i| = \prod_{k=1}^K |A(h_k)|,$$

which explodes with the depth of the tree. In a game where player 2 answers each of player 1's three moves with S or N , player 2 has $2^3 = 8$ strategies, even though only one answer is ever used.

Two simplifications are standard:

- **Reduced normal form.** Strategies that differ only at information sets made unreachable by the player's *own* earlier moves are payoff-equivalent and can be merged. A player who ends the game at the root need not say what they would do later. Many textbooks still list them, because subgame perfection *does* care about those plans.
- **Simultaneous moves inside a tree.** Moves made without observing each other are drawn in any order, with the later mover's nodes joined in one information set. The choice of who is drawn first is cosmetic.

The normal form is the right tool for finding *all* Nash equilibria, including those supported by unreasonable plans off the path of play. The extensive form is the right tool for discarding them (Chapter 9).

8.8. Exercises

Problems from Badia and Marchioro (BM), Chapters 3 and 4. Their subgame-perfect equilibria are found in Chapter 9.

Exercise 8.1

(BM 3.1) A student U chooses project quality $x \in \{0, 1, 2\}$; the professor P then accepts (S) or rejects (N). If accepted, each gets $m - e_j$ with mark $m = 18 + 6x$, $e_U = 4^x$ and $e_P = 15$; a rejection gives 0 to both. Draw the tree, write the normal form and find all pure NE.

Solution

Accepted payoffs (U, P): $x = 0$: (17, 3); $x = 1$: (20, 9); $x = 2$: (14, 15). P has one information set per x , hence $2^3 = 8$ strategies (an answer to 0, 1 and 2). The normal form is 3×8 ; marking best responses gives **7 pure NE**: (0, SNS), (0, SNN), (1, SSS), (1, SSN), (1, NSS), (1, NSN), (2, NNS). Several rely on P rejecting projects that are never submitted. (2, NNS), for instance, is the “strict professor” who accepts only top quality.

Exercise 8.2

(BM 3.2) Miss 1 and Mister 2 start with $u_1 = 1$, $u_2 = -1$. Miss 1 plays A (multiply both by 3) or B (multiply both by 2, then add 5 to u_2). Mister 2 observes and either Drops (D, game ends) or Continues (C), after which Miss 1 plays A or B once more. Draw the tree and count the pure strategies.

Solution

After A: (3, -3); after B: (2, 3). Continuing from (3, -3) leads to (9, -9) by A or (6, -1) by B; continuing from (2, 3) leads to (6, 9) by A or (4, 11) by B. Miss 1 moves at three nodes (the root and the two continuation nodes): $2^3 = 8$ strategies. Mister 2 moves at two: $2^2 = 4$ strategies. In reduced normal form, Miss 1's plans at the continuation node after the move she did not make can be merged, leaving $2 \times 2 = 4$ classes.

Exercise 8.3

(BM 3.4) Javier and Krista buy a house. In September J buys Atlas (A: J 10, K 5) or waits (X). In October K buys Bellevue (B: 15, 28) or Cicada (C: 5, 12), or waits (X). In November J must buy Daffodil (D: 18, 15), Ember (E: 8, 35) or Floral (F: 40, 10). Write the normal form and find all pure NE.

Solution

J has $2 \times 3 = 6$ strategies (AD, AE, AF, XD, XE, XF) and K has 3 (B, C, X). Rows AD, AE, AF give (10, 5) in every column. Against B, C, X: XD gives (15, 28), (5, 12), (18, 15); XE gives (15, 28), (5, 12), (8, 35); XF gives (15, 28), (5, 12), (40, 10). The pure NE are (AD, C), (AE, C), (AF, C), (XD, B) and (XF, B). In the first three Krista's choice of C is a (non-credible) threat that makes waiting unattractive to J.

Exercise 8.4

(BM 4.6) The Count chooses Sword or Pistol; then the Duke chooses First blood or Last blood. With swords the Count wins with 0.4. With pistols the Count wins with 0.7, the Duke with 0.2, and both are shot with 0.1. At first blood the loser survives dishonoured (-20); at last blood the loser dies (-100); being shot is -100; winning is +10. Compute the expected payoffs, the normal form and the NE.

8. Extensive-form games

Solution

Expected payoffs (Count, Duke): $SF (-8, -2)$, $SL (-56, -34)$, $PF (-7, -22)$, $PL (-23, -78)$. For instance, PF gives the Duke $0.2 \cdot 10 + 0.7(-20) + 0.1(-100) = -22$. (The printed solution has slightly different Duke payoffs; the conclusions are the same.) The Duke's strategies say what to answer to S and to P: FF, FL, LF, LL. The pure NE are (S, FL) , (P, FF) and (P, LF) . (S, FL) rests on the Duke threatening last blood if pistols are chosen.

Exercise 8.5

(BM 4.7) David picks a cake A, B or C; Emily, seeing it, shops at the Farmer's market (F) or Grocery (G). The cake succeeds with probability $p_{AF} = 0.5$, $p_{AG} = 0.2$, $p_{BF} = 0.8$, $p_{BG} = 0.4$, $p_{CF} = 0.7$, $p_{CG} = 0.9$. Emily gets 1 for a success; David gets 3, 2, 1 times that for A, B, C. Write the "flat" simultaneous version and the true sequential normal form, and compare.

Solution

Flat: $(A, F) = (1.5, 0.5)$, $(A, G) = (0.6, 0.2)$, $(B, F) = (1.6, 0.8)$, $(B, G) = (0.8, 0.4)$, $(C, F) = (0.7, 0.7)$, $(C, G) = (0.9, 0.9)$. A is strictly dominated by B; the flat game has NE (B, F) and (C, G) . Sequentially, Emily's strategies are pairs (answer to B, answer to C), ignoring A: the NE are (B, FF) , (B, FG) and (C, GG) . The extra NE of the sequential form are again sustained by off-path plans.

From the course material:

Exercise 8.6

Two players start with two peas each and alternate, A first. A move transfers to the opponent a positive number of peas never transferred before by anyone. A player who cannot move loses. Draw the tree and determine the winner.

Solution

Write states as (peas of A, peas of B; amounts used). A can open with 1 or 2.

- **A gives 2** $\rightarrow (0, 4; \{2\})$. B replies by giving 1 $\rightarrow (1, 3; \{1, 2\})$: A's only amount is 1, already used, so A loses.
- **A gives 1** $\rightarrow (1, 3; \{1\})$. B replies by giving 2 $\rightarrow (3, 1; \{1, 2\})$: A's only new feasible amount is 3 $\rightarrow (0, 4; \{1, 2, 3\})$; B gives 4 $\rightarrow (4, 0; \{1, 2, 3, 4\})$, and A cannot give a new amount 4, so A loses. (B must avoid giving 3 instead, which would let A win by giving 2.)

Every opening loses against correct play: **B wins**. This is backward induction on a game of perfect information (Chapter 9).

Exercise 8.7

In the sequential Battle of the Sexes, write Brian's behavioural strategy equivalent to the mixed strategy $p_{rr} = 0.1$, $p_{rs} = 0.5$, $p_{sr} = 0.2$, $p_{ss} = 0.2$.

Solution

$\sigma_B(r \mid x_1) = p_{rr} + p_{rs} = 0.6$ and $\sigma_B(r \mid x_2) = p_{rr} + p_{sr} = 0.3$. (Many mixed strategies map to the same behavioural strategy: only these two marginals matter for outcomes.)

8.9. Sources

Prof. Badia, *Game Theory* slides, deck Gth2008 (dynamic games, extensive form, information sets, strategies, behavioural strategies, perfect recall, absent-minded driver, extensive normal form); exercise sheet gt20d. Tadelis, *Game Theory: An Introduction*, Chapter 7. Original references: H. Kuhn, *Extensive games and the problem of information* (1953); M. Piccione and A. Rubinstein, *Games Econ. Behav.* **20**, 3 (1997). Exercises: L. Badia and T. Marchioro, *Game Theory: A Handbook of Problems and Exercises*, Esculapio (2022), Chapters 3–4.

9. Credibility and subgame perfection

Nash equilibrium can be applied to dynamic games through their normal form, but it misses something essential: it treats a strategy as a commitment made before the game starts, so it accepts threats that the threatening player would never carry out when the moment comes. This chapter identifies the problem, solves perfect-information games by backward induction, and introduces Selten's refinement — **subgame-perfect equilibrium** — which rules out non-credible threats. Stackelberg leader–follower games close the chapter.

9.1. Nash equilibria of dynamic games

The normal form of the sequential Battle of the Sexes (Chapter 8) has three pure Nash equilibria:

- (R, rr) : Ann goes to R; Brian “always goes to R”;
- (R, rs) : Ann goes to R; Brian “does what Ann does”;
- (S, ss) : Ann goes to S; Brian “always goes to S”.

These strategies are chosen as if Brian moved first. (R, rr) and (R, rs) produce the same outcome: they coincide **on** the equilibrium path but differ **off** it. Out of equilibrium, rs is smarter; rr prescribes the irrational reply r to S — which never comes into play.

Definition 9.1 (Equilibrium path). Given a Nash equilibrium in behavioural strategies, its **equilibrium path** is the set of nodes reached with positive probability.

Why specify behaviour off the path at all? Because equilibria are sustained by beliefs about what others would do *both on and off* the path. (S, ss) is a NE because Ann's deviation to R is “punished”: Brian threatens to go to S anyway. But is the threat credible? Once Ann is at the romance movie, Brian's best reply is r .

Example 9.1 (*Ultimatum game*)

A proposer P splits 10 candies: D (9 for P, 1 for R), F (5–5) or G (2–8). The responder R accepts (Y) or refuses (N), in which case both get nothing. A strategy of R is a triple (x_D, x_F, x_G) , e.g. “accept D and F but not G ”. Every pair “P offers $x/10 - x$ ” with “R refuses whenever P's share exceeds x ” is a Nash equilibrium — the responder's threat makes many splits stable.

9.2. Backward induction

In a dynamic game of **perfect information** every information set is a singleton. Take two players: player 1 chooses $a_1 \in A_1$; player 2 observes a_1 and chooses $a_2 \in A_2(a_1)$; payoffs are $u_1(a_1, a_2)$, $u_2(a_1, a_2)$. When it is their turn, player 2 knows exactly where they are and solves

$$R_2(a_1) = \arg \max_{a_2 \in A_2(a_1)} u_2(a_1, a_2).$$

Thanks to complete information, player 1 anticipates this reaction and solves $\max_{a_1} u_1(a_1, R_2(a_1))$. The resulting outcome $(a_1^*, R_2(a_1^*))$ is a Nash equilibrium in pure strategies.

Theorem 9.1 (Zermelo / Kuhn). *Every finite game of perfect information has a backward-induction solution, which is a pure-strategy Nash equilibrium and is sequentially rational. If all terminal payoffs are different for each player, it is unique.*

A consequence: every finite game of perfect information — tic-tac-toe, chess, go — has an optimal way to be played, and its value (win, draw, loss) is determined.

Example 9.2 (*A trust game*)

A chooses L (ending the game with payoffs $(2, 0)$) or R . Then B chooses L' (payoffs $(1, 1)$) or R' . Then A chooses L'' (payoffs $(5, 0)$) or R'' (payoffs $(3, 2)$).

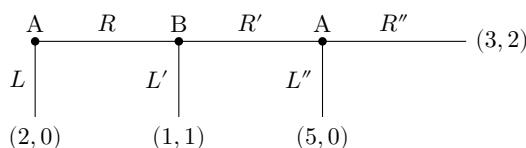


Figure: the trust game; payoffs are (A, B) .

At the last node A prefers L'' ($5 > 3$). Anticipating this, B prefers L' ($1 > 0$). Anticipating that, A plays L ($2 > 1$). The outcome $(2, 0)$ is Pareto dominated by $(3, 2)$: rational players do not trust each other. A could promise to play R'' , but nothing makes the promise binding.

Imperfect information, two stages. If players 1 and 2 move simultaneously, then players 3 and 4 observe (a_1, a_2) and move simultaneously, apply the same logic: for each (a_1, a_2) , let players 3 and 4 play a NE $(a_3^*(a_1, a_2), a_4^*(a_1, a_2))$ of the second-stage game; players 1 and 2 then play a NE of the first-stage game with payoffs $u_j(a_1, a_2, a_3^*(\cdot), a_4^*(\cdot))$.

9.3. Subgame-perfect equilibrium

Definition 9.2 (Subgame). A (proper) **subgame** is a part of the tree that starts at a single node and contains all its successors, such that no information set is cut: if x belongs to the subgame and $x' \in h_i(x)$, then x' belongs to the subgame too. The whole game is a subgame of itself.

A node inside a non-singleton information set cannot start a subgame — the player there does not know they are there.

Definition 9.3 (Subgame-perfect equilibrium (Selten, 1965)). A strategy profile is a **subgame-perfect (Nash) equilibrium** (SPE) if it induces a Nash equilibrium in every subgame.

An SPE is a NE that passes an additional test, so SPE is a **refinement** of NE. Every finite extensive-form game has an SPE (apply Nash's theorem to the smallest subgames, replace them with their equilibrium payoffs, and proceed backwards). For finite games of perfect information, the SPE are exactly the backward-induction solutions.

Definition 9.4 (Credibility). If player 1 knows that action a_1 triggers the response $R_2(a_1)$, then a strategy of player 2 of the form “if a_1 then $a_2 \neq R_2(a_1)$ ” is a **non-credible threat**. SPE eliminates them.

Example 9.3 (*Credibility in the Battle of the Sexes*)

Brian's strategy ss threatens to go to S even if Ann chooses R . In the subgame after R , playing s is not a best response; so (S, ss) is not an SPE. The unique SPE is (R, rs) . An extreme version: there is no sci-fi movie, S means "stay at home", and after R Brian can comply (r) or harm himself (k , payoffs $(-10, -99)$). Threatening k might scare Ann into staying home, but carrying it out would be irrational: a non-credible threat.

Example 9.4 (*Ultimatum game, solved*)

Whatever the offer, accepting gives the responder something rather than nothing, so in every subgame R accepts. Anticipating this, P offers D . Among the many NE, the unique SPE is "P offers D ; R accepts everything". Refusing a low offer is a non-credible threat.

Warning

Experimental ultimatum games do not look like this: humans routinely refuse "insulting" offers. The SPE prediction is correct for the stated payoffs; the discrepancy says that real payoffs include fairness or spite, not that backward induction is wrong.

9.4. Stackelberg games

Von Stackelberg (1934) modelled an incumbent (leader) and an outsider (follower). Take a static bi-matrix game and let player 1 move first, observed by player 2. The backward-induction outcome is the **Stackelberg equilibrium**.

- In the Battle of the Sexes with Ann as leader, the Stackelberg equilibrium is (R, R) ; Brian gets his minimax 1.
- In Odds & Evens moving first is a disadvantage, unless the leader commits to a mixture: the Stackelberg outcome is both players mixing 50/50, each getting the minimax 0.

Example 9.5 (*Leader advantage*)

Carl \ Joe	F	G	H
R	2, 2	3, 1	0, 0
S	1, 6	5, 4	6, 4
T	0, 1	4, 3	6, 2

(R, F) is the pure NE, with payoffs $(2, 2)$. **Carl as leader:** Joe's best responses are F to R and to S , G to T ; Carl's payoffs are then 2, 1, 4, so he leads with T : outcome (T, G) , payoffs $(4, 3)$. Joe gets 3, above his minimax 2. **Joe as leader:** Carl's best replies are R to F , S to G , and $\{S, T\}$ to H . Ties need a convention; with a *generous follower* (who breaks ties in the leader's favour) and a *generous leader* (who breaks ties in the follower's favour), Joe leads with H and Carl answers S : outcome (S, H) , payoffs $(6, 4)$. Carl, the follower, gets 6, far above his minimax 2.

Proposition 9.2 (Leader and follower payoffs). *With ties broken in the leader's favour, the leader's Stackelberg payoff is at least their payoff in any pure Nash equilibrium of the static game (they could*

always lead with their NE strategy, to which the follower's NE strategy is a best reply). The follower's payoff is at least their minimax.

The follower is not necessarily worse off in general games. In **zero-sum** games, however, the leader's gain is the follower's loss — strange at first sight, since the follower has *more* information. In single-agent optimisation knowledge is power; in games **ignorance can be bliss**: the leader anticipates the follower's informed reaction and restricts the follower's useful options. This “first-mover advantage” returns in the Stackelberg duopoly of Chapter 11.

9.5. Centipede games and the limits of backward induction

In a **centipede game** two players alternately choose to *stop*, taking the larger share of a growing pot, or *continue*, letting the pot grow and passing the move. Each stop is arranged so that the mover gains more by stopping now than by being stopped by the opponent at the next node. Backward induction then unravels from the last node: the last mover stops, so the one before stops, and so on. The unique SPE stops at the root, although continuing would make both far better off.

Experiments find that most subjects continue for several rounds. The prediction requires *common knowledge of rationality at every node*, including nodes that rationality says are never reached. A player who observes a continuation (an irrational move, by hypothesis) may reasonably doubt that the opponent will be rational later. Backward induction is most convincing in short games and for automated agents. Several exercises (house hunting, project hand-over, parties, joint projects) are centipede-like: whether the game unravels depends on whether stopping beats waiting at each node.

9.6. Commitment and the order of moves

Turning a simultaneous game into a sequential one changes what is credible.

- The **leader** can always do at least as well as in some Nash equilibrium of the simultaneous game: by playing that equilibrium strategy, the follower's best response reproduces it. Often the leader does strictly better (first-mover advantage, as in Stackelberg).
- Moving first can also **hurt**: in Matching Pennies, being observed is fatal. Information about the leader's move is valuable to the follower whenever the game has a competitive flavour.
- **Commitment** is the power to make a threat credible by removing your own options: burning bridges, contracts, reputations, automated policies. Schelling's paradox: limiting your own choices can raise your payoff, because it changes the opponent's best response.

In a game with ties at terminal nodes, backward induction needs a **tie-breaking** rule. The usual conventions are a *generous follower*, who breaks ties in the leader's favour, and a *generous leader*. Without such a rule the SPE need not be unique.

9.7. Exercises

Problems from Badia and Marchioro (BM), Chapters 3 and 4.

Exercise 9.1

(BM 3.1, continued) Find all SPE of the student–professor game. Which NE involve non-credible threats?

Solution

Every answer S gives P a positive payoff ($3, 9, 15 > 0$), so in every subgame P accepts: P plays SSS . Anticipating this, U compares 17, 20, 14 and chooses $x = 1$. The unique SPE is $(1, SSS)$. The NE $(1, SSN)$, $(1, NSS)$, $(1, NSN)$ have the same outcome but irrational plans off the path. $(0, SNS)$, $(0, SNN)$ and $(2, NNS)$ change the outcome through the non-credible threat of rejecting project 1 (the “strict professor”).

Exercise 9.2

(BM 3.2, continued) Solve the Miss 1 / Mister 2 game by backward induction and give the SPE strategies. Is $((A, A, A), (D, D))$ a NE?

Solution

At both final nodes Miss 1 plays A ($9 > 6$ and $6 > 4$). Mister 2 then compares: after A, D gives -3 and C gives -9 , so he drops; after B, D gives 3 and C gives 9, so he continues. Miss 1 compares 3 (A) with 6 (B). SPE: Miss 1 ($B; A, A$) (root, after-A node, after-B node), Mister 2 (D, C), with outcome $(6, 9)$. $((A, A, A), (D, D))$ is a NE (payoffs $(3, -3)$; Miss 1 switching to B gets 2 under D), but not an SPE: dropping after B is a non-credible threat.

Exercise 9.3

(BM 3.3) Lou asks Karen to go shopping on day 1. If she says No, Lou either Gives up or Requests again on day 2, and likewise on day 3, after which the game ends. Shopping on day d gives Karen d and Lou $5 - 2d$; not shopping gives 0. Find the SPE and a NE that is not subgame perfect.

Solution

Day 3: Karen says Yes ($3 > 0$), giving Lou -1 . Day 2: Lou Gives up ($0 > -1$). Karen on day 2 says Yes ($2 > 0$). Day 1: Karen compares Yes (1) with No, after which Lou Requests ($1 > 0$) and she gets 2: she says No. The SPE is Karen (N, Y, Y), Lou (R, G), outcome day 2 with $(2, 1)$. Non-SPE NE: Lou (G, G) and Karen ($Y, \cdot, \cdot$). The threat of giving up immediately makes Karen accept on day 1, but Lou would not carry it out.

Exercise 9.4

(BM 3.4, continued) Find the SPE of the house-buying game.

Solution

November: J buys F (40). October: K compares B (28), C (12), and waiting (10, since J then buys F): she buys B. September: J compares A (10) with waiting (15). SPE (XF, B) , outcome Bellevue, $(15, 28)$. (XD, B) has the same outcome but J's plan to buy D in November is not optimal: it is a NE, not an SPE.

Exercise 9.5

(BM 3.5) Yvonne (with a car) and Zoe (by bus) choose Beach or Disco simultaneously. Yvonne gets 4 at the beach and 1 at the disco; Zoe 1 and 7; each gets $+5$ if together. If they are apart, Yvonne alone may Change location (costing her 3) or Stay. Find the SPE outcome and Yvonne's strategy.

Solution

After (B, D) : Change gives Yvonne $1+5-3 = 3$, Stay gives 4, so she stays, $(4, 7)$. After (D, B) : Change gives $4+5-3 = 6$, Stay gives 1, so she changes, $(6, 6)$. The reduced simultaneous game is $(B, B) = (9, 6)$, $(B, D) = (4, 7)$, $(D, B) = (6, 6)$, $(D, D) = (6, 12)$. D is dominant for Zoe, and Yvonne's best response is D . Outcome (D, D) with $(6, 12)$; Yvonne's SPE strategy is $(D; S \text{ after } (B, D), C \text{ after } (D, B))$. Yvonne's flexibility hurts her: because she can always fix a mismatch, Zoe never needs to compromise.

Exercise 9.6

(BM 3.6) Wife (7am and 7pm) and Husband (2pm) can each, at their turn, fix Food (benefit 30 to both, cost 10 to the doer), Clean (benefit 50, cost 20) or do Nothing (+10 to themselves). A chore cannot be repeated. The Wife's 7am chores have 10% larger cost and benefit. Find the SPE.

Solution

Backward induction over the three-level tree gives the path: Wife does **Nothing** at 7am, Husband fixes **Food**, Wife **Cleans** at 7pm, with payoffs $(10 + 30 + 50 - 20, 30 + 50 - 10) = (70, 70)$. At 7pm the Wife does whichever chore remains if its benefit net of cost beats 10 (clean: $30 > 10$; food: $20 > 10$). Anticipating this, the Husband prefers food to cleaning, and the Wife prefers to wait rather than pay the premium at 7am. The full SPE also specifies the Husband's and the Wife's moves in the 8 unreached subgames.

Exercise 9.7

(BM 3.7) Anita and Ben simultaneously rush to the closet (5 candies) or the dining room (10 candies). In different rooms each keeps their candies; in the same room Anita proposes a split, giving Ben at least one candy, and Ben accepts or cries (both get 0). Reduce the game by backward induction and find the SPE and a non-subgame-perfect NE.

Solution

In each ultimatum subgame Ben accepts anything positive, so Anita offers 1: $(4, 1)$ in the closet and $(9, 1)$ in the dining room. Reduced game: $(C, C) = (4, 1)$, $(C, D) = (5, 10)$, $(D, C) = (10, 5)$, $(D, D) = (9, 1)$. D is dominant for Anita and Ben answers C . SPE: Anita plays D and offers 1 in either room; Ben plays C and accepts every offer. Non-SPE NE: Anita plays C , Ben plays D and, in either bargaining subgame, accepts only splits leaving Anita a single candy. Outcome (C, D) with $(5, 10)$. Anita does not deviate to D , since she would then get at most 1. It is a NE because the threat is never tested, but crying when offered a positive amount is not credible.

Exercise 9.8

(BM 3.8) Jeremy (odd days) and Kayla (even days) work on a project worth 2^t on day t . At the end of each day the worker Submits, getting 2^t (the other gets 60%), or Hands over. After day 10 the project is worth 0. Find the SPE and a non-subgame-perfect NE.

Solution

On day 10 Kayla submits ($1024 > 0$). On day 9 Jeremy compares 512 (submit) with $0.6 \times 1024 = 614.4$: he hands over. Inductively, at every $t < 10$ handing over gives $0.6 \cdot 2^{t+1} = 1.2 \cdot 2^t > 2^t$. SPE: always hand over before day 10; Kayla submits on day 10. Payoffs: Kayla 1024, Jeremy 614.4. This centipede does *not* unravel, because the pot grows fast enough. A non-SPE NE: Jeremy submits on day 9, and

Kayla's plan is to hand over even on day 10 (letting the deadline expire). Jeremy then rationally stops at 9; Kayla's irrational plan is never tested.

Exercise 9.9

(BM 3.9) In weeks $0, 1, \dots, 5$ the Sorority (even weeks) and Fraternity (odd weeks) alternately decide to throw a Party or stop (N). A party costs the organiser 10 and gives both sections fun $25 \cdot 0.6^j$ in week j . Find the SPE. What if there is no deadline?

Solution

Cumulative payoffs (S, F) if the game stops after k parties: $k = 0$: (0, 0); 1: (15, 25); 2: (30, 30); 3: (29, 39); 4: (34.4, 34.4); 5: (27.64, 37.64); 6: (29.58, 29.58). Week 5: F stops ($37.64 > 29.58$). Week 4: S stops ($34.4 > 27.64$). Week 3: F stops ($39 > 34.4$). Week 2: S stops ($30 > 29$). Week 1: F parties ($30 > 25$). Week 0: S parties ($30 > 0$). SPE: S plays (P, N, N) and F plays (P, N, N) at their three nodes. Two parties are thrown, S stops in week 2, and the payoffs are (30, 30). (The printed solution misreads its table and ends in week 4.) Without a deadline the same stationary profile, stopping from week 2 on, remains an SPE. From week 2 the organiser's fun $25 \cdot 0.6^j \leq 9$ is below the cost 10, so if the other will stop next, organising never pays (one-shot deviation principle).

Exercise 9.10

(BM 3.11) Pirates of ranks $1, \dots, n$: pirate 1 may take a pearl; whenever pirate k holds it, pirate $k + 1$ may kill him and take it. Each prefers (alive with pearl) $\succ$ (alive without) $\succ$ (dead). Solve for $n = 5$ and $n = 8$.

Solution

The captain n always takes the pearl (no one above him). So pirate $n - 1$ never takes it (he would be killed), so pirate $n - 2$ can take it safely, and so on: pirate j takes the pearl iff $n - j$ is even. For $n = 5$ (odd), pirate 1 takes the pearl and pirate 2 looks away. For $n = 8$ (even), pirate 1 leaves the pearl. The SPE specifies the rule for every pirate, though only one or two ever move. The parity argument is typical of backward induction in long games.

Exercise 9.11

(BM 4.6, continued) Which NE of the duel game are subgame perfect?

Solution

After S the Duke prefers F ($-2 > -34$); after P he prefers F ($-22 > -78$). So the Duke plays FF, and the Count compares -8 (S) with -7 (P). The unique SPE is (P, FF). (S, FL) relies on the threat of a duel to the death with pistols, which the Duke would not carry out. (P, LF) has the right outcome but an irrational plan after S.

Exercise 9.12

(BM 4.7, continued) Which NE of the cake game are subgame perfect?

Solution

Emily answers each cake with the shop that maximises its success probability: F after A and after B, G after C. David then compares 1.5 (A), 1.6 (B), 0.9 (C). The SPE is David B and Emily (F, F, G), outcome (1.6, 0.8). Of the sequential NE (B, FF) , (B, FG) , (C, GG) (strategies for B and C), only (B, FG) is subgame perfect.

From the course material:

Exercise 9.13

In the entry game, an entrant chooses Out (payoffs $(0, 2)$) or Enter; after entry the incumbent chooses Accept $((1, 1))$ or Fight $((-1, -1))$. Find all pure NE and the SPE.

Solution

Normal form: (Out, Fight) gives $(0, 2)$ and is a NE (entering against Fight gives -1 ; the incumbent is indifferent since its move is off-path). (Enter, Accept) is a NE with $(1, 1)$. Backward induction: after entry Accept beats Fight $(1 > -1)$, so the entrant enters. The unique SPE is (Enter, Accept); (Out, Fight) rests on a non-credible threat.

Exercise 9.14

Two investors each put 5000 € in a fund that pays x € each after 4 years if untouched. At the end of years 1, 2, 3 each may independently withdraw, getting $6000 + 500t$ € for themselves (the other gets nothing); simultaneous withdrawal splits that amount; any withdrawal ends the game. Find the SPE for $x = 7200$ and $x = 8000$.

Solution

Solve backwards. **Year 3:** withdrawing gives 7500 (alone) or 3750 (both); keeping gives x if the other keeps, 0 if the other withdraws. For $x = 7200$, Withdraw strictly dominates $(7500 > 7200, 3750 > 0)$: both withdraw, 3750 each. For $x = 8000$ the year-3 stage is a coordination game with NE (Keep, Keep) $\rightarrow 8000$ and (Withdraw, Withdraw) $\rightarrow 3750$. **Year 2:** withdrawing alone gives 7000. With continuation value 3750 (case $x = 7200$, or $x = 8000$ with the bad year-3 NE) Withdraw dominates; with continuation 8000, Keep–Keep is a NE (and so is Withdraw–Withdraw at 3500). **Year 1** repeats the logic with 6500. Hence for $x = 7200$ the unique SPE has both withdrawing at year 1 (3250 each), while for $x = 8000$ there are SPE where both keep until the end (payoff 8000) as well as SPE where both withdraw at the first opportunity.

9.8. Sources

Prof. Badia, *Game Theory* slides, deck Gth2009 (Nash equilibria of dynamic games, equilibrium path, backward induction, subgames, SPE, credibility) and the Stackelberg part of Gth2010; exercise sheet gt20d. Tadelis, *Game Theory: An Introduction*, Chapter 8. Original reference: R. Selten, *Spieltheoretische Behandlung eines Oligopolmodells mit Nachfrageträgheit*, Z. Ges. Staatswiss. **121**, 301 (1965). Exercises: L. Badia and T. Marchioro, *Game Theory: A Handbook of Problems and Exercises*, Esculapio (2022), Chapters 3–4.

10. Multistage and repeated games

Many interactions consist of several rounds, each producing a partial payoff: tournaments, rounds of cards, partial exams, network sessions. This chapter studies **multistage games** (a sequence of static games played by the same players) and their most important special case, **repeated games** (the same stage game over and over). The central message is that the future can discipline the present: threats of punishment and promises of reward make cooperative behaviour — impossible in a one-shot Prisoner's Dilemma — part of a subgame-perfect equilibrium.

10.1. Why exponential discounting

Aggregating payoffs across periods requires a discounting rule. A single-person example shows why it must be exponential.

A player has a budget $K = 1$ to spend over three periods with per-period utility $u(x) = \log(1 + x)$ and total value $v = u(x_1) + \delta u(x_2) + \delta^2 u(x_3)$. The first-order conditions $\frac{1}{1+x_1} = \frac{\delta}{1+x_2} = \frac{\delta^2}{1+x_3}$ give

$$x_1 = \frac{3 - \delta - \delta^2}{1 + \delta + \delta^2}, \quad x_2 = \frac{-1 + 3\delta - \delta^2}{1 + \delta + \delta^2}, \quad x_3 = \frac{-1 - \delta + 3\delta^2}{1 + \delta + \delta^2},$$

an equal split for $\delta = 1$ and $(0.6393, 0.3115, 0.0492)$ for $\delta = 0.8$ (an interior solution requires $3\delta^2 - \delta - 1 \geq 0$, i.e. $\delta \gtrsim 0.77$). At period 2, with $1 - x_1$ left, the player re-optimises $u(x_2) + \delta u(x_3)$ and finds $x_2 = (2 - x_1 - \delta)/(1 + \delta)$: for $x_1 = 0.6393$ this is again 0.3115. Exponential discounting is **time consistent** — the plan made today is still optimal tomorrow.

Now try **quasi-hyperbolic** discounting, $v = u(x_1) + \delta u(x_2) + \delta u(x_3)$ (the future discounted by a single factor). The plan is $x_1 = (3 - 2\delta)/(1 + 2\delta)$, $x_2 = x_3 = (2\delta - 1)/(1 + 2\delta)$, i.e. $(0.5385, 0.2308, 0.2308)$ for $\delta = 0.8$. But at period 2 the player re-optimises with $u(x_2) + \delta u(x_3)$ and chooses $(0.3675, 0.0940)$: the plan is **inconsistent**. A fully rational player anticipates this struggle between the present self and the future self; solving it by backward induction leads to $x_1 = 1$ — overconsumption to prevent future misuse. From now on, discounting is always exponential.

10.2. Multistage games

Definition 10.1 (Multistage game). A **multistage game** is a finite sequence of T normal-form **stage games** $G_1, \dots, G_T$, played by the same players, with the outcome of each stage observed before the next. Stage payoffs $u_i^{(t)}$ are aggregated with a common, commonly known discount factor:

$$u_i = \sum_{t=1}^T \delta^{t-1} u_i^{(t)}.$$

Each stage is a simultaneous-move game, so the whole is a dynamic game of complete but imperfect information. A strategy specifies the first-stage action and the action at each later stage *as a function of the history* of outcomes.

Example 10.1 (*Prisoner–Revenge*)

Al and Bob first play a Prisoner’s Dilemma; out of jail, each may join a **G**ang or stay a **L**oner:

Al \ Bob	m	f
M	4, 4	−1, 5
F	5, −1	1, 1

Al \ Bob	g	
L	0, 0	−4, −1
G	−1, −4	−3, −3

The second stage has two NE: the “friendly” (L, ℓ) and the “gang” (G, g). Each player already has $2 \times 2^4 = 32$ pure strategies.

Theorem 10.1 (Equilibria of multistage games). 1. If s^{t*} is a NE of stage game G_t for each t , then playing $s^{1*}, \dots, s^{T*}$ regardless of history is an SPE.
 2. In any NE (even non-subgame-perfect) of a multistage game, a NE of G_T is played in the last stage.
 3. If every stage game has a unique NE, the multistage game has a unique SPE: play those NE regardless of history.

(2) holds because at the last stage there is no future to influence, so players simply best-respond. Statement (1) removes any strategic link between stages: in Prisoner–Revenge, “ F then L whatever happened” and “ F then G whatever happened” are SPE. But if the last stage has **multiple** equilibria, non-equilibrium play can be sustained earlier.

Proposition 10.2 (Strategic connection). In Prisoner–Revenge, let both players use “play M in stage 1; then play Loner if the first outcome was (M, m) , otherwise Gang”. This is an SPE iff $\delta \geq \frac{1}{3}$.

Proof. A NE is played in every second-stage subgame, so it suffices to check the first stage. Against this strategy, M yields $4 + 0 \cdot \delta$ and F yields $5 - 3\delta$; M is a best response iff $\delta \geq \frac{1}{3}$. $\square$

The construction is **carrot and stick**: play the desired (non-equilibrium) action now; reward compliance with the good equilibrium (carrot); punish deviation with the bad one (stick). δ must be large enough for the difference to matter — for $\delta = 0$ the stick is not a credible deterrent. The same trick can support *any* first-stage profile: (F, m) is sustainable if Bob prefers $-1 + 0 \cdot \delta$ to $1 - 3\delta$, i.e. $\delta \geq \frac{2}{3}$.

Theorem 10.3 (One-stage deviation principle). A strategy is optimal if no alternative strategy improves the payoff at any information set; it is one-stage unimprovable if no change of the action at a single information set does. In finite multistage games (and in infinite ones with discounting), a **one-stage unimprovable strategy is optimal**.

Proof idea. Suppose s_i is one-stage unimprovable but some s'_i does better. Take the profitable deviation with the fewest differences and consider its *last* deviation: in the subgame starting there, s'_i differs from s_i at a single information set and improves the payoff — contradiction. $\square$

To verify an SPE, it is enough to check one-shot deviations — exactly what backward induction does.

10.3. Finitely repeated games

Definition 10.2 (Repeated game). $G(T, \delta)$ is the multistage game in which the same stage game G is played T times, with discount δ . For $T = \infty$ we need $\delta < 1$; δ may also be read as the probability that the game continues one more round.

Corollary 10.4 (Unique stage equilibrium). *If G has a unique NE s^* , then $G(T, \delta)$ with $T < \infty$ has a unique SPE: play s^* at every stage.*

In the twice-repeated Prisoner's Dilemma (with 4, 0, 5, 1 and $\delta = 1$) the second stage always ends at (F, F) , so the first stage is a Prisoner's Dilemma with every payoff raised by 1 — and (F, F) is played twice. Repetitions of games with a unique equilibrium are uninteresting.

Multiple stage equilibria change everything. Add a third strategy:

	M	F	H
M	4, 4	0, 5	0, 0
F	5, 0	1, 1	0, 0
H	0, 0	0, 0	3, 3

Now (F, F) and (H, H) are both NE. Consider “play M ; then H if the first outcome was (M, M) , otherwise F ”. The first stage becomes the game with (M, M) raised by 3 and everything else by 1, in which (M, M) , with payoffs $(7, 7)$, is a NE ($7 > 6$). Cooperation is supported in the first stage even though (M, M) is not an equilibrium of G . Note that this needs no communication — only shared anticipation.

Remark (Punishing the punisher). Punishing with (F, F) instead of (H, H) hurts the punisher too, which invites renegotiation (“let’s forget it and play (H, H) ”). The construction is more robust with punishments that *benefit* the punisher: add strategies P, Q with (P, P) giving $(4, \frac{1}{2})$ and (Q, Q) giving $(\frac{1}{2}, 4)$. Then a deviation by Bob is punished by (P, P) — good for Al — and vice versa. Cooperation is easier when punishments exist, and better still when several punishment options are available.

In finitely repeated games cooperation is therefore limited: the last stage is always played selfishly, and cooperative play requires multiple stage equilibria.

10.4. Infinitely repeated games

With $T = \infty$ there is no last stage and backward induction cannot start. Surprisingly, this leads to a *stronger* conclusion: no external punishment equilibrium is needed.

Definition 10.3 (Grim trigger). In the Prisoner's Dilemma, the **grim-trigger** strategy plays M at stage 1 and at every stage $t > 1$ plays M if all previous outcomes were (M, M) ; otherwise it plays F forever.

Proposition 10.5 (Cooperation in the infinitely repeated Prisoner's Dilemma). *With stage payoffs $(M, M) = (4, 4)$, $(M, F) = (0, 5)$, $(F, F) = (1, 1)$, both players using grim trigger is an SPE of $G(\infty, \delta)$ iff $\delta \geq \frac{1}{4}$.*

Proof. (NE) Suppose Al uses grim trigger. If Bob ever deviates, Al plays F forever, so Bob's best continuation is F forever too. Deviating now yields

$$V_{\text{dev}} = 5 + \delta + \delta^2 + \dots = 5 + \frac{\delta}{1 - \delta},$$

while cooperating forever yields $V_{\text{coop}} = 4/(1 - \delta)$. $V_{\text{coop}} \geq V_{\text{dev}}$ iff $\delta \geq \frac{1}{4}$. (SPE) Subgames come in two classes: after a history of (M, M) only, the subgame is identical to the whole game, where grim trigger is a NE; after any deviation, grim trigger prescribes (F, F) forever, which is a NE of that subgame. $\square$

The argument generalises.

Definition 10.4 (Feasible and individually rational payoffs). A payoff vector is **feasible** if it is a convex combination of pure-strategy payoff vectors of G (achievable by alternating outcomes, or with a public randomisation device).

Theorem 10.6 (Friedman's folk theorem). Let G be a finite static game of complete information, $(e_1, \dots, e_n)$ the payoffs of a NE of G , and $(x_1, \dots, x_n)$ feasible payoffs with $x_j > e_j$ for all j . Then for δ close enough to 1, $G(\infty, \delta)$ has an SPE with average payoffs $(x_1, \dots, x_n)$.

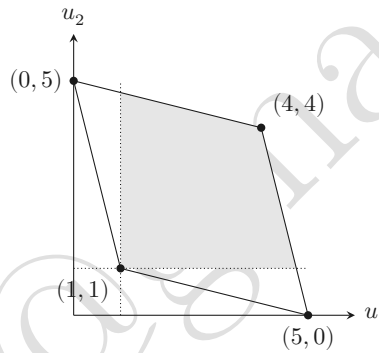


Figure: feasible payoffs of the Prisoner's Dilemma (the quadrilateral) and the payoffs above the NE $(1, 1)$ that the folk theorem sustains (shaded).

Developments of the theorem:

- **Minimax punishment.** In the Prisoner's Dilemma the NE payoff coincides with the security payoff 1, but in general $\maximin \leq u(\text{NE})$, and harsher (still credible) punishments can push the threshold down to the minimax payoffs $(r_1, \dots, r_n)$.
- **Small δ .** A lower discount factor weakens punishments; in some games (not the Prisoner's Dilemma) punishments harsher than reverting to the NE are available — the Cournot carrot-and-stick of Chapter 11.

10.5. Tit for tat and reputation

Grim trigger holds a grudge forever. With two actions (cooperate / defect), **tit for tat** (TFT) plays at stage t whatever the opponent played at $t - 1$: it punishes immediately but forgives after one round (eye for an eye, live-and-let-live, reciprocal altruism in biology). TFT was the winner of Axelrod's famous tournaments, but it is not subgame perfect: after a single unilateral deviation two TFT players alternate defection forever (a "death spiral"), which is not a NE of that subgame. *Tit for two tats* forgives the first defection and punishes the second; it avoids death spirals but is exploited by aggressive strategies.

Example 10.2 (*Trust and certified reputation*)

In a one-shot trust game A either does **Not** trust (payoffs (0, 0)) or **Trusts**, after which B **Collaborates** (1, 1) or **Defects** (−1, 2). Backward induction predicts no trust. In the repeated game, A can use a grim trigger (“trust as long as B has always collaborated”), and B’s best response is to collaborate if $\frac{1}{1-\delta} \geq 2$, i.e. $\delta \geq \frac{1}{2}$.

To strengthen B’s incentives, introduce a **guarantor** G who holds an insurance of 2 from B, keeps it if B defects, and otherwise returns it minus a fee of 0.1. With G, defecting no longer pays and trust is the SPE path. Why would G not simply keep the insurance? One-shot, keeping it is dominant. In the repeated game, G also has a reputation to protect: keeping the insurance yields 2 once, honest service yields $0.1/(1-\delta)$, which is better iff $\delta \geq 0.95$.

10.6. Beyond Nash reversion

Friedman’s theorem punishes deviators by reverting to a stage NE forever. Two refinements widen and sharpen the picture.

Definition 10.5 (Minmax payoff). Player i ’s **minmax** (pure-strategy) payoff is $\underline{v}_i = \min_{s_{-i}} \max_{s_i} u_i(s_i, s_{-i})$: the lowest payoff the others can hold i to, if i best-responds. A payoff vector is **individually rational** if $x_i > \underline{v}_i$ for all i .

No equilibrium of the repeated game can give a player less than $\underline{v}_i$ on average: they could always best-respond at every stage. Conversely:

Theorem 10.7 (Folk theorem (Fudenberg and Maskin, 1986)). *In an infinitely repeated game (satisfying a mild full-dimensionality condition), every feasible and individually rational payoff vector is the average payoff of some SPE, for all δ close enough to 1.*

The gain over Friedman lies between the minmax and the NE payoffs. Punishments harsher than the stage NE are possible, provided they are temporary and the punishers are themselves rewarded for carrying them out (“stick and carrot”, Abreu). With a stronger stick, a lower δ suffices.

Checking an equilibrium. By the one-stage deviation principle, a profile is an SPE iff no player gains by deviating once, in any history, and then following the profile. Two kinds of history must be checked:

- **on the path:** cooperating gives $R/(1-\delta)$; the best one-shot deviation gives T now plus the discounted punishment phase;
- **off the path:** the punishment itself must be a best response. Grim trigger passes, because it prescribes a stage NE. Tit for tat fails for some δ : after a single deviation it prescribes alternating (C, D) , (D, C) , ..., and returning to cooperation may pay more. So tit for tat is a NE but generally not an SPE.

Alternating outcomes. Payoffs outside the pure-outcome set (a fair split of an asymmetric outcome) are reached by alternation. With $\delta < 1$, the player who is favoured first gets slightly more: $(a + \delta b)/(1 - \delta^2)$ versus $(b + \delta a)/(1 - \delta^2)$.

10.7. Exercises

Problems from Badia and Marchioro (BM), Chapter 5. Several printed solutions contain slips; the corrected values are given here.

Exercise 10.1

(BM 5.1) A guitarist and a pianist play two songs; for each song each Sings (S) or just Plays (J). $(S, S) = (5, 5)$; $(S, J) = (1, 6)$; $(J, S) = (7, 1)$; $(J, J) = (0, 0)$ (guitarist first). Payoffs add over the two songs. (a) Find the stage NE. (b) Find the SPE that repeats the same stage outcome and gives the highest total utility. (c) Is there an SPE in which the first song is a duet?

Solution

(a) Pure NE (S, J) and (J, S) , and a mixed one: the guitarist sings with $1/2$, the pianist with $1/3$, with payoffs $(7/3, 3)$. (b) Repeating (J, S) gives $(14, 2)$, total 16; repeating (S, J) gives $(2, 12)$, total 14. So (J, S) twice. (c) **No**. A deviation from (S, S) is detected and attributed: the guitarist deviating yields (J, S) , the pianist (S, J) . For the guitarist, $5 + v_G(\text{reward}) \geq 7 + v_G(\text{punishment})$ needs a gap ≥ 2 ; the only stage NE giving him at least 3 is (J, S) . For the pianist the gap must be ≥ 1 , so the reward must give him at least 2, which rules out (J, S) (he gets 1). No second-stage NE rewards both. (An exhaustive check over all assignments of the three stage NE confirms it. The printed solution overlooks the guitarist's own incentive to deviate.)

Exercise 10.2

(BM 5.2) At Christmas Ashley and Brook each give a gift (G, costing 10, worth 100 to the receiver) or not (N). At New Year's Eve they choose Downtown or Home: $(D, D) = (50, 50)$, $(H, H) = (0, 0)$, mismatch $(-100, -100)$. The second stage is discounted by δ . Find a trivial SPE, and the smallest δ for an SPE with gifts.

Solution

Stage 1 is a Prisoner's Dilemma: $(G, G) = (90, 90)$, $(G, N) = (-10, 100)$, NE (N, N) . Stage 2 has NE (D, D) and (H, H) . Trivial SPE: (N, N) followed by either NE regardless of history. Carrot and stick: play G ; then D if both gave, H otherwise. Deviating gains $100 - 90 = 10$ now and loses 50δ later, so the SPE requires $\delta \geq 1/5$.

Exercise 10.3

(BM 5.3) Ann and Brian play daily: both do chores $(-1, -1)$; only one does: the cleaner -2 , the other 5; neither $(0, 0)$; $\delta = 0.9$. (a) Stage NE? (b) Show that "Ann always cleans" is not sustainable. (c) Show that alternating (Brian on odd days, Ann on even days) is, with a grim trigger.

Solution

(a) N is dominant: NE (N, N) with $(0, 0)$, which is also each player's minmax. (b) Ann would average $-2 < 0$: not individually rational, so no equilibrium can sustain it. (c) Grim trigger: follow the alternation; after any deviation play N forever. Values: Ann (lazy first) $\frac{5-2\delta}{1-\delta^2} = 16.8$, Brian $\frac{-2+5\delta}{1-\delta^2} = 13.2$. The binding check is whoever must clean today: cleaning gives $\frac{-2+5\delta}{1-\delta^2}$, deviating gives 0 forever, so the alternation is an SPE iff $\delta \geq 0.4$. It holds at $\delta = 0.9$. (The printed numerical values use $1/(1-\delta^2) = 1.2345$ instead of 5.263; the conclusion is unaffected.)

Exercise 10.4

(BM 5.4) Carl and Diana either Study or watch Movies in the library: $(S, S) = (10, 10)$, $(S, M) = (0, 18)$, $(M, S) = (15, 0)$, $(M, M) = (7.5, 9)$ (Carl first). (a) NE of $G(3)$ with $\delta = 1$? (b) For which δ does grim trigger sustain (S, S) forever? (c) Add a punishment P giving $(-10, -10)$ whenever used. In $G'(2)$, when can (S, S) be played?

Solution

(a) (M, M) is the unique stage NE (dominant), so in $G(3)$ it is played every night. (b) Diana: $\frac{10}{1-\delta} \geq 18 + \frac{9\delta}{1-\delta}$, so $\delta \geq 8/9$. Carl: $10 \geq 15(1-\delta) + 7.5\delta$, so $\delta \geq 2/3$. Grim trigger needs $\delta \geq 8/9$. (c) G' has NE (M, M) and (P, P) (P is weakly dominated, but a NE). Play (S, S) in round 1, then (M, M) if round 1 was (S, S) and (P, P) otherwise. Diana: $10 + 9\delta \geq 18 - 10\delta$, so $\delta \geq 8/19$; Carl: $10 + 7.5\delta \geq 15 - 10\delta$, so $\delta \geq 2/7$. Hence $\delta \geq 8/19$. Cooperation is possible only in round 1, since the last round must be a stage NE.

Exercise 10.5

(BM 5.5) A warrior and a mage first choose Cast (C) or Draw (D): different actions give 2 each, equal actions 0, and 1 is subtracted from whoever uses their disliked action (C for the warrior, D for the mage). Then a troll appears: both Attack (5, 5), a lone attacker dies (-100) while the other gets 10, both Run $(0, 0)$. Find all SPE.

Solution

Stage 1: $(C, C) = (-1, 0)$, $(C, D) = (1, 1)$, $(D, C) = (2, 2)$, $(D, D) = (0, -1)$. Stage 2 has a dominant action Run, with unique NE (R, R) . Since the last stage has a unique NE, nothing can be conditioned on stage 1. The SPE are: any stage-1 NE, then (R, R) after every history. The stage-1 NE are (C, D) , (D, C) and the mixed one, in which the warrior Casts with $3/4$ and the mage Casts with $1/4$ (each mixture makes the *other* indifferent; the printed $1/2$ is a slip).

Exercise 10.6

(BM 5.6) Two chipmunks search an Oak or a Pine: $(O, O) = (2, 2)$, $(O, P) = (3, 5)$, $(P, O) = (5, 3)$, $(P, P) = (1, 1)$. Then a wolf comes: both Help $(-1, -1)$, a lone helper is eaten (-100) while the other gets 0, both Flee $(-4, -4)$. Find all SPE.

Solution

Stage 2: Flee is dominant ($0 > -1$, $-4 > -100$), so (F, F) is the unique NE. Stage 1 NE: (O, P) , (P, O) , and both playing O with 0.4. SPE: one of these, then (F, F) after every history. As in the previous problem, a unique last-stage NE leaves no room for rewards or punishments.

Exercise 10.7

(BM 5.7) A knight and a thief simultaneously Attack the warlock or Save the princess. Both attack: $(0, 0)$; both save: $(-100, -100)$. Otherwise the hero who attacked may then Grab the wand and kill his ally (G) or be Fair (F). The knight values killing at -20 , the thief at 20 (the victim gets -100); a fair split gives 10 each. Reduce the game, find the SPE, and discuss (AG, AG) .

Solution

At the second move the knight plays F ($10 > -20$) and the thief plays G ($20 > 10$). Reduced game: $(A, A) = (0, 0)$, $(A, S) = (10, 10)$, $(S, A) = (-100, 20)$, $(S, S) = (-100, -100)$. A is dominant for the knight, and the thief answers S . SPE (AF, SG) : the knight slays and is fair, the thief saves, and the thief's betrayal plan is never used. (AG, AG) is a NE: both attack, and the princess burns. Switching to S gives -100 because of the other's announced G , and that threat is not credible for the knight.

Exercise 10.8

(BM 5.8) In the infinitely repeated game $(M, M) = (1, 1)$, $(M, F) = (-2, 3)$, $(F, F) = (0, 0)$, describe tit for tat, find when it sustains (M, M) , and say whether it is an SPE.

Solution

Tit for tat: play M first, then copy the opponent's previous move. Against tit for tat, defecting once and returning gives $3 - 2\delta$ over two periods versus $1 + \delta$; defecting forever gives 3 versus $\frac{1}{1-\delta}$. Both require $\delta \geq 2/3$. It is not an SPE in general: after a single deviation it locks the players into alternating (M, F) , (F, M) . The punisher's echo is worth $\frac{3-2\delta}{1-\delta^2}$, while forgiving restores $\frac{1}{1-\delta} = \frac{1+\delta}{1-\delta^2}$. For $\delta > 2/3$ forgiving is better, so tit for tat fails the one-shot deviation test off the path exactly when it works on the path (except at $\delta = 2/3$).

Exercise 10.9

(BM 5.9) Stage game (rows/columns M, F, H): $(7, 7)$ $(-2, 9)$ $(-1, -1)$ / $(9, -2)$ $(2, 2)$ $(0, 0)$ / $(-1, -1)$ $(0, 0)$ $(6, 6)$, repeated twice. (a) Stage NE? (b) An SPE that plays (M, M) first? (c) The condition on δ ?

Solution

(a) M is strictly dominated by F . Pure NE (F, F) and (H, H) ; mixed: each plays F with $3/4$ and H with $1/4$, with payoff 1.5 (the printed $\frac{1}{4}F + \frac{3}{4}H$, payoff 5 , inverts the weights). (b) Play M ; then (H, H) if the first outcome was (M, M) , and (F, F) otherwise. (c) Deviating to F gains 2 now and loses $6 - 2 = 4$ later: $9 + 2\delta \leq 7 + 6\delta$, so $\delta \geq 1/2$.

Exercise 10.10

(BM 5.10) Siblings play every afternoon: $(F, F) = (1, 1)$, $(B, F) = (3, -3)$, $(B, B) = (0, 0)$. Find strategies sustaining friendship forever. Do they work for $\delta = 0.6$?

Solution

Grim trigger: cooperating gives $\frac{1}{1-\delta}$, the best deviation $3 + 0$, so $\delta \geq 2/3$. Tit for tat: a one-period deviation needs $1 + \delta \geq 3 - 3\delta$ ($\delta \geq 1/2$), but permanent defection gives 3 , so again $\delta \geq 2/3$. At $\delta = 0.6$ **neither** sustains cooperation ($\frac{1}{0.4} = 2.5 < 3$). The printed solution's conclusion for $\delta = 0.6$ is wrong. For $\delta \geq 2/3$, grim trigger is an SPE; tit for tat is only a NE.

From the course material:

Exercise 10.11

Consider the stage game below repeated twice with $\delta = 1$. Find its NE, and an SPE of the repeated game in which (M, M) is played in the first stage. For which δ does it remain an SPE?

	M	F	H
M	7, 7	-2, 9	-1, -1
F	9, -2	2, 2	0, 0
H	-1, -1	0, 0	6, 6

Solution

Pure NE of the stage game: (F, F) and (H, H) (check: against F , F gives $2 > -2, 0$; against H , H gives $6 > -1, 0$). Strategy: play M ; then H if the first outcome was (M, M) , else F . Deviating from M to F in stage 1 yields $9 + 2\delta$; complying yields $7 + 6\delta$. The strategy is an SPE iff $7 + 6\delta \geq 9 + 2\delta$, i.e. $\delta \geq \frac{1}{2}$; with $\delta = 1$ it holds.

Exercise 10.12

In the infinitely repeated Prisoner's Dilemma with stage payoffs $(M, M) = (R, R)$, $(F, M) = (T, S)$, $(F, F) = (P, P)$ with $T > R > P > S$, find the threshold on δ for grim trigger to be an SPE.

Solution

Cooperating: $R/(1 - \delta)$. Deviating: $T + \delta P/(1 - \delta)$. Grim trigger is an SPE iff $R \geq (1 - \delta)T + \delta P$, i.e. $\delta \geq \frac{T - R}{T - P}$. With $T = 5$, $R = 4$, $P = 1$ this gives $\frac{1}{4}$.

10.8. Sources

Prof. Badia, *Game Theory* slides, decks **Gth2011** (time inconsistency, multistage games, Prisoner–Revenge, one-stage deviation principle) and **Gth2012** (finitely and infinitely repeated games, grim trigger, Friedman's theorem, tit for tat, reputation and guarantors); exercise sheet **gt20d**. Tadelis, *Game Theory: An Introduction*, Chapters 9–10. Original references: J. Friedman, *A non-cooperative equilibrium for supergames*, Rev. Econ. Stud. **38**, 1 (1971); R. Axelrod, *The Evolution of Cooperation*, Basic Books (1984). Exercises: L. Badia and T. Marchioro, *Game Theory: A Handbook of Problems and Exercises*, Esculapio (2022), Chapter 5. D. Fudenberg and E. Maskin, *The folk theorem in repeated games with discounting or with incomplete information*, Econometrica **54**, 533 (1986).

11. Bargaining and dynamic duopolies

Two important economic interactions unfold over time. In **bargaining**, players alternate offers to split a surplus that shrinks while they argue. In **dynamic oligopoly**, firms either move in sequence (Stackelberg) or compete repeatedly, which opens the door to tacit collusion. Both are solved with the tools of Chapters 9–10: backward induction, subgame perfection, and trigger strategies.

11.1. Alternating-offer bargaining

Two players must split a pie of size 1: player 1 gets x , player 2 gets $1 - x$. There are two approaches: the *axiomatic* Nash bargaining solution (Chapter 15) and the *strategic* model below (Ståhl, Rubinstein).

Definition 11.1 (Alternating-offer bargaining). At stage $t = 1, 2, \dots$ a **proposer** offers a split and a **responder** accepts (the game ends) or refuses (the game moves to stage $t + 1$). Player 1 proposes at odd stages, player 2 at even stages. If agreement on $(x, 1 - x)$ is reached at stage t , payoffs are $(\delta^{t-1}x, \delta^{t-1}(1 - x))$ with $0 < \delta < 1$: each round of delay wastes a fraction $1 - \delta$ of the pie. If no agreement is reached by the deadline T , both get 0.

This is *not* a repeated game: an acceptance terminates it.

Deadline $T = 1$ is the ultimatum game: many NE, but the unique SPE has the proposer keeping everything ($x = 1$) and the responder accepting.

Odd deadline T . Player 1 makes the last offer; at stage T player 2 accepts anything, so player 1 takes the whole pie. At stage $T - 1$, player 2 knows that refusing gives player 1 δ (in stage- $(T - 1)$ units), so offers exactly $x = \delta$, which is accepted. Iterating backwards, the SPE has immediate agreement with

$$u_1 = \frac{1 + \delta^T}{1 + \delta}, \quad u_2 = \frac{\delta - \delta^T}{1 + \delta}.$$

Proposition 11.1 (Immediate agreement). *In every SPE of the finite-horizon alternating-offer game, agreement is reached in the first round.*

Delay is doubly costly: it shrinks the pie and merely leads to another, symmetric round of proposer–responder interaction, which rational players anticipate.

Theorem 11.2 (Rubinstein, 1982). *With an infinite horizon the alternating-offer game has a unique SPE: player 1 offers and player 2 immediately accepts*

$$u_1 = \frac{1}{1 + \delta}, \quad u_2 = \frac{\delta}{1 + \delta}.$$

The limit $T \rightarrow \infty$ of the finite formulas gives the same numbers, but uniqueness must be proved without backward induction. *Proof sketch (Shaked–Sutton)*. Every subgame starting with a proposal is a copy of the whole game with the roles possibly swapped, so let M and m be the supremum and infimum of the

proposer's SPE payoff. A responder who refuses becomes the next proposer, worth between δm and δM today. Hence the responder accepts any offer above δM , so the proposer gets at least $1 - \delta M$: $m \geq 1 - \delta M$. And the responder never accepts less than δm , so $M \leq 1 - \delta m$. Combining, $M \leq 1 - \delta m \leq 1 - \delta(1 - \delta M)$, i.e. $M(1 - \delta^2) \leq 1 - \delta$, so $M \leq 1/(1 + \delta)$; symmetrically $m \geq 1/(1 + \delta)$. Thus $M = m = 1/(1 + \delta)$. $\square$

The first-mover advantage $1/(1 + \delta) - \delta/(1 + \delta) = (1 - \delta)/(1 + \delta)$ vanishes as $\delta \rightarrow 1$: with patient players the split tends to 50/50.

11.2. Bargaining power: patience and proposal rights

The Rubinstein argument generalises when players discount differently, with factors δ_1, δ_2 . Let x be what player 1 gets when proposing and y what player 2 gets when proposing. In the unique SPE each proposer offers the responder exactly the discounted value of becoming proposer next round:

$$1 - x = \delta_2 y, \quad 1 - y = \delta_1 x \quad \Rightarrow \quad x = \frac{1 - \delta_2}{1 - \delta_1 \delta_2}, \quad 1 - x = \frac{\delta_2(1 - \delta_1)}{1 - \delta_1 \delta_2}.$$

Patience is power: as $\delta_2 \rightarrow 1$ with δ_1 fixed, player 2 takes almost everything, even though player 1 proposes first. If both $\delta_i = e^{-r_i \Delta}$ and the time between offers $\Delta \rightarrow 0$, player 1's share tends to $\frac{r_2}{r_1 + r_2}$: the asymmetric Nash bargaining solution of Chapter 15 with weights inversely proportional to the impatience rates.

Legislative bargaining (Baron and Ferejohn, 1989). n legislators (odd) divide a budget by majority rule. Each round a randomly chosen member proposes a division; if a majority accepts, it is implemented; otherwise a new proposer is drawn and the budget is discounted by δ . In the stationary SPE the proposer buys the cheapest majority: offers δ/n (the continuation value) to $(n - 1)/2$ others, gives nothing to the rest, and keeps

$$1 - \frac{n - 1}{2} \cdot \frac{\delta}{n}.$$

Agreement is immediate, the proposer captures much more than an equal share, and minimum winning coalitions form — features observed in real legislatures and committees.

Stationary strategies and wars of attrition. In infinite-horizon games whose subgames are scaled copies of the whole game (all Rubinstein-like problems), one looks for **stationary** equilibria: the action depends only on the current role (proposer or responder, mover or waiter), not on the date. The value of each role then solves a pair of equations like those above. When *ending* the game benefits the other player more than yourself, the game is a **war of attrition**: each wants the other to concede. Then there are typically two asymmetric pure equilibria (one side always concedes) and a mixed one in which each concedes with a constant probability per period, so delay is wasteful but possible.

11.3. Stackelberg duopoly

Return to the Cournot setting (Chapter 4: price $a - Q$, unit cost c), but now a **leader** (firm 1) chooses q_1 first and a **follower** (firm 2) observes it before choosing q_2 .

By backward induction the follower plays its best response

$$R_2(q_1) = \frac{a - q_1 - c}{2},$$

and the leader maximises $u_1(q_1, R_2(q_1)) = q_1(a - q_1 - R_2(q_1) - c) = q_1(a - q_1 - c)/2$, giving

$$q_1^* = \frac{a - c}{2}, \quad q_2^* = \frac{a - c}{4}, \quad u_1^* = \frac{(a - c)^2}{8}, \quad u_2^* = \frac{(a - c)^2}{16}.$$

Compared with Cournot ($q = (a - c)/3$ each, profit $(a - c)^2/9$), the leader produces more and earns more; the follower produces and earns less. Total output $3(a - c)/4$ exceeds Cournot's $2(a - c)/3$.

Remark (Threats and information). The follower might threaten “choose the Cournot quantity or I will flood the market”. The threat is non-credible: flooding hurts the follower too. More interesting is the role of information. If the follower moved *without* observing q_1 , a chain of anticipations (“the leader will play $(a - c)/2$, so I play $(a - c)/4$, so the leader should play $3(a - c)/8$, ...”) leads back to Cournot. It is precisely the follower’s *observation* of q_1 — common knowledge to both — that hands the leader the first-mover advantage. In multi-agent problems more information can make a player worse off.

11.4. Collusion in a repeated Cournot duopoly

At the Cournot equilibrium each firm earns $u^C = (a - c)^2/9$, while sharing the monopoly output would give each $u^m/2 = (a - c)^2/8$. The one-shot game cannot sustain this, but by the folk theorem an infinitely repeated one can.

Proposition 11.3 (Grim-trigger collusion). *Consider the strategy: produce $q^m/2 = (a - c)/4$ at $t = 1$ and as long as both firms have always done so; after any deviation produce the Cournot quantity $q^C = (a - c)/3$ forever. Both firms using it is an SPE iff $\delta \geq 9/17$.*

Proof. After a deviation, (q^C, q^C) forever is a NE of every subgame. On the cooperative path, the most profitable one-shot deviation against $q^m/2$ is $\arg\max_q q(a - q - (a - c)/4 - c) = 3(a - c)/8$, earning $u^D = 9(a - c)^2/64$. Cooperation is optimal iff

$$\frac{(a - c)^2/8}{1 - \delta} \geq \frac{9(a - c)^2}{64} + \frac{\delta}{1 - \delta} \frac{(a - c)^2}{9} \iff \delta \geq \frac{9}{17}. \quad \square$$

Less ambitious collusion. For $\delta < 9/17$ the firms can still beat Cournot by targeting some $q^* \in [q^m/2, q^C]$ with a grim trigger. Against q^* , the best deviation $q^D = (a - q^* - c)/2$ earns $u^D = (a - q^* - c)^2/4$, while cooperation earns $u^* = q^*(a - 2q^* - c)$ per period. The smallest sustainable q^* makes the incentive constraint $u^*/(1 - \delta) \geq u^D + \delta u^C/(1 - \delta)$ binding:

$$q^*(\delta) = \frac{(a - c)(9 - 5\delta)}{3(9 - \delta)},$$

which moves continuously from q^C at $\delta = 0$ to $q^m/2$ at $\delta = 9/17$.

Carrot and stick. Unlike the Prisoner’s Dilemma, the Cournot game offers punishments *harsher* than reverting to the NE. Abreu’s two-phase strategy uses a **reward** action R (produce $q^m/2$) and a **punishment** action P (produce some $x > q^C$):

- start with R ;
- play R if both played R in the previous period, or if both played P ;
- otherwise play P .

A deviator is punished for one period with low prices, and the punishment itself is enforced because deviating from it restarts it. One can check that with $\delta = \frac{1}{2}$ and $x = 2(a - c)/5$ the strategy is an SPE: the monopoly split is sustained even though $\frac{1}{2} < 9/17$.

! Important

Competition authorities fine firms that meet to fix prices or quantities. The repeated-game analysis shows why this is not enough: a grim-trigger agreement needs no meeting and no communication. It can be reached “in the CEO’s head”, as a mutually understood equilibrium of the repeated game — tacit collusion.

11.5. Exercises

Problems from Badia and Marchioro (BM), Chapters 3 and 4.

Exercise 11.1

(BM 3.10) Amelie (odd days) and Bernard (even days) work on a project worth v_1 on day 1 and $v_{j+1} = \delta v_j$ afterwards. On their day, the mover either Calls the end (C), getting $0.4v_j$ while the other gets $0.6v_j$, or Keeps working (K). Show that in any stationary SPE the project ends by day 2, and find for which δ Amelie ends it.

Solution

Every subgame starting on an odd (even) day is the day-1 game scaled by δ^{j-1} , with roles swapped on even days. So a stationary strategy depends only on whether one is the mover. If the mover’s rule were “keep” for both, payoffs would be $0 < 0.4$, so somebody calls by day 2. Amelie compares $0.4v_1$ (call now) with $0.6\delta v_1$ (Bernard calls tomorrow). **For** $\delta < 2/3$ calling is better for every mover, and the unique SPE is that Amelie ends the project on day 1. **For** $\delta > 2/3$ being the non-caller is worth waiting for, and there are two pure stationary SPE: Amelie always keeps and Bernard always calls (the project ends on day 2), or Bernard always keeps and Amelie always calls (it ends on day 1). This is a war of attrition, in which each wants the other to concede. (The printed solution states the inequality the wrong way round.)

Exercise 11.2

(BM 4.10, parts 2–3) In the game below (static NE (A, X) , Chapter 3), find the Stackelberg outcome when M leads, and when N leads (break ties generously: a follower favours the leader, and a leader favours the follower).

	X	Y	Z
A	2, 2	3, 1	0, 0
B	1, 6	7, 4	9, 4
C	0, 1	5, 3	9, 2

Solution

M leads. N answers $A \rightarrow X$, $B \rightarrow X$, $C \rightarrow Y$, so M compares 2, 1, 5 and plays C. Outcome (C, Y) with (5, 3). Both gain over the static NE (2, 2): the leader never does worse than a NE (it could replay A), and here the follower also gets more than its pure minmax 2. **N leads.** M answers $X \rightarrow A$, $Y \rightarrow B$, and $Z \rightarrow B$ or C (tie at 9; the generous follower picks B, which gives N 4 rather than 2). N then compares 2 (X), 4 (Y), 4 (Z). The tie is broken generously toward Z, which gives M 9 instead of 7. Outcome (B, Z) with (9, 4). Sequential play removes the coordination failure of the static game.

From the course material:

Exercise 11.3

In the alternating-offer game with deadline $T = 2$ (player 1 proposes first, player 2 proposes last), find the SPE.

Solution

At stage 2 player 2 proposes and keeps everything: payoffs $(0, \delta)$. At stage 1 player 1 offers player 2 exactly δ and keeps $1 - \delta$; player 2 accepts. SPE payoffs $(1 - \delta, \delta)$: with an even deadline the last proposer has the advantage when $\delta > \frac{1}{2}$.

Exercise 11.4

Verify the carrot-and-stick strategy for $a - c = 1$, $\delta = \frac{1}{2}$, $x = \frac{2}{5}$.

Solution

Reward profit $\frac{1}{8}$; punishment profit (both at 0.4, price 0.2) is 0.08. Value on the reward path $V_R = \frac{1}{8} / (1 - \frac{1}{2}) = 0.25$. Value at the start of a punishment phase with compliance $V_P = 0.08 + \frac{1}{2} V_R = 0.205$. **Reward phase:** best deviation $q = \frac{3}{8}$ earns $\frac{9}{64} \approx 0.1406$, then punishment: $0.1406 + \frac{1}{2} \cdot 0.205 = 0.243 < 0.25$. **Punishment phase:** best deviation against 0.4 is $q = 0.3$, earning 0.09, then punishment restarts: $0.09 + \frac{1}{2} \cdot 0.205 = 0.1925 < 0.205$. No profitable one-shot deviation exists, so by the one-stage deviation principle the strategy is an SPE.

11.6. Sources

Prof. Badia, *Game Theory* slides, deck Gth2013 (dynamic bargaining, Stackelberg duopoly, Cournot collusion with grim trigger and carrot-and-stick). Tadelis, *Game Theory: An Introduction*, Chapters 8.3.2, 10.6 and 11. Original references: A. Rubinstein, *Perfect equilibrium in a bargaining model*, *Econometrica* **50**, 97 (1982); A. Shaked and J. Sutton, *Econometrica* **52**, 1351 (1984); D. Abreu, *Extremal equilibria of oligopolistic supergames*, *J. Econ. Theory* **39**, 191 (1986). Exercises: L. Badia and T. Marchioro, *Game Theory: A Handbook of Problems and Exercises*, Esculapio (2022), Chapters 3–4. D. Baron and J. Ferejohn, *Bargaining in legislatures*, *Am. Polit. Sci. Rev.* **83**, 1181 (1989).

Part IV.

Part IV: Incomplete information



12. Bayesian games

Until now every player knew everyone's payoffs. In reality a firm does not know a competitor's costs, a bidder does not know the others' valuations, a network node does not know whether its neighbour is selfish or malicious. Harsanyi's insight was that such **incomplete information** can be absorbed into the familiar framework: let Nature draw each player's private characteristics — their **type** — from a commonly known distribution, and let players form beliefs about types just as they form beliefs about actions. This chapter defines static Bayesian games and Bayesian Nash equilibrium and works through the standard examples.

12.1. Types and the common prior

Definition 12.1 (Game of incomplete information). A game has **incomplete information** if some player does not know another player's payoff function. Each possible “version” of a player is called a **type**.

Harsanyi (1967–68) turned incomplete into *imperfect* information by adding a preliminary move of Nature:

1. Nature draws a type vector $(t_1, \dots, t_n)$ from a probability distribution ϕ ;
2. each player i observes **only** their own type t_i ;
3. players choose their actions;
4. payoffs, which may depend on the types, are realised.

The result is a dynamic game in which players do not observe all of Nature's move. Types need not describe preferences only: they can capture what a player knows about the world or about other players.

Example 12.1 (Entry with a possibly crazy incumbent)

An entrant (player 1) stays **Out**, getting 0 while the incumbent gets 2, or **Enters**. After entry the incumbent (player 2) **Accepts**, giving (1, 1), or **Fights**. A *rational* incumbent gets -1 from fighting (so does the entrant); a *crazy* incumbent enjoys fighting and gets 2. Nature makes the incumbent rational with probability p ; the entrant does not observe the type.

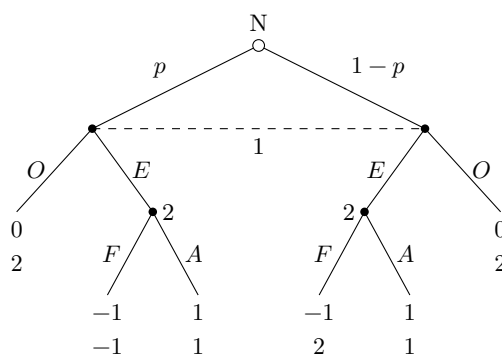


Figure: the entry game with a rational (left) or crazy (right) incumbent; the entrant's two nodes form one information set.

To best-respond, the entrant needs beliefs about the incumbent's type. We assume that players do not know the opponents' types but know *how they are drawn*:

Definition 12.2 (Common prior assumption). The distribution ϕ of type vectors is **common knowledge**.

12.2. Type-contingent strategies

A strategy must now say what each *type* does. In the entry game the entrant has one type and two strategies; the incumbent moves at two information sets (one per type) and has four strategies xy : x for the rational type, y for the crazy type. Expected payoffs follow from the prior; for instance under (E, AF) the entrant gets $p \cdot 1 + (1 - p)(-1) = 2p - 1$ and the incumbent $p \cdot 1 + (1 - p) \cdot 2 = 2 - p$:

$1 \setminus 2$	AA	AF	FA	FF
O	0, 2	0, 2	0, 2	0, 2
E	1, 1	$2p - 1, 2 - p$	$1 - 2p, 1 - 2p$	$-1, 2 - 3p$

For $p = \frac{2}{3}$ the entries of row E are $(1, 1)$, $(\frac{1}{3}, \frac{4}{3})$, $(-\frac{1}{3}, -\frac{1}{3})$, $(-1, 0)$, and the game has three pure Nash equilibria: (O, FA) , (O, FF) and (E, AF) . Only (E, AF) is sequentially rational: each incumbent type does what is best *for that type* once entry has occurred (a rational incumbent accepts, a crazy one fights). If $p < \frac{1}{2}$, then $2p - 1 < 0$ and the sensible outcome becomes (O, AF) : the incumbent always plays AF , and the entrant decides on the basis of the prior.

Remark (Type agents). The entrant averages over the incumbent's types — natural. The incumbent, oddly, “takes expectations over its own types”: the real players (rational 2, crazy 2) are replaced by a single meta-player choosing a plan for both. This **type-agent representation** gives the right best responses because a plan maximises the expected payoff iff each type's component maximises that type's payoff (the types' choices do not interact). The common prior is a strong assumption — it must hold for players to be able to conjecture the outcome — and its realism must be checked in every application.

12.3. Static Bayesian games

Definition 12.3 (Static Bayesian game). A static Bayesian game is

$$G = \{\mathcal{N}; A_1, \dots, A_n; T_1, \dots, T_n; \phi_1, \dots, \phi_n; u_1, \dots, u_n\},$$

where A_i are action sets, T_i type spaces, $\phi_i(t_{-i} | t_i)$ player i 's beliefs about the others' types, and $u_i(a_1, \dots, a_n; t_i)$ type-dependent payoffs.

- **Private values** means u_i depends only on t_i ; with **common (interdependent) values** u_i depends on the whole type vector.
- Types can also restrict actions: if action H is available only with probability q , give player i two types and set all payoffs of H to $-\infty$ for the second type.
- With a common prior $\phi(t_1, \dots, t_n)$, beliefs are posteriors by Bayes' rule, $\phi_i(t_{-i} | t_i) = \phi(t_i, t_{-i}) / \phi(t_i)$. Types are independent if $\phi = \prod_i \phi(t_i)$, in which case beliefs equal the marginals.
- In principle one should model beliefs about beliefs about beliefs...; Harsanyi showed that the common prior collapses this hierarchy to first-order beliefs.

Definition 12.4 (Bayesian strategy). A pure strategy of player i is a map $s_i : T_i \rightarrow A_i$; a mixed strategy is a distribution over such maps.

Why specify what *every* type does, when only one type is real? Not for player i — but for the others, who need to form beliefs about i 's behaviour. (The same reason led us to specify off-path behaviour in dynamic games.) From the opponents' viewpoint, a player using a type-dependent pure strategy with randomly drawn type is indistinguishable from one using a mixed strategy.

Definition 12.5 (Bayesian Nash equilibrium). A profile $s^* = (s_1^*, \dots, s_n^*)$ is a **Bayesian Nash equilibrium** (BNE) if for every player i and every type $t_i \in T_i$, the action $s_i^*(t_i)$ maximises i 's expected payoff given their beliefs:

$$s_i^*(t_i) \in \arg \max_{a_i \in A_i} \sum_{t_{-i}} \phi_i(t_{-i} | t_i) u_i(a_i, s_{-i}^*(t_{-i}); t_i).$$

Equivalently, $\mathbb{E}[u_i(s_i^*(t_i), s_{-i}^*(t_{-i}); t_i) | t_i] \geq \mathbb{E}[u_i(a_i, s_{-i}^*(t_{-i}); t_i) | t_i]$ for all a_i : nobody wants to change given what they know, and what they don't know they estimate. With continuous types the sums become integrals.

12.4. Harsanyi's purification of mixed strategies

Mixed strategies were interpreted as beliefs (Chapter 5). Harsanyi made this precise: a mixed equilibrium is the limit of **pure** equilibria of slightly perturbed games of incomplete information.

Perturb the Battle of the Sexes: Ann's payoff at (R, R) becomes $2 + c$ and Brian's at (S, S) becomes $2 + d$, where c, d are independent, uniform on $[0, x]$, and privately known. Look for threshold strategies: Ann plays R iff $c > C$; Brian plays S iff $d > D$.

Ann does not know d ; Brian plays R with probability D/x . Her expected payoffs are $(2 + c)D/x$ from R and $1 - D/x$ from S, so she plays R iff $c \geq x/D - 3$; hence $C = x/D - 3$. Symmetrically $D = x/C - 3$. Thus $C = D$ and

$$C^2 + 3C - x = 0 \implies C = \frac{-3 + \sqrt{9 + 4x}}{2}.$$

Ann plays R with probability

$$1 - \frac{C}{x} = \frac{2x + 3 - \sqrt{9 + 4x}}{2x} \xrightarrow{x \rightarrow 0} \frac{2}{3},$$

which is exactly her probability in the mixed NE. As the perturbation vanishes, players who never actually randomise generate, from each other's viewpoint, the mixed equilibrium.

12.5. Examples

Example 12.2 (Chicken with strict parents)

Two youngsters drive towards each other. A **Chicken** (who swerves) gets 0; a **Driver** gets respect 8 against a chicken; if both drive they split the respect (4 each) but crash and receive a punishment P from their parents: $P = 16$ for **Hard** parents, $P = 0$ for **Lenient** ones, each with probability $\frac{1}{2}$, independently. Each youngster knows only their own parents' type. A strategy xy is the action of the L-type and of the H-type. Expected payoffs:

	CC	CD	DC	DD
CC	0, 0	0, 4	0, 4	0, 8
CD	4, 0	-1, -1	-1, 3	-6, 2
DC	4, 0	3, -1	3, 3	2, 2
DD	8, 0	2, -6	2, 2	-4, -4

The BNE is (DC, DC) : lenient-parent kids drive, hard-parent kids swerve. Check type by type: against DC the opponent drives with probability $\frac{1}{2}$; an L-type gets $\frac{1}{2} \cdot 4 + \frac{1}{2} \cdot 8 = 6 > 0$ by driving, an H-type gets $\frac{1}{2}(4 - 16) + \frac{1}{2} \cdot 8 = -2 < 0$.

Example 12.3 (Committee voting)

A jury of two decides whether to **A**cquit or **C**onvict; conviction requires both votes. The defendant is guilty (G) with common prior $q > \frac{1}{2}$. Each juror gets payoff 1 for a correct verdict, 0 otherwise. Without further information C is dominant.

Now each juror receives a private, conditionally independent signal $t_i \in \{t_G, t_I\}$ with $\Pr[t_G | G] = \Pr[t_I | I] = p > \frac{1}{2}$ — a binary symmetric channel. A lone juror would update

$$\Pr[G | t_G] = \frac{qp}{qp + (1-q)(1-p)} > q, \quad \Pr[G | t_I] = \frac{q(1-p)}{q(1-p) + (1-q)p},$$

and the latter is below $\frac{1}{2}$ iff $p > q$: when the signal is more informative than the prior, a lone juror follows it.

With two jurors, is “follow the signal” (CA : convict on t_G , acquit on t_I) an equilibrium? The key is **pivotality**: a vote matters only if the other juror votes C , i.e. (under CA) only if the other received t_G . Conditioning on that event,

$$\Pr[G | t_1 = t_I, t_2 = t_G] = \frac{qp(1-p)}{qp(1-p) + (1-q)(1-p)p} = q > \frac{1}{2},$$

so a juror with an innocent signal should *convict*: CA is not a best response to itself. Reasoning about the pivotal event biases jurors toward conviction, even against their own informative signal. (For $p \leq q$, (CC, CC) is a BNE; for $p > q$ the symmetric equilibrium has t_I -jurors convicting with a probability strictly between 0 and 1, as in Feddersen and Pesendorfer’s analysis of the unanimity rule.)

Example 12.4 (Study group)

Two students share a project. Each either puts **E**ffort (cost c) or **S**hirks. The project succeeds if at least one puts effort, giving student i value t_i^2 , with $t_i \sim \mathcal{U}[0, 1]$ privately known:

	E	S
E	$t_A^2 - c, t_B^2 - c$	$t_A^2 - c, t_B^2$
S	$t_A^2, t_B^2 - c$	0, 0

With a continuum of types there are infinitely many strategies, but best responses are of **threshold** form: E gives $t_i^2 - c$ regardless, S gives $t_i^2 \Pr[s_j(t_j) = E]$, so i prefers E iff $t_i^2 \geq c/(1 - \Pr[s_j(t_j) = E])$. (A lone student would put effort iff $t_i^2 \geq c$.) If j uses threshold θ_j (effort iff $t_j \geq \theta_j$), then $\Pr[E_j] = 1 - \theta_j$ and i 's best threshold is $\theta_i = \sqrt{c/\theta_j}$. The symmetric BNE has $\theta^2 = c/\theta$, i.e.

$$\theta^* = c^{1/3} \quad (c < 1).$$

Since $c^{1/3} > c^{1/2}$, each student works less often than alone: free-riding on the partner.

12.6. Two classic applications

Cournot with a private cost. Firm 1 has unit cost c , known to all. Firm 2's cost is c_H with probability θ and c_L otherwise, known only to firm 2; demand is $a - Q$. Each type of firm 2 best-responds to q_1 , and firm 1 best-responds to firm 2's *expected* output:

$$q_2(c_2) = \frac{a - c_2 - q_1}{2}, \quad q_1 = \frac{a - c - \mathbb{E}[q_2]}{2} \implies q_1^* = \frac{a - 2c + \bar{c}_2}{3}, \quad \bar{c}_2 = \theta c_H + (1 - \theta)c_L,$$

and $q_2^*(c_2) = \frac{2a + 2c - 3c_2 - \bar{c}_2}{6}$. Compared with complete information, the high-cost type produces *more* (by $(c_H - \bar{c}_2)/6$) and the low-cost type less. Firm 1 cannot tailor its output to the true cost, so each type of firm 2 faces an "average" rival.

Adverse selection: the market for lemons (Akerlof, 1970). Used cars have quality $\theta \sim \mathcal{U}[0, 1]$, known to the seller, who values a car at θ ; buyers value it at $\frac{3}{2}\theta$. Trade would be efficient for *every* car. At a price p only owners with $\theta \leq p$ sell, so the average quality on offer is $p/2$ and buyers are willing to pay $\frac{3}{2} \cdot \frac{p}{2} = \frac{3}{4}p < p$. No price clears the market, and only $\theta = 0$ trades. Private information can destroy a market completely. Warranties, inspections, reputation and signalling (Chapter 13) exist to repair it.

Remark (A taxonomy of dynamic Bayesian games). When the uninformed player moves **first** and the informed one reacts (a *screening* or Bayesian entry game), the uninformed player can only use the prior. Backward induction on each type's decision node gives a subgame-perfect answer, and no belief updating is needed. When the informed player moves **first** (a *signaling* game), the action may reveal the type, and the responder's beliefs after each action must be specified. This is where perfect Bayesian equilibrium (Chapter 13) is required.

12.7. Exercises

Problems from Badia and Marchioro (BM), Chapter 6: static Bayesian games and screening games.

Exercise 12.1

(BM 6.1) Stevie is Good (G) or Naughty (N). If naughty, he gets nothing: Stevie 0, parents -10 . If good, the parents give a Sweater (-10 for Stevie, 0 for them) or a Train ($+40$ for Stevie, $+20$ for them), but the train costs them 50 unless Santa exists. Stevie believes Santa exists with probability p ; the parents know. Write the type-agent normal form and find the pure BNE for $p = 0.9$. Which are subgame perfect?

Solution

The parents' strategies are pairs (Santa, no Santa). Row G: SS $(-10, 0)$, ST $(40 - 50p, 30p - 30)$, TS $(50p - 10, 20p)$, TT $(40, 50p - 30)$; row N: $(0, -10)$ throughout. For $p = 0.9$: G gives $(-10, 0)$, $(-5, -3)$, $(35, 18)$, $(40, 15)$. BNE: (N, SS) , (N, ST) , (G, TS) . Only (G, TS) is subgame perfect: giving a sweater when the train is free is not optimal. Stevie behaves because he believes in Santa: against TS, being good is worth $50p - 10$, so for $p < 0.2$ he would be naughty.

Exercise 12.2

(BM 6.3) An opportunistic transmitter T either Buys a licensed channel (-10) ; the incumbent S gets 100) or Accesses S's band. S then Combats it (T gets -20 ; S gets 90 with high power, 50 without) or does nothing (T 20, S 80). S has high power with probability $p = 0.7$. Find the BNE and the SPE.

Solution

S's strategies are (high, low). Row A: CC $(-20, 78)$, CD $(-8, 87)$, DC $(8, 71)$, DD $(20, 80)$; row B: $(-10, 100)$. BNE: (A, CD) and (B, CC) . Sequential rationality requires C when strong $(90 > 80)$ and D when weak $(80 > 50)$, so CD. Then T compares -8 (A) with -10 (B) and accesses. SPE (A, CD) . In (B, CC) the weak incumbent's threat to fight is not credible. This is the Bayesian entry game in a cognitive-radio setting.

Exercise 12.3

(BM 6.4) Henchmen Franco and Gino each act Conspicuously or Inconspicuously. At least one C gives a reward of 10 each, but both C gets them arrested (-30) . With probability $p = 0.3$ Franco is an undercover cop, who gets 20 if both are arrested and 0 otherwise. Find the BNE.

Solution

Franco's strategies are (cop, crook). For $p = 0.3$ the payoffs (Gino, Franco) are: Gino C against CC, CI, IC, II gives $(-30, -15)$, $(-2, 13)$, $(-18, -21)$, $(10, 7)$; Gino I gives $(10, 7)$, $(3, 0)$, $(7, 7)$, $(0, 0)$. Gino's best responses are I, I, I, C. Franco's best response to C is CI, and to I it is CC or IC. Pure BNE: (CC, I) and (IC, I) . In general, Gino plays I, the crook plays C, and the cop is indifferent, so any mixture of the cop's actions is part of a BNE. With a single information set for Gino, every BNE is subgame perfect. The PBE beliefs on Gino's four nodes are $(0.3\alpha, 0.3(1 - \alpha), 0.7, 0)$.

Exercise 12.4

(BM 6.5) Short-sighted John Greets or Ignores a girl who is Camilla with probability $p = 0.3$. Ignoring gives John -20 , and the girl -50 if she is Camilla (0 otherwise). After a greeting she Waves (John 10; the girl 30 if Camilla, 10 otherwise) or stays Silent (John -50 and Camilla 10 if she is Camilla; 0 and 0 otherwise). Find the BNE.

Solution

Row G against SS, SW, WS, WW: $(-15, 3)$, $(-5, 10)$, $(3, 9)$, $(10, 23)$; row I: $(-20, -15)$ throughout. Waving is optimal for both types, and John's best response to WW is G. Ignoring is never a best response (every G entry exceeds -20). The unique BNE (G, WW) is also the SPE, and John's belief is simply the prior.

Exercise 12.5

(BM 6.11) Emma (Aquamarine or Beige) and Frank (Azure or Brown) dress for a wedding. Matching azure gives 3 each, matching brown 2, mismatch 0. With probability $p = 0.8$ Frank's suits are dirty: a dirty brown suit costs both 1; a dirty azure suit costs Frank 3 and Emma 5. Find the pure BNE.

Solution

Frank's strategies are (dirty, clean). Emma A against AA, AB, BA, BB: $(-1, 0.6)$, $(-1.6, 0)$, $(-0.2, -0.2)$, $(-0.8, -0.8)$; Emma B: $(-4, -2.4)$, $(-3.6, -2)$, $(0.8, 0.8)$, $(1.2, 1.2)$. Against A, both of Frank's types match with A (dirty: $0 > -1$; clean: $3 > 0$), and against AA Emma prefers A ($-1 > -4$). Against B, both types play B, and Emma prefers B. **Two** pure BNE: (A, AA) and (B, BB) , the second Pareto dominant. (The printed solution lists only (B, BB) .) Incomplete information has turned a coordination game into one where the matching colour is worse when dirty, but coordination on azure still survives.

Exercise 12.6

(BM 6.13) Alberto and Gianluca apply to Coimbra or Santander. Gianluca gets 2 (together in C), 1 (together in S), 0 apart. With probability p Alberto is chasing a girl in Santander: then S is dominant, giving him 2 (C gives 0). Otherwise he has Gianluca's preferences. Find the pooling BNE, and the values of p for which a separating BNE exists.

Solution

Against Alberto's pooling SS, Gianluca plays S ($1 > 0$), and S is a best response for both of Alberto's types: (SS, S) is a BNE for every p . If Gianluca plays C, Alberto's best response is SC (the chaser goes to S, the other to C). Against SC Gianluca gets $2(1 - p)$ from C and p from S. So (SC, C) , a **separating** BNE, exists iff $p \leq 2/3$. With a small chance of romance, the friends can still coordinate on the better destination.

Exercise 12.7

(BM 6.14) Jane Keeps her €10 000 or buys a Car (€8000; worth €5000 after a year). Simultaneously the bank Leaves the money (5% interest) or buys Subprime bonds: a good bank (probability $p = 0.2$) earns 15%, a bad bank loses everything. The bank's payoff is the interest G if positive, $-G/5$ otherwise. Find the BNE.

Solution

In k€: for Jane, K-L gives 10.5, K-S gives 11.5 (good bank) or 0 (bad bank); C-L gives 7.1, C-S gives 7.3 or 5. The bank's payoffs are 0.5, 1.5, 2 and 0.1, 0.3, 0.4 respectively. For both types of bank, S is strictly better whatever Jane does, so SS is dominant. Against SS, Jane gets $11.5p = 2.3$ from K and $5 + 2.3p = 5.46$ from C. The unique BNE is (C, SS) : Jane spends her money because the bank would gamble it away. The incentive structure ($-G/5$ rewards losses) is what makes every type of bank reckless.

Exercise 12.8

(BM 6.17) A thief Enters or Flees; simultaneously the sheriff Patrols or Quits. The sheriff is armed with probability 0.3. Flee: thief 0, and the sheriff -10 if patrolling, 0 if quitting. Enter against Quit: thief 10, sheriff 0. Enter against Patrol: if armed, thief -100 and sheriff 20; if unarmed, both -20 . Find the BNE.

Solution

Sheriff strategies are (armed, unarmed). Thief E against PP, PQ, QP, QQ: $(-44, -8)$, $(-23, 6)$, $(-11, -14)$, $(10, 0)$; F: $(0, -10)$, $(0, -3)$, $(0, -7)$, $(0, 0)$. An unarmed sheriff never patrols (-20 or $-10 < 0$), so PP and QP are out, and there is no pure BNE (QQ invites entry, PQ deters it). The armed sheriff mixes P with ρ , the thief enters with ε . Armed indifference: $20\varepsilon - 10(1 - \varepsilon) = 0$, so $\varepsilon = 1/3$. Thief indifference: $0.3(10 - 110\rho) + 0.7 \cdot 10 = 0$, so $\rho = 10/33$. The unarmed sheriff stays home; the armed one patrols with probability $10/33$.

From the course material:

Exercise 12.9

Two interns independently choose an operating system, M or W. Each prefers M with probability p and W with probability $1 - p$ (independently, privately known, p common knowledge). Using the favourite OS gives 1, the other 0; choosing the same OS as the partner adds 2. For which p is “follow your type” a BNE?

Solution

Suppose the partner follows their type: they choose M with probability p . An M-type gets $1 + 2p$ from M and $0 + 2(1 - p)$ from W; M is optimal iff $1 + 2p \geq 2 - 2p$, i.e. $p \geq \frac{1}{4}$. A W-type gets $1 + 2(1 - p)$ from W and $2p$ from M; W is optimal iff $3 - 2p \geq 2p$, i.e. $p \leq \frac{3}{4}$. “Follow your type” is a BNE iff $\frac{1}{4} \leq p \leq \frac{3}{4}$; outside this range the minority type conforms to the likely choice of the majority.

Exercise 12.10

In the study-group game with $c = 0.125$, compute the equilibrium probability that the project fails.

Solution

$\theta^* = 0.125^{1/3} = 0.5$. Each student shirks with probability 0.5; the project fails iff both shirk: probability 0.25. (A single student working alone would shirk only if $t < \sqrt{0.125} \approx 0.354$.)

12.8. Sources

Prof. Badia, *Game Theory* slides, decks Gth2014 (Bayesian games, types, common prior, BNE, purification) and Gth2015 (Chicken, committee voting) and the study-group example from Gth2017; exercise sheet gt20b. Tadelis, *Game Theory: An Introduction*, Chapter 12. Original references: J. Harsanyi, *Games with incomplete information played by “Bayesian” players*, Management Sci. **14** (1967–68); J. Harsanyi, Int. J. Game Theory **2**, 1 (1973); T. Feddersen and W. Pesendorfer, Am. Polit. Sci. Rev. **92**, 23 (1998). Exercises: L. Badia and T. Marchioro, *Game Theory: A Handbook of Problems and Exercises*, Esculapio (2022), Chapter 6. G. Akerlof, *The market for “lemons”*, Q. J. Econ. **84**, 488 (1970).

13. Perfect Bayesian equilibrium and signaling

Static Bayesian games are solved by Bayesian Nash equilibrium; dynamic games of complete information by subgame-perfect equilibrium. When both features are present — a player with private information moves, and others react after observing the move — neither concept is enough: Nature’s initial move merges all subtrees into a single subgame, so subgame perfection loses its bite. This chapter introduces **beliefs** at every information set and the resulting **perfect Bayesian equilibrium**, then applies it to signaling games, reputation building, and cheap talk.

13.1. Why SPE is not enough

Game	Solution concept
static, complete information	Nash equilibrium
dynamic, complete information	subgame-perfect equilibrium (backward induction if perfect information)
static, incomplete information	Bayesian Nash equilibrium
dynamic, incomplete information	perfect Bayesian equilibrium

Entry with an uncertain entrant. Let the *entrant* now have two types: **Competitive** (probability $p = \frac{1}{2}$), with the payoffs of Chapter 12 (entry is met by Accept $(1, 1)$ or Fight $(-1, -1)$), or **Weak**, for whom entering is always bad: Accept gives $(-1, 1)$, Fight $(-2, 0)$. Staying out gives $(0, 2)$ to either type. The incumbent observes entry but not the type, so it cannot apply backward induction. The entrant’s strategies are OO, OE, EO, EE (action of C, action of W):

$1 \setminus 2$	F	A
OO	0, 2	0, 2
OE	-1, 1	$-\frac{1}{2}, \frac{3}{2}$
EO	$-\frac{1}{2}, \frac{1}{2}$	$\frac{1}{2}, \frac{3}{2}$
EE	$-\frac{3}{2}, -\frac{1}{2}$	0, 1

There are two pure NE: (EO, A) — only the competitive entrant enters and the incumbent accepts — and (OO, F) — nobody enters because the incumbent threatens to fight. The second is not sequentially rational: *whoever* entered, fighting is worse than accepting for the incumbent ($-1 < 1$ against C, $0 < 1$ against W). Yet (OO, F) is an SPE, because the only subgame is the whole game. We need a refinement that can evaluate behaviour at information sets that are off the equilibrium path.

13.2. Beliefs and perfect Bayesian equilibrium

Definition 13.1 (Equilibrium path in Bayesian games). Given a BNE, an information set is **on the equilibrium path** if it is reached with positive probability (given the type distribution), and **off the path** otherwise. In (OO, F) the incumbent's information set is off the path.

Definition 13.2 (System of beliefs). A **system of beliefs** μ assigns to every information set a probability distribution over its nodes: $\mu(x)$ is the probability that the player believes they are at node x when the information set $h(x)$ is reached.

On the path, beliefs should be conditional probabilities, $\mu(x) = \Pr[x] / \Pr[h(x)]$ by Bayes' rule.

Definition 13.3 (Perfect Bayesian equilibrium). A **perfect Bayesian equilibrium** (PBE) is a pair (s^*, μ) of a strategy profile and a system of beliefs such that:

1. every player has beliefs at every information set where they move;
2. on the equilibrium path, beliefs are derived from s^* and the prior by Bayes' rule;
3. off the equilibrium path, beliefs are arbitrary;
4. given the beliefs, strategies are **sequentially rational**: at every information set the player's continuation strategy is a best response.

A PBE is not just a strategy profile: the beliefs are part of it.

Example 13.1 (*Entry, solved*)

Let x_1 (x_2) be the incumbent's node after entry by C (by W), and let the types enter with probabilities q_C, q_W . By Bayes' rule

$$\mu(x_1) = \frac{p q_C}{p q_C + (1-p) q_W}.$$

(EO, A) is a PBE with $\mu(x_1) = 1$: the incumbent knows that an entrant is competitive and accepts. (OO, F) is not: Bayes' rule cannot be applied ($q_C = q_W = 0$), but for *any* belief Fight is not a best response, so requirement 4 fails. Compare with off-path choices in SPE.

Theorem 13.1 (Fully reached information sets). *If a profile s^* (possibly mixed) reaches every information set with positive probability, then s^* together with the beliefs derived by Bayes' rule is a PBE if and only if s^* is a BNE.*

13.3. Refinements: sequential equilibrium

The definition above is sometimes called **weak PBE** because requirement 3 lets off-path beliefs be anything, even inconsistent with the structure of the game.

Example 13.2 (*An odd PBE*)

Nature picks node x_1 or x_2 with probability $\frac{1}{2}$ each; player 1 *does not observe it* and chooses U (payoff 1 for player 1, game over) or D. After D, player 2 (who doesn't observe Nature either) chooses L, giving (2, 2) at both nodes, or R, giving (0, 3) at x_1 and (0, 0) at x_2 . If player 1 plays D, Bayes' rule gives $\mu(x_1) = \frac{1}{2}$; R is then worth $1.5 < 2$, so L, and D is optimal for player 1: the PBE (D, L) . But if player 1 never plays D, any belief is admissible; with $\mu(x_1) > \frac{2}{3}$, R is optimal, and against R player 1 prefers U. So (U, R) is also a (weak) PBE — supported by the belief that player 1's non-move

somehow reveals Nature's choice, which player 1 does not even observe.

Definition 13.4 (Sequential equilibrium (Kreps–Wilson, 1982)). An assessment (s^*, μ) is **consistent** if it is the limit of a sequence (s^k, μ^k) of *fully mixed* strategy profiles s^k and the beliefs μ^k derived from them by Bayes' rule. A **sequential equilibrium** is a consistent, sequentially rational assessment.

In the odd example, every fully mixed profile gives $\mu^k(x_1) = \frac{1}{2}$, so consistency forces $\mu(x_1) = \frac{1}{2}$ and eliminates (U, R) . Sequential equilibrium can still be insufficient and is much harder to check; in practice the solution concept is chosen case by case — these distinctions are also useful to argue about how “rational” human behaviour is.

13.4. Signaling games

When the typed player moves first, the move itself may reveal information. If instead only the second mover has types, the first can only guess the reaction from the prior: a **screening** game (think of a quiz show with a prize behind a screen), where SPE is enough.

Definition 13.5 (Signaling game). A two-player dynamic Bayesian game in which:

1. Nature chooses the type of player 1 (the **sender**);
2. player 1 observes the type and chooses an action (the **signal**), with at least as many actions as types;
3. player 2 (the **receiver**) observes the signal, not the type, updates beliefs, and acts;
4. player 2 cares about player 1's type (common values).

Definition 13.6 (Separating, pooling and hybrid equilibria). In a **separating** equilibrium different types send different signals, so the signal reveals the type. In a **pooling** equilibrium all types send the same signal, which carries no information. **Hybrid** (semi-separating, partially pooling) equilibria have some types randomising so that some signals are partially informative.

Separating equilibria have easy beliefs (each signal points to one type); in pooling equilibria the on-path signal leaves the prior unchanged, while the unused signal is off the path and its beliefs must be chosen to support the equilibrium.

13.4.1. A recipe for pure PBE in signaling games

With two sender types and two messages, there are only four sender strategies to try:

1. **Fix a candidate:** pooling (both types send the same message) or separating (they send different ones).
2. **Beliefs on the path** follow from Bayes' rule: after a separating message the type is known; after a pooling message the belief is the prior.
3. **Beliefs off the path** (after a message nobody sends) are free in a PBE. Find the range of beliefs that makes the receiver's prescribed response optimal.
4. **Receiver:** check that the response to each message is a best reply to the belief.
5. **Sender:** check that no type gains by switching message, given the receiver's responses.

A pooling candidate survives if *some* off-path belief deters deviation. A separating candidate needs no off-path beliefs, but each type must prefer its own message.

Revealing (verifiable) actions. Some messages cannot lie: a medical test, a public audit, a costly campaign that exposes the product. After such an action the receiver's information set is a singleton and backward induction applies directly. Only the non-revealing message needs a belief. The game then shrinks: the receiver has one real decision, and the sender asks whether revealing is worth its cost for each type. Typically the “good” types reveal and the “bad” types stay silent, so silence itself becomes informative (*unraveling*).

13.5. Worked example: a coffee for Brian

Brian is invited for coffee by his colleague Zoe. His partner Ann is **J**ealous with probability 0.8 or **E**asygoing with probability 0.2. Ann may stay **S**ilent or **T**rash Zoe; Brian observes the signal and **C**omes to the coffee or **D**eclines.

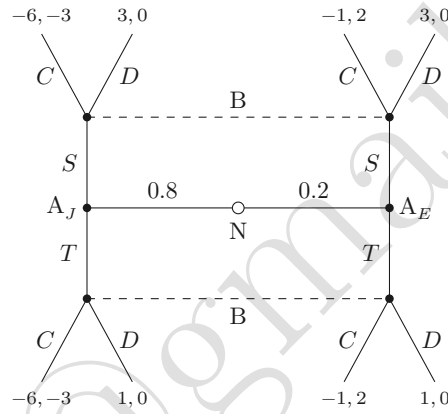


Figure: the signaling game in “butterfly” form; payoffs are (Ann, Brian). Brian’s information sets join the nodes after the same signal.

Jealous Ann is deeply hurt if Brian goes; easygoing Ann is only mildly annoyed; trashing Zoe costs Ann 2; Brian enjoys the coffee (2) only if Ann is easygoing and suffers (−3) if she is jealous.

Step 1: the normal form. Ann’s strategy lists the signal of each type (*ST*: silent if jealous, trash if easygoing); Brian’s lists the reply to each signal (*CD*: come after *S*, decline after *T*). Careful: Ann’s pair is indexed by *types*, Brian’s by *signals*, so filling e.g. (*TS, CD*) requires matching Brian’s reply to the signal each type sends.

Ann \ Brian	CC	CD	DC	DD
SS	−5, −2	−5, −2	3, 0	3, 0
ST	−5, −2	−4.6, −2.4	2.2, 0.4	2.6, 0
TS	−5, −2	0.6, 0.4	−4.2, −2.4	1.4, 0
TT	−5, −2	1, 0	−5, −2	1, 0

There are five Nash equilibria: three pure, (*SS, DD*), (*SS, DC*), (*TT, CD*), and two mixed, (*TT*, $\frac{1}{2}CD + \frac{1}{2}DD$) and ($\frac{1}{6}SS + \frac{5}{6}TS$, $\frac{2}{9}CD + \frac{7}{9}DD$).

Step 2: beliefs. Let μ_S (μ_T) be Brian’s belief that Ann is jealous after *S* (after *T*). Coming is worth $-3\mu + 2(1 - \mu)$ to Brian, declining 0: he comes iff $\mu \leq 0.4$.

- **No separating PBE.** If Ann played ST , beliefs would be $\mu_S = 1$, $\mu_T = 0$ and Brian would reply DC ; but against DC Ann prefers SS . Similarly TS induces CD , against which Ann prefers TT .
- **PBE 1:** (SS, DD) with $\mu_S = 0.8$ (the prior, since both types are silent) and off-path $\mu_T \geq 0.4$ (so that declining after T is optimal). Ann has no reason to trash Zoe since Brian declines anyway.
- **PBE 2:** (SS, DC) with $\mu_S = 0.8$, $\mu_T \leq 0.4$: Brian still declines in practice, but would come if Ann trashed Zoe, because he would read it as a sign that she is easygoing.
- **PBE 3:** (TT, CD) with $\mu_T = 0.8$ and off-path $\mu_S \leq 0.4$.
- **PBE 4:** Ann plays TT ; Brian declines after T ($\mu_T = 0.8$); off the path, Brian believes $\mu_S = 0.4$ — which makes him indifferent — and comes with probability q after S. Ann must not want to deviate to S: the easygoing type requires $-q + 3(1 - q) \leq 1$, i.e. $q \geq \frac{1}{2}$. So this is a continuum of PBE (with $q = 1$ recovering PBE 3); note that (TT, DD) itself is *not* an equilibrium.
- **PBE 5 (semi-separating):** easygoing Ann is always silent; jealous Ann is silent with probability $\frac{1}{6}$. T is sent only by the jealous type, so $\mu_T = 1$ and Brian declines. After S, Bayes' rule gives

$$\mu_S = \frac{0.8 \cdot \frac{1}{6}}{0.8 \cdot \frac{1}{6} + 0.2} = 0.4,$$

exactly the value that makes Brian indifferent, so he can mix. He comes with probability $\frac{2}{9}$ after S, which makes jealous Ann indifferent between S ($-6 \cdot \frac{2}{9} + 3 \cdot \frac{7}{9} = 1$) and T (1).

💡 Tip

The recipe for signaling games: (1) write the normal form, minding the type/signal indexing; (2) find all NE; (3) for each, compute on-path beliefs by Bayes' rule and find off-path beliefs that make the receiver's prescribed replies optimal — if none exist, the NE is not a PBE.

13.6. Education as a signal and the intuitive criterion

Spence's (1973) job-market model shows how a costly action can transmit information even when it is useless in itself. A worker is of High productivity (2) with probability p or Low (1). Before being hired, the worker chooses an education level $e \geq 0$, which costs e to a Low type and $e/2$ to a High type. Competitive employers pay the expected productivity given their belief after observing e .

- **Separating PBE.** High chooses e^* and Low chooses 0; the wage is 2 after e^* and 1 otherwise. Low must not mimic, $1 \geq 2 - e^*$; High must prefer to signal, $2 - e^*/2 \geq 1$. Any $e^* \in [1, 2]$ works, supported by the belief "anyone with $e < e^*$ is Low".
- **Pooling PBE.** Both choose $e = 0$ and receive $\bar{w} = 1 + p$, with pessimistic beliefs off the path.

Perfect Bayesian equilibrium allows any belief after an unexpected action, hence the multiplicity. The **intuitive criterion** (Cho and Kreps, 1987) asks: *could this deviation possibly benefit this type?* If a deviation can only benefit one type, the responder should believe it comes from that type.

- In the separating family, an education level $e \in (1, e^*)$ would never pay for a Low worker (even with wage 2, $2 - e < 1$), but would pay for a High one. So the beliefs " $e < e^*$ means Low" are unreasonable, and only the **least-cost separating** equilibrium $e^* = 1$ survives (the Riley outcome).
- The pooling equilibrium always fails the criterion (for $p < 1$). Any $e \in (1 - p, 2(1 - p))$ cannot benefit Low even at wage 2 ($2 - e < 1 + p$), but if read as High it gives High $2 - e/2 > 1 + p$.

The lessons generalise: signals work because they are **costlier for the types you want to separate from** (single crossing), and equilibrium refinements select the cheapest credible signal. In networks the same logic underlies proof-of-work, costly authentication and reputation deposits.

13.7. Reputation building

In a finitely repeated Prisoner's Dilemma the unique SPE is to defect always. A little uncertainty about types changes this dramatically (Kreps, Milgrom, Roberts and Wilson, 1982).

Stage payoffs are $(C, C) = (1, 1)$, $(C, D) = (-1, 2)$, $(D, C) = (2, -1)$, $(D, D) = (0, 0)$. Player B is strategic. Player A is a **grim-trigger** type with probability p (cooperates until B defects, then defects forever) or strategic with probability $1 - p$.

- **Two rounds.** In round 2 strategic players defect. In round 1 strategic A defects, and B, cooperating, gets $p(1) + (1 - p)(-1)$ in round 1 plus $2p$ in round 2 (exploiting the grim-trigger type, who still cooperates): total $4p - 1$. Defecting gives $2p$ (round 1) and 0 after. B cooperates in round 1 iff $p > \frac{1}{2}$.
- **Three rounds.** If $p > \frac{1}{2}$, the unique PBE has *both* B and the strategic A cooperating in round 1. Strategic A pools with the grim-trigger type: by cooperating he keeps B's belief at $p > \frac{1}{2}$, so B keeps cooperating in round 2, and A earns $1 + 2 + 0$ instead of $2 + 0 + 0$ from revealing himself. B, in turn, cooperates in round 1 even if he expected the strategic A to defect, since $5p - 1 > 2p$ for $p > \frac{1}{3}$.

Strategic players **build a reputation**: they defect at the end, but to an outside observer they cooperate. In general, in a T -stage game the number of rounds in which a strategic player defects is bounded by a constant depending on p but not on T .

The tough negotiator. In an ultimatum game the proposer offers a **High** share (payoffs $(2, 1)$ if accepted) or a **Moderate** one $((1, 2))$; refusal gives $(0, 0)$. The unique SPE is $(H, \text{always accept})$. Now let the responder be a “jerk” with probability p , who finds H worse than nothing (payoff -1). In one round the proposer offers H iff $2(1 - p) > 1$, i.e. $p < \frac{1}{2}$. With a second round (after a refusal the proposer offers again, payoffs discounted by δ) the game becomes intractable in normal form — the responder has 2^{12} strategies — but sequential rationality reduces it. With $\delta = 0.9$ and $p = 0.1$ the unique PBE is semi-separating: H is always offered first; the jerk always refuses; a normal responder refuses with probability $\frac{1}{9}$ (so that after a refusal $\mu_H = \Pr[\text{jerk} \mid \text{refusal}] = \frac{1}{2}$, making the proposer indifferent between H, worth $2\delta(1 - \mu_H)$, and M, worth δ); in round 2 the proposer repeats H with probability $\frac{8}{9}$ (making the normal responder indifferent: $\delta[\frac{8}{9} \cdot 1 + \frac{1}{9} \cdot 2] = 1$). Even an unlikely “tough” type gives the responder bargaining power: a normal responder sometimes mimics it.

13.8. Cheap talk

In **cheap talk** (Crawford and Sobel, 1982), Nature reveals a state t to player 1 only; player 1 sends a message that costs nothing and has no direct effect on payoffs; player 2 then acts. The question is how much information costless messages can transmit when interests are not aligned.

Example 13.3 (Where to live)

Player 2 found a job in city A and chooses where to live, $a \in \{1, \dots, 5\}$ ($1 = \text{city A}$, $5 = \text{city B}$ where player 1 lives). The best location depends on traffic conditions $t \in \{1, 3, 5\}$, equally likely and known only to player 1. Player 2's payoff is $5 - (t - a)^2$; player 1 is biased towards having the friend close: $5 - (t + b - a)^2$ with $b = 1.1$.

1. **No truthful PBE.** If messages were believed, the type $t = 3$ would claim “5”: the friend would choose $a = 5$, giving player 1 $5 - 0.81 = 4.19$ instead of $5 - 1.21 = 3.79$.
2. **A babbling PBE** always exists: messages are uninformative and player 2 chooses $a = 3$ from the prior.
3. **A partially informative PBE:** player 1 reports $t = 1$ truthfully but pools $t \in \{3, 5\}$ into “high”. Player 2 then chooses $a = 1$ or $a = 4$. Type 1 prefers the truth (3.79 vs. 1.39); types 3

and 5 prefer “high”.

With a continuous state $t \in [0, 1]$ and quadratic losses, there is never a truthful equilibrium, there is always a babbling one, and a **two-message (threshold) equilibrium** exists iff the bias is small enough ($b < \frac{1}{4}$ for uniform t). Applications include legislative committees and expert advisers: with the decision maker’s payoff $-(t - a)^2$, $t \sim \mathcal{U}[-w, w]$, and the adviser’s $-(t + b - a)^2$, informative advice (under open or closed rules) is possible only if $b \leq w$.

Note

The “negative” results of cheap talk justify a common engineering practice: protocols exchange **coarse** information — congested / not congested, low / high battery — rather than exact values. With slightly misaligned incentives, more detail would not be credible anyway.

13.9. Exercises

Problems from Badia and Marchioro (BM), Chapter 6.

Exercise 13.1

(BM 6.6) Mark buys trousers Immediately at €50 or Waits, in which case Nadia Keeps the price or Lowers it to €30 at the end of the week. Mark values them at €100. Late transactions are discounted by $\delta = 0.8$. Nadia’s cost is €20 with probability 0.75 (and lowering the price then also brings €20 of goodwill, not discounted) or €35. Find the BNE and the SPE.

Solution

Nadia’s strategies are (cost 20, cost 35). Waiting gives Mark 40 (K) or 56 (L). It gives Nadia 24 (K) or 28 (L) at cost 20, and 12 (K) or -4 (L) at cost 35. Buying immediately gives Mark 50 and Nadia 30 or 15. Normal form, Mark W against KK, KL, LK, LL: (40, 21), (44, 17), (52, 24), (56, 20); Mark I: (50, 26.25). BNE: (I, KK) , (I, KL) , (W, LK) . Sequential rationality requires L at cost 20 and K at cost 35, so Nadia plays LK. Then Mark waits ($52 > 50$). The unique SPE is (W, LK) . (The printed payoffs contain small arithmetic slips; the equilibria are unchanged.)

Exercise 13.2

(BM 6.7) Janine brings food from Home (payoff 2; stand 0) or Eats at the stand. The worker there is Alfred (0.8) or Brianna (0.2) and Gives a sandwich (Janine 5, worker 1) or Denies it (Janine 0, worker 0; but Brianna, who dislikes Janine, gets 5). Find the BNE and the SPE.

Solution

The stand’s strategies are (Alfred, Brianna). E against GG, GD, DG, DD gives (5, 1), (4, 1.8), (1, 0.2), (0, 1); H gives (2, 0). BNE: (E, GD) , (H, DG) , (H, DD) . Sequential rationality: Alfred gives, Brianna denies, so GD, and Janine eats ($4 > 2$). SPE (E, GD) ; the other two rest on Alfred threatening to deny. (The printed expected payoffs of the stand are slightly off: 1.8, not 2.8.)

Exercise 13.3

(BM 6.8) Riccardo is infected with probability 0.2. At the border he takes a Swab (revealing his status; cost 10) or Talks. The officer Admits or Denies. Riccardo gets 20 if admitted, 0 otherwise. The officer gets 5 for admitting a healthy traveller, -50 for an infected one, 0 for denying. Find the PBE and classify it.

Solution

After a swab the officer knows the type: he admits the healthy (Riccardo 10) and denies the infected (-10). After T, let μ be the officer's belief that Riccardo is infected; admitting pays $5 - 55\mu$, so he admits iff $\mu \leq 1/11$. The infected type never swabs ($-10 < 0$). If the healthy type talked too (pooling), $\mu = 0.2 > 1/11$ and the officer would deny, so the healthy type would rather swab ($10 > 0$). PBE: infected Talks, healthy Swabs, the officer believes $\mu(T) = 1$ and Denies after T, and admits after a negative swab. It is **separating**: the costly swab is the signal only the healthy type is willing to send.

Exercise 13.4

(BM 6.9) John and Mary exchange gifts at Christmas and on Valentine's Day (two rounds, $\delta = 1$): a gift costs the giver 5 and is worth 12 to the receiver. Mary is strategic. With probability p John is a tit-for-tat type (gives at Christmas, then copies Mary); otherwise he is strategic. (a) Payoffs if $p = 0$? (b) Probability that John does not give at Christmas? (c) When does Mary give at Christmas?

Solution

(a) A twice-repeated Prisoner's Dilemma: (N, N) twice, payoffs 0. (b) Only the strategic John withholds at Christmas: probability $1-p$. (c) Mary never gives on Valentine's Day (the last round). If she withholds at Christmas she gets $12p$ (the tit-for-tat John's Christmas gift). If she gives, she gets $p(12-5+12) - 5(1-p) = 24p-5$: tit-for-tat John rewards her on Valentine's Day. She gives iff $24p-5 > 12p$, i.e. $p > 5/12$. A modest probability of a reciprocating type makes even a selfish player cooperate early: the seed of the reputation results of Kreps–Milgrom–Roberts–Wilson.

Exercise 13.5

(BM 6.10) A clerk Stays put (bandit 100, clerk 0) or Punches a bandit whose gun is fake with probability p . After a punch the bandit Reacts or Flees. Fake gun and react: both -10 . Real gun and react: clerk -100 , bandit 80. Flee: clerk 10, bandit 0. Find the BNE as a function of p and the SPE.

Solution

The bandit's strategies are (fake, real). P against RR, RF, FR, FF: clerk $90p-100$, $10-20p$, $110p-100$, 10. S gives 0 against everything. Sequential rationality: flee with a fake gun ($0 > -10$) and react with a real one ($80 > 0$), i.e. FR. The clerk punches iff $110p-100 \geq 0$, i.e. $p \geq 10/11$. BNE: (S, RR) for all p ; (S, RF) for $p \geq 1/2$; (S, FR) for $p \leq 10/11$; (P, FR) for $p \geq 10/11$. Only those with FR are SPE. The clerk should play the hero only if he is almost sure the gun is fake.

Exercise 13.6

(BM 6.12) A start-up's gadget is Effective or Useless (probability $\frac{1}{2}$ each). It runs a Campaign (cost 20, reveals the type) or keeps a Low profile. The incumbent Fights or Yields. Yield: 50 each. Fight: (80, 0) if Effective, (0, 80) if Useless. Find the PBE.

Solution

After a campaign the incumbent yields to an effective gadget ($50 > 0$) and fights a useless one ($80 > 50$), so C yields 30 (effective) or -20 (useless). After L, with belief $\mu = \Pr(\text{effective})$, fighting pays $80(1 - \mu)$ against 50 for yielding. Normal form (start-up strategies (effective, useless), incumbent's answer to L): CC (5, 65) for both answers; CL (15, 65) / (40, 50); LC (30, 40) / (15, 65); LL (40, 40) / (50, 50). LL is strictly dominant. With $\mu = 1/2$ the incumbent yields ($50 > 40$). **Pooling** PBE: both types keep a low profile, the incumbent believes the prior after L and yields. The campaign is too expensive to be worth revealing even good news.

Exercise 13.7

(BM 6.16) A student's green pass is valid (0.9) or fake (0.1). The student Displays or Hides it; the professor Admits or sends Online, unable to tell fakes apart. Valid: admitted (10, 10); displayed but sent online: professor -50 ; otherwise 0. Fake: admitted after display ($-100, -100$); admitted without display ($-50, -200$); displayed and sent online: student -200 ; otherwise 0. Find the pure PBE.

Solution

Student strategies (valid, fake), professor strategies (after D, after H). The pure NE are (DH, AO) and (HH, OO) . **Separating** (DH, AO) : valid passes are displayed, fakes hidden. Beliefs $\mu(\text{valid} | D) = 1$ and $\mu(\text{valid} | H) = 0$; admitting after D and sending online after H are optimal. **Pooling** (HH, OO) : nobody displays, and $\mu(\text{valid} | H) = 0.9$ makes O optimal ($0 > 0.9 \cdot 10 - 0.1 \cdot 200 = -11$). D is off the path, so its belief is free. Admitting after D pays $110\mu - 100$ and online pays -50μ , so O after D is optimal iff $\mu(\text{valid} | D) \leq 5/8$. The pooling PBE needs pessimistic off-path beliefs. By the intuitive criterion, displaying could only benefit the valid type, so the pooling equilibrium is not reasonable.

From the course material:

Exercise 13.8

A young cowboy Y, insulted by an outlaw O, can let it slide ($(-20, 0)$) or challenge O. O can apologise ($(10, -10)$) or duel. If O is a sharpshooter (prior p) the duel gives $(-100, 20)$; otherwise each wins with probability $\frac{1}{2}$, the winner getting 20 and the loser -100 . Find the PBE for $p = 0.2$.

Solution

For a non-sharpshooter the duel is worth $\frac{1}{2} \cdot 20 + \frac{1}{2}(-100) = -40$ to both players. Sequential rationality: a sharpshooter duels ($20 > -10$); a fake apologises ($-10 > -40$). Then challenging gives Y $p(-100) + (1 - p)(10) = 0.2(-100) + 0.8 \cdot 10 = -12 > -20$, so Y challenges. The unique PBE (indeed the unique SPE of the type-agent tree) is: Y challenges; a sharpshooter O duels, a fake O apologises. Y would let it slide only if $-100p + 10(1 - p) < -20$, i.e. $p > 3/11 \approx 0.27$. This is a screening game — the uninformed player moves first and O knows his own type — so beliefs are trivial and SPE suffices; in normal form there is also a NE in which Y lets it slide and O threatens to duel whatever his type, but it is not sequentially rational.

Exercise 13.9

In the coffee game, check that (TT, DD) is not a Nash equilibrium and explain in words why PBE 4 needs Brian to come after S with probability at least $\frac{1}{2}$.

Solution

Against DD , Ann gets 3 from SS and 1 from TT : she deviates to silence. In PBE 4 Brian's reply to the off-path signal S must deter *both* types from switching to S . The jealous type gets $-6q + 3(1 - q) \leq 1$ iff $q \geq \frac{2}{9}$; the easygoing type gets $-q + 3(1 - q) \leq 1$ iff $q \geq \frac{1}{2}$. The binding constraint is the easygoing type's, hence $q \geq \frac{1}{2}$.

13.10. Sources

Prof. Badia, *Game Theory* slides, decks **Gth2015** and **Gth2016** (dynamic Bayesian games, PBE, sequential equilibrium, signaling games, the coffee example) and **Gth2017** (reputation building, the tough negotiator, cheap talk); exercise sheet **gt20b**. Tadelis, *Game Theory: An Introduction*, Chapters 15–18. Original references: D. Kreps and R. Wilson, *Sequential equilibria*, *Econometrica* **50**, 863 (1982); D. Kreps, P. Milgrom, J. Roberts and R. Wilson, *J. Econ. Theory* **27**, 245 (1982); V. Crawford and J. Sobel, *Strategic information transmission*, *Econometrica* **50**, 1431 (1982). Exercises: L. Badia and T. Marchioro, *Game Theory: A Handbook of Problems and Exercises*, Esculapio (2022), Chapter 6. M. Spence, *Job market signaling*, *Q. J. Econ.* **87**, 355 (1973); I.-K. Cho and D. Kreps, *Signaling games and stable equilibria*, *Q. J. Econ.* **102**, 179 (1987).

Part V.

Part V: Mechanisms and cooperation

14. Auctions and mechanism design

So far the game was given and we predicted its outcome. **Mechanism design** reverses the question: a designer wants a certain outcome — the good goes to whoever values it most, a protocol allocates bandwidth efficiently — and must *design the game* so that self-interested players with private information produce it in equilibrium. Auctions are the most important example, and the second-price auction shows that a well-designed game can make telling the truth a dominant strategy. The revelation principle then tells us where to look for good mechanisms.

14.1. Sealed-bid auctions

Definition 14.1 (Sealed-bid auction). One seller and N bidders; bidder i privately values the object $v_i \geq 0$. Each bidder submits a bid $b_i \geq 0$ without seeing the others. The highest bid wins; ties are broken by sharing (or by a coin flip).

14.1.1. First-price auction

The winner pays their own bid. With W tied winners,

$$u_i(b) = \begin{cases} v_i - b_i & \text{if } i \text{ is the only highest bidder,} \\ (v_i - b_i)/W & \text{if } i \text{ is among } W > 1 \text{ highest bidders,} \\ 0 & \text{otherwise.} \end{cases}$$

- Bidding $b_i \geq v_i$ always yields $u_i \leq 0$, so it is weakly dominated by bidding below v_i .
- Bidding $b_i = 0$ wins only if everyone bids 0, and is weakly dominated by a small positive bid.
- Between two bids $0 < b_i < b'_i < v_i$ nothing can be said in general: against opponents bidding 0 the lower bid is better, against opponents bidding $(b_i + b'_i)/2$ the higher one is.

Rational bidders **shade** their bids below their value. With incomplete information one computes the Bayesian Nash equilibrium: for two bidders with values independent and uniform on $[0, 1]$, the symmetric BNE is $b(v) = v/2$. Indeed, if the opponent bids $v_j/2$, bidding b wins with probability $\Pr[v_j/2 < b] = 2b$, and $\max_b (v - b) 2b$ gives $b = v/2$. (With N bidders, $b(v) = \frac{N-1}{N}v$.)

14.1.2. Second-price (Vickrey) auction

Proposed by Vickrey (1961): the highest bidder wins but pays the **second-highest** bid $S = \max_{j \neq i} b_j$:

$$u_i(b) = \begin{cases} v_i - S & \text{if } b_i > S, \\ (v_i - S)/W & \text{if } b_i = S \text{ with } W \text{ tied winners,} \\ 0 & \text{if } b_i < S. \end{cases}$$

Theorem 14.1 (Truthful bidding is weakly dominant). *In a second-price auction, bidding $b_i = v_i$ weakly dominates every other bid.*

Proof. Compare v_i with a lower bid $b_i < v_i$, for each value of S :

- $S < b_i$: both bids win alone and pay S : same payoff $v_i - S$.
- $S = b_i$: b_i ties and gets $(v_i - S)/W$; v_i wins alone and gets $v_i - S > (v_i - S)/W$.
- $b_i < S < v_i$: b_i loses (0), v_i wins ($v_i - S > 0$).
- $S \geq v_i$: b_i loses (0); v_i loses or ties at zero surplus (0).

Now a higher bid $b_i > v_i$:

- $S < v_i$: both win and pay S : same payoff.
- $S = v_i$: both get 0.
- $v_i < S$: v_i loses (0); b_i either loses (0) or wins/ties paying $S > v_i$ (negative payoff).

In every case v_i is at least as good, and strictly better in some. $\square$

So $(v_1, \dots, v_n)$ is a Nash equilibrium (in dominant strategies — no beliefs needed), and the second-price auction makes bidders reveal their true values. In a first-price auction, by contrast, bidding v_i always gives 0 and is dominated.

Efficiency. Social surplus is maximised if the object goes to the bidder with the highest value. In the second-price auction this bidder wins and pays S : the bidder gets $v - S$, the seller S , total v — Pareto efficient. For the first-price auction with symmetric bidders the BNE is also efficient (bids are increasing in values), and with independent private values the two formats yield the same expected revenue: for two uniform bidders, $\mathbb{E}[\max_i v_i]/2 = \frac{1}{3} = \mathbb{E}[\min_i v_i]$ (the **revenue equivalence theorem**).

14.2. Revenue equivalence and reserve prices

First-price bidders shade their bids and second-price bidders do not, but the winner of a second-price auction pays only the second-highest value. Which raises more money? With N bidders and i.i.d. values uniform on $[0, 1]$, both give expected revenue $\frac{N-1}{N+1}$. This is no coincidence:

Theorem 14.2 (Revenue equivalence (Myerson, 1981; Riley and Samuelson, 1981)). *With risk-neutral bidders and independent private values from a common continuous distribution, every auction format in which (i) the highest-value bidder always wins and (ii) a bidder with the lowest possible value expects zero payment yields the **same expected revenue** and the same expected payment for each type.*

The reason is the envelope theorem. A type's equilibrium utility grows with its value at a rate equal to its winning probability, which is fixed by (i). Payments are then pinned down by (ii). First-price, second-price, English, Dutch, all-pay and even third-price auctions (where bidders bid *above* their values) all raise $\frac{N-1}{N+1}$ in the uniform case. Revenue equivalence fails with risk-averse bidders (first-price then raises more, as shading is risky), with asymmetric bidders, and with correlated or common values.

Reserve prices. The seller can beat every efficient format by refusing to sell below a reserve r . With two uniform bidders and a second-price auction, revenue is $R(r) = \frac{1+3r^2-4r^3}{3}$, maximised at $r^* = \frac{1}{2}$ with $R = \frac{5}{12} > \frac{1}{3}$. The seller gives up some efficient trades (when both values are below $\frac{1}{2}$) to extract more from a lone high bidder. In general r^* solves $r - \frac{1-F(r)}{f(r)} = 0$ (zero *virtual value*), independently of the number of bidders.

14.3. Common values and the winner's curse

When the object has the same, unknown value for everyone (an oil tract, a spectrum licence), each bidder holds a noisy estimate. Winning is bad news: it means your estimate was the most optimistic. A bidder who bids as if their own estimate were the truth overpays systematically — the **winner's curse**. Rational bidders condition on winning, $\mathbb{E}[V \mid \text{my signal, I win}]$, and shade more as the number of competitors grows. Revenue equivalence breaks down here: open ascending auctions, which reveal others' information as bidders drop out, raise more revenue than sealed bids (the *linkage principle*).

14.4. Vickrey–Clarke–Groves auctions

A **VCG auction** extends the second-price idea to several items. Each bidder submits a value for every *combination* of items; the seller chooses the allocation maximising the total declared value (possibly a hard combinatorial problem); then each winner pays for the harm they cause to the others:

$$p_i = \underbrace{V(\mathcal{N} \setminus \{i\})}_{\text{others' best total value without } i} - \underbrace{V_{-i}(x^*)}_{\text{others' value in the chosen allocation}}.$$

Example 14.1 (Two items, three bidders)

A bids 5 for a single item, B bids 4 for a single item, C bids 7 for the pair only. The optimal allocation gives one item to A and one to B (total 9 > 7). Without A, the best the others could do is C taking both (7); in the chosen allocation the others get 4 (B). So A pays 7 – 4 = 3. Symmetrically B pays 7 – 5 = 2, and C, who wins nothing and harms nobody, pays 0.

VCG mechanisms are truthful: bidding one's true valuations is weakly dominant, for the same reason as in the Vickrey auction — a bidder's report affects only the allocation, never the price conditional on the allocation, and the allocation maximises the bidder's own utility plus a term they cannot affect. Yet VCG is rare in practice: revenue can be low (even zero), it is non-monotone (adding bidders can lower revenue), and it is vulnerable to collusion and shill bidding.

14.5. Mechanism design

Players $1, \dots, n$ must choose among public alternatives X . Their preferences depend on private types $t = (t_1, \dots, t_n) \in T = T_1 \times \dots \times T_n$, the **state of the world**; types are private, the prior is common knowledge. Payoffs are quasi-linear in money,

$$v_i(x, t_i, m_i) = u_i(x, t_i) + m_i,$$

where m_i is a monetary transfer.

Definition 14.2 (Choice rule). A **choice rule** is a map $f : T \rightarrow Y$, with $Y = \{(x, m_1, \dots, m_n) : x \in X, \sum_i m_i \leq 0\}$. It consists of a **decision rule** $x(t)$ and a **transfer rule** $m(t)$; the budget constraint says the designer does not inject money.

The designer would like to implement f , but does not know t , and players might lie.

Definition 14.3 (Mechanism and implementation). A **mechanism** $\Gamma = (A_1, \dots, A_n; g)$ specifies an action set A_i for each player and an outcome function $g : A_1 \times \dots \times A_n \rightarrow Y$. In the induced Bayesian game a pure strategy is $s_i : T_i \rightarrow A_i$. The mechanism **implements** f (in BNE) if it has a Bayesian Nash

equilibrium s^* with $g(s^*(t)) = f(t)$ for every $t \in T$.

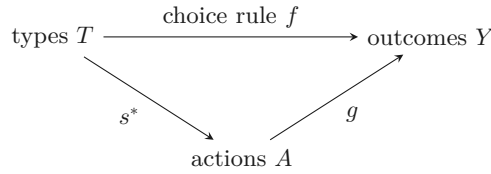


Figure: a mechanism implements f if equilibrium play s^* followed by the outcome function g reproduces f .

14.6. The revelation principle

Definition 14.4 (Direct revelation mechanism). A mechanism is a **direct revelation mechanism** if $A_i = T_i$ for every i and $g = f$: the designer simply asks players to report their types and applies the choice rule. f is **truthfully implementable** in BNE if truth-telling, $s_i^*(t_i) = t_i$ for all i and t_i , is a BNE of its direct revelation mechanism.

Theorem 14.3 (Revelation principle). A choice rule f is implementable by some mechanism if and only if it is truthfully implementable in Bayesian Nash equilibrium.

Proof. ($\Leftarrow$) The direct mechanism is a mechanism. ($\Rightarrow$) Let Γ implement f through the equilibrium s^* . Build a direct mechanism that, on reports $\hat{t}$, computes $g(s^*(\hat{t}))$: it “plays the equilibrium on behalf of” each player. If some type gained by misreporting $\hat{t}_i$, the same type would have gained in Γ by playing $s_i^*(\hat{t}_i)$ instead of $s_i^*(t_i)$, contradicting that s^* is an equilibrium. $\square$

The consequence is powerful: to know whether *any* clever mechanism can achieve f , it suffices to check whether truth-telling is an equilibrium of the direct one. If the direct mechanism fails, no other mechanism will work. VCG mechanisms are exactly direct mechanisms in which truth-telling is even dominant.

Tip

In ICT, “mechanisms” are protocols: a scheduler asking users to declare their traffic demand, a spectrum auction, a crowdsensing platform paying for reported measurements. The revelation principle says one can restrict attention to protocols that ask for the private information directly and pay so that reporting it honestly is in each user’s interest.

14.7. Exercises

Badia and Marchioro’s handbook has no auction problems; these are from Tadelis (T), Chapters 13–14.

Exercise 14.1

(T 13.2) In a second-price auction with complete information (everyone knows all values), is bidding one’s value still weakly dominant? Are there other NE?

Solution

Yes: dominance compares payoffs profile by profile, so information about the others is irrelevant. But there are many other NE. For example, the bidder with the *lowest* value bids above the highest value

and everyone else bids 0. The low bidder wins at price 0; anyone else could win only by outbidding, paying more than their value. Such equilibria are in weakly dominated strategies and are inefficient.

Exercise 14.2

(T 13.3) Two bidders with values in $\{0, 1, 2\}$ (equally likely, independent) face a second-price auction with reserve r : if only one bid reaches r that bidder pays r ; if none does, the good is not sold. (a) Is truthful bidding still dominant? (b)–(c) Expected revenue for $r = 0$ and $r = 1$? (d) The optimal r ?

Solution

(a) Yes: the reserve acts like an extra bid by the seller. (b) Revenue is the lower value: $\mathbb{E}[\min] = 1 \cdot \frac{3}{9} + 2 \cdot \frac{1}{9} = \frac{5}{9}$. (c) With $r = 1$: if both values are ≥ 1 (probability $4/9$), the minimum is paid, contributing $\frac{5}{9}$; if exactly one is ≥ 1 ($4/9$), that bidder pays 1, contributing $\frac{4}{9}$. Total 1. (d) For $r \in (1, 2]$ revenue is $\frac{1}{9} \cdot 2 + \frac{4}{9}r$, maximised at $r = 2$: $\frac{10}{9}$, double the no-reserve revenue. (Above 2 nothing sells.) The reserve extracts the full value from a lone high bidder, at the cost of never selling to two low ones.

Exercise 14.3

(T 13.4) Two bidders with values uniform on $[0, 1]$; second-price auction with reserve r . Show that $r = 0$ is not optimal and find the optimal r .

Solution

Revenue: if both values exceed r the seller gets $\min(v_1, v_2)$; if exactly one does, r . Integrating, $R(r) = \frac{1+3r^2-4r^3}{3}$. $R'(r) = 2r - 4r^2 > 0$ for $0 < r < \frac{1}{2}$, so a small reserve raises revenue (the loss, not selling when both values are tiny, is second order). The optimum is $r^* = \frac{1}{2}$ with $R = \frac{5}{12}$, versus $\frac{1}{3}$ without a reserve; a simulation with 10^6 draws gives 0.4167.

Exercise 14.4

(T 13.5) First-price auction, two bidders with values 4 or 8 (probability $\frac{1}{2}$ each). Show that the low type bids 4 and the high type mixes on $[4, 6]$; find the mixture and the seller's revenue, and compare with an English auction.

Solution

A low type bidding below 4 would be overbid by the other low type at no cost, so $b_L = 4$ (zero surplus). The high type wins against a low rival by bidding just above 4, earning $8 - 4 = 4$ with probability $\frac{1}{2}$, so its equilibrium payoff is 2. A bid b must give the same: $(8 - b)(\frac{1}{2} + \frac{1}{2}F(b)) = 2$, so

$$F(b) = \frac{b-4}{8-b}, \quad b \in [4, 6].$$

The support ends where $F = 1$, at $b = 6$. (Tadelis states “uniformly”, but uniform bids are not optimal: e.g. $b = 5$ would give 2.25.) Expected high bid $8 - 4 \ln 2 = 5.23$, expected maximum of two high bids 5.55. Revenue: $\frac{1}{4} \cdot 4 + \frac{1}{2} \cdot 5.23 + \frac{1}{4} \cdot 5.55 = 5$. An English auction sells at the second-highest value: $\frac{3}{4} \cdot 4 + \frac{1}{4} \cdot 8 = 5$. Revenue equivalence holds even with discrete types here.

Exercise 14.5

(T 13.6) In a third-price auction ($n \geq 3$ bidders, values uniform on $[0, 1]$; the highest bid wins and pays the third-highest bid), verify that $s(\theta) = \frac{n-1}{n-2}\theta$ is a symmetric BNE, explain overbidding, and compute the revenue.

Solution

Bidding above value is harmless in the event that matters: winning is good whenever the price (the third-highest bid) is below your value. The second-highest bid is irrelevant to the price, which raises the value of winning. Checking the first-order condition of the expected payoff under the candidate strategy confirms $s(\theta) = \frac{n-1}{n-2}\theta$. Revenue: $\frac{n-1}{n-2} \mathbb{E}[\theta_{(3)}] = \frac{n-1}{n-2} \cdot \frac{n-2}{n+1} = \frac{n-1}{n+1}$, the same as first- and second-price auctions, as revenue equivalence predicts.

Exercise 14.6

(T 14.1, Brothers) Brother 1 ranks $A \succ B \succ C$. Brother 2 is type θ' ($C \succ B \succ A$) or θ'' ($B \succ A \succ C$). The parents ask brother 2 to report his type and implement $f(\theta') = B$, $f(\theta'') = A$. Is truth-telling dominant? What if $f(\theta') = C$, $f(\theta'') = A$? Or $f \equiv B$?

Solution

- (a) Type θ'' gets A by telling the truth but B , which he prefers, by lying. Not incentive compatible.
- (b) θ' gets his favourite C ; θ'' gets A truthfully versus C by lying, and prefers A . Truth-telling is dominant.
- (c) A constant choice is trivially incentive compatible, but it ignores the information. This is the revelation principle in miniature: a social choice function is implementable iff no type gains by imitating another.

From the course material:

Exercise 14.7

Three bidders with values 10, 7, 4 participate in a second-price auction. What are the dominant-strategy outcome and the seller's revenue? Show that bidder 2 cannot gain by bidding 12.

Solution

Truthful bids: bidder 1 wins and pays 7; revenue 7, bidder 1's surplus 3. If bidder 2 bids 12, he wins and pays 10 $>$ 7 = v_2 : payoff $-3 <$ 0.

Exercise 14.8

In a first-price auction with N bidders whose values are i.i.d. uniform on $[0, 1]$, verify that $b(v) = \frac{N-1}{N}v$ is a symmetric BNE and compute the expected revenue.

Solution

If the others bid $\frac{N-1}{N}v_j$, bidding b wins iff all $v_j < \frac{N}{N-1}b$, with probability $(\frac{Nb}{N-1})^{N-1}$. Maximising $(v-b)b^{N-1}$ gives $(N-1)(v-b) = b$, i.e. $b = \frac{N-1}{N}v$. Revenue: $\frac{N-1}{N} \mathbb{E}[\max v] = \frac{N-1}{N} \cdot \frac{N}{N+1} = \frac{N-1}{N+1}$, equal to the expected second-highest value — revenue equivalence.

14.8. Sources

Prof. Badia, *Game Theory* slides, deck Gth2018 (first- and second-price auctions, VCG, mechanism design, revelation principle). Tadelis, *Game Theory: An Introduction*, Chapters 13–14. Original references: W. Vickrey, *Counterspeculation, auctions, and competitive sealed tenders*, J. Finance **16**, 8 (1961); E. Clarke, *Public Choice* **11**, 17 (1971); T. Groves, *Econometrica* **41**, 617 (1973); R. Myerson, *Incentive compatibility and the bargaining problem*, *Econometrica* **47**, 61 (1979). Exercises from Tadelis, Chapters 13–14. R. Myerson, *Optimal auction design*, *Math. Oper. Res.* **6**, 58 (1981).

15. Coalitions and axiomatic bargaining

Non-cooperative game theory models each player's choices and asks what is self-enforcing. Cooperation then has to *emerge* — through repetition, reputation or punishment. Cooperative game theory takes a different stance: it assumes that players can sign binding agreements and asks which agreements are *stable* and which are *fair*. Its two main instruments are **coalitional games**, where groups of players generate value to be shared, and **axiomatic bargaining**, where a sensible split is characterised by a list of desirable properties rather than by a bargaining procedure.

15.1. Coalitional games

Definition 15.1 (Coalitional game with transferable utility). A coalitional (TU) game is a pair $(\mathcal{N}, v)$: a set of players $\mathcal{N}$ and a **characteristic function** $v : 2^{\mathcal{N}} \rightarrow \mathbb{R}$ with $v(\emptyset) = 0$. A **coalition** is any $S \subseteq \mathcal{N}$; $v(S)$ is the total value its members can generate on their own. Each player's action is the coalition to join (a loner joins $\{i\}$), and the value of a coalition is split among its members as payoffs x_i with $\sum_{i \in S} x_i = v(S)$.

Transferable utility means $v(S)$ can be split arbitrarily (think of money). In **non-transferable utility** (NTU) games, coalition members know in advance what each gets, and $v(S)$ is replaced by a set of achievable payoff vectors.

Definition 15.2 (Simple games). A coalitional game is **simple** if $v(S) \in \{0, 1\}$ for every S : coalitions are *winning* or *losing* (weighted voting, committees). A **veto player** belongs to every winning coalition; a **dictator** is a veto player whose singleton coalition is winning.

The set $\mathcal{N}$ is the **grand coalition**. Whether it forms — whether everybody cooperates — depends on how $v(\mathcal{N})$ is split: equal shares, proportional shares, the Shapley value, the nucleolus. A player who deems their share too small may leave.

Definition 15.3 (Superadditivity). $(\mathcal{N}, v)$ is **superadditive** if $v(S_1 \cup S_2) \geq v(S_1) + v(S_2)$ for all disjoint S_1, S_2 .

In a superadditive game cooperation is always beneficial and the grand coalition maximises total value; the question is how to split it so that nobody wants to leave.

15.2. The core

Definition 15.4 (Core). The **core** is the set of allocations $(x_1, \dots, x_n)$ such that

$$\sum_{i \in \mathcal{N}} x_i = v(\mathcal{N}) \quad (\text{efficiency}), \quad \sum_{i \in S} x_i \geq v(S) \quad \forall S \subseteq \mathcal{N} \quad (\text{coalitional rationality}).$$

No coalition can do better by breaking away: a non-empty core means a **stable grand coalition**. Drawbacks: the core is often empty; sometimes it is very large; and core allocations may be unfair.

Definition 15.5 (Balanced game). A collection of weights $\alpha(S) \geq 0$, $S \neq \emptyset$, is **balanced** if $\sum_{S \ni i} \alpha(S) = 1$ for every player i — think of $\alpha(S)$ as the fraction of time each member of S spends in S . The game is **balanced** if $\sum_S \alpha(S) v(S) \leq v(N)$ for every balanced collection.

Theorem 15.1 (Bondareva–Shapley). A TU game has a non-empty core if and only if it is balanced.

The proof is LP duality: the core is non-empty iff the minimum of $\sum_i x_i$ subject to $\sum_{i \in S} x_i \geq v(S)$ is at most $v(N)$, and the dual of that LP is the maximisation over balanced weights.

Definition 15.6 (Convex game). (N, v) is **convex** if $v(S_1 \cup S_2) + v(S_1 \cap S_2) \geq v(S_1) + v(S_2)$ for all S_1, S_2 — equivalently, a player's marginal contribution grows with the coalition it joins.

Sufficient conditions for a non-empty core: the game is convex (Shapley, 1971); or it is simple with at least one veto player (give everything to the veto players). In practice one can also try a natural allocation, such as the equal split, and check whether it lies in the core.

15.3. The Shapley value

The core is about stability; the **Shapley value** is about fairness. Shapley (1953) characterised a unique allocation rule by four axioms: **efficiency** ($\sum_i \phi_i = v(N)$), **symmetry** (interchangeable players get the same), **null player** (a player contributing nothing to any coalition gets 0), and **additivity** ($\phi(v+w) = \phi(v) + \phi(w)$).

Theorem 15.2 (Shapley value). The unique rule satisfying the four axioms is

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)],$$

player i 's **average marginal contribution** over all $n!$ orders in which the grand coalition can be assembled.

Example 15.1 (A veto player)

Three players; any coalition containing player 1 and at least one other is worth 1; everything else is worth 0. Player 1's marginal contribution is 1 whenever he is not first in the order (4 of the 6 orders), so $\phi_1 = \frac{2}{3}$; players 2 and 3 each contribute only when they join player 1 alone, $\phi_2 = \phi_3 = \frac{1}{6}$. The core, instead, is $\{(1, 0, 0)\}$: since $x_1 + x_2 \geq 1$ and $x_1 + x_3 \geq 1$ with $x_1 + x_2 + x_3 = 1$, both x_2 and x_3 must be 0. Fairness and stability can disagree. (For convex games the Shapley value always lies in the core.)

Non-superadditive games. When cooperation is not always beneficial (coordination costs, interference, overhead), the grand coalition need not form. The questions become how many coalitions emerge, how they form, and whether the resulting partition is stable. A constructive answer is the **merge-and-split** process: starting from any partition, coalitions merge if all members benefit, and split if some sub-coalition benefits; under suitable preference orders the process converges to a stable partition.

i Note

Applications of coalitional games in ICT include content sharing in P2P networks, spectrum sharing among operators, cooperative transmission and CoMP clustering in cellular networks, vehicular communication, localisation and cooperative crowdsensing.

15.3.1. Two standard families

Power in committees. In a weighted voting game $[q; w_1, \dots, w_n]$ a coalition wins iff its total weight reaches the quota q . The Shapley value of this simple game is the **Shapley–Shubik power index**: the fraction of orderings in which a player is *pivotal*, turning a losing coalition into a winning one. Power is not proportional to weight. In $[4; 3, 2, 1, 1]$ the weight-2 player is exactly as powerful as each weight-1 player, since any coalition it makes winning, a weight-1 player would too. The **Banzhaf index** instead counts, over unordered coalitions, how often a player is critical. A simple game has a non-empty core iff it has veto players, and then the core gives everything to them.

Cost sharing (airport games). Players need a facility of different sizes, with costs $c_1 \leq c_2 \leq \dots \leq c_n$, and a coalition pays the largest cost among its members, $C(S) = \max_{i \in S} c_i$. The Shapley value has a transparent form: the cost of each increment $c_k - c_{k-1}$ is shared equally among the $n - k + 1$ players who need it,

$$\phi_i = \sum_{k=1}^i \frac{c_k - c_{k-1}}{n - k + 1}, \quad c_0 = 0.$$

Small users pay only for the part of the runway they use. The same rule is used to share network infrastructure costs.

15.4. Nash bargaining

Return to two players splitting a surplus — a **Rich** and a **Poor** player sharing one unit, each keeping their share only if they agree. Chapter 11 modelled the negotiation as an alternating-offer game. Nash (1950) proposed instead to characterise the *outcome* axiomatically, without modelling any game at all.

Let $S \subset \mathbb{R}^2$ be a compact, convex set of feasible utility pairs and $d = (d_1, d_2)$ the **disagreement point**: what each gets if no agreement is reached — the worst threat a player can make.

Definition 15.7 (Nash’s axioms). A bargaining solution $s^*(S, d)$ should satisfy:

1. **Feasibility / individual rationality:** $s^* \in S$ and $s^* \geq d$;
2. **Pareto efficiency:** no $s \in S$ improves on s^* for one player without hurting the other;
3. **Symmetry:** if $d_1 = d_2$ and S is symmetric, then $s_1^* = s_2^*$;
4. **Invariance** to positive affine transformations of either player’s utility;
5. **Independence of irrelevant alternatives:** if $U \subset S$ is compact and convex and $s^*(S, d) \in U$, then $s^*(U, d) = s^*(S, d)$.

Theorem 15.3 (Nash bargaining solution). *The unique solution satisfying axioms 1–5 maximises the Nash product:*

$$s^* = \arg \max_{s \in S, s \geq d} (s_1 - d_1)(s_2 - d_2).$$

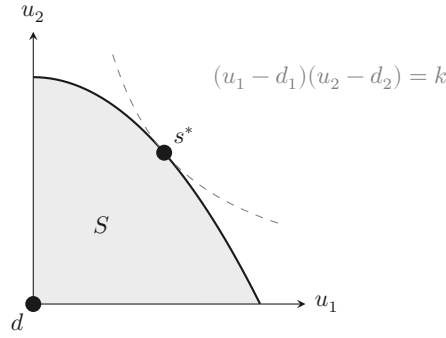


Figure: the Nash bargaining solution is the point of the Pareto frontier touched by the highest level curve of the Nash product.

Generalisations cover more than two players, asymmetric bargaining power (dropping symmetry gives the *generalised* NBS $\max \prod_i (s_i - d_i)^{\beta_i}$), and with $d = 0$ the NBS coincides with **proportional fairness**, widely used in resource allocation. A notable link: the Rubinstein SPE of Chapter 11 converges to the Nash bargaining solution as $\delta \rightarrow 1$.

Is the NBS fair? Fairness is subjective, and bargaining power matters. The rich player can tolerate rejecting a fair deal and claim more; the poor player accepts because even a little helps. As a solution concept the NBS provides Pareto optimality, accounts for bargaining power (sometimes unjustly), requires a convex utility region, and is static — it says nothing about the dynamics of reaching the agreement.

15.5. Kalai–Smorodinsky solution

Kalai and Smorodinsky (1975) replaced IIA, which they found unappealing, with **individual monotonicity**: if the feasible set is enlarged in a way that raises the maximum utility attainable by player i , player i should not lose. Together with axioms 1–4 this characterises the unique solution

$$s^{KS} = d + \lambda^*(I - d), \quad \lambda^* = \max\{\lambda : d + \lambda(I - d) \in S\},$$

where $I = (I_1, I_2)$ is the **ideal point** of individual maximal utilities. Graphically, s^{KS} is where the segment from d to I meets the Pareto frontier: each player gets the same *fraction* of their maximal possible gain.

The KS solution extends to more players, asymmetric cases and non-convex S , and rewards most the players with the largest potential gains. Neither solution is “right”: both are points on the Pareto frontier, and the choice between them is philosophical. Constructive procedures that converge to each exist, as do dynamic bargaining models.

Example 15.2 (Interference channel)

Two transmitter–receiver pairs share a frequency-selective channel and interfere with each other. The non-cooperative Nash equilibrium is inefficient: each transmits at full power and beamforms only to maximise its own rate. Using the NE rates as the disagreement point, the NBS (or KS) picks a Pareto-efficient operating point on the rate region — whose shape, not the bargaining formula, is the hard part of the problem. In multimedia transmission the Pareto frontier of a resource-sharing problem is easy to find, but QoE utility functions make the region complex and fairness delicate.

15.6. Exercises

Further problems (not covered by the handbook).

Exercise 15.1

(Weighted voting) Four parties hold weights 3, 2, 1, 1 and decisions need 4 votes. Compute the Shapley–Shubik and Banzhaf indices and decide whether the core is non-empty.

Solution

Of the 24 orderings, A (weight 3) is pivotal in 12 and each of B, C, D in 4: Shapley–Shubik $(\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$. Banzhaf counts: A is critical in 6 coalitions, each other party in 2, giving normalised $(\frac{1}{2}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6})$ here as well. B 's extra weight buys no power. There is no veto player ($\{B, C, D\}$ wins without A), so the core is empty: any allocation leaves some winning coalition with less than 1.

Exercise 15.2

(Airport game) Three airlines need runways costing 1, 2, 3; a shared runway must be as long as the largest need. Find the Shapley cost shares and check that they are in the core of the cost game.

Solution

Increment 1 is shared by three, increment 2–1 by two, increment 3–2 by one: $\phi = (\frac{1}{3}, \frac{1}{3} + \frac{1}{2}, \frac{1}{3} + \frac{1}{2} + 1) = (\frac{1}{3}, \frac{5}{6}, \frac{11}{6})$, summing to 3. Core of a cost game: no coalition pays more than its stand-alone cost. $\{1\}$: $\frac{1}{3} \leq 1$; $\{2\}$: $\frac{5}{6} \leq 2$; $\{3\}$: $\frac{11}{6} \leq 3$; $\{1, 2\}$: $\frac{7}{6} \leq 2$; $\{1, 3\}$: $\frac{13}{6} \leq 3$; $\{2, 3\}$: $\frac{8}{3} \leq 3$. In the core: airport games are concave cost games, for which the Shapley value is always in the core.

From the course material:

Exercise 15.3

Three firms can share a facility: $v(\{i\}) = 0$, $v(\{i, j\}) = 4$ for every pair, $v(\mathcal{N}) = 9$. Is the core non-empty? Compute the Shapley value.

Solution

By symmetry try $x = (3, 3, 3)$: every pair gets $6 \geq 4$, so the core is non-empty (it is the set of $x \geq 0$ summing to 9 with every pair getting at least 4, i.e. $x_i \leq 5$). The game is symmetric, so the Shapley value is $(3, 3, 3)$, in the core. (The game is convex: marginal contributions 0, 4, 5 increase with coalition size.)

Exercise 15.4

Two players split one unit with disagreement point $d = (0.2, 0)$ and linear utilities. Find the Nash bargaining solution and compare it with the Kalai–Smorodinsky solution.

Solution

On the frontier $s_2 = 1 - s_1$: maximise $(s_1 - 0.2)(1 - s_1)$, giving $s_1 = 0.6$, $s^* = (0.6, 0.4)$: each player gets their disagreement payoff plus half the surplus 0.8. Ideal point: $I_1 = 1$ (with $s_2 = 0 \geq d_2$), $I_2 = 0.8$ (with $s_1 = 0.2$). The KS segment $(0.2 + 0.8\lambda, 0.8\lambda)$ meets $s_1 + s_2 = 1$ at $\lambda = 0.5$: $s^{KS} = (0.6, 0.4)$. For this linear frontier the two coincide.

15.7. Sources

Prof. Badia, *Game Theory* slides, deck Gth2018 (coalitional games, core, Bondareva–Shapley, convex games, merge-and-split, Nash bargaining, Kalai–Smorodinsky, applications). Original references: L. Shapley, *A value for n -person games* (1953); O. Bondareva (1963) and L. Shapley (1967) on balancedness; L. Shapley, *Cores of convex games*, Int. J. Game Theory **1**, 11 (1971); J. Nash, *The bargaining problem*, Econometrica **18**, 155 (1950); E. Kalai and M. Smorodinsky, Econometrica **43**, 513 (1975); A. Leshem and E. Zehavi, *Game theory and the frequency selective interference channel*, IEEE Signal Proc. Mag. **26**, 28 (2009).